AutoHaul™ Project

Construction Activities for Rail Infrastructure Adjacent to the Tom Price Rail Line within the Millstream-Chichester National Park

Mining Proposal

L47/47 and L47/228

October, 2012

Robe River Mining Co. Pty Ltd

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EXECUTIVE SUMMARY

Robe River Mining Co. Pty Ltd (**the Company**), a wholly owned subsidiary of Rio Tinto, proposes to automate mainline locomotive operations (AutoHaul™) from the ports to mines, excluding the Deepdale line west of Emu siding.

This Mining Proposal (**MP**) has been prepared to request authority to use Miscellaneous Licences (**L**) L47/47 and L47/228 to undertake the works associated with AutoHaul™ (the **Project**) located adjacent to the Tom Price Rail Line within the Millstream-Chichester National Park (**MCNP**). Due to the construction and commissioning schedule of the overall project, the works are proposed to commence in Q1 2013 and be completed during Q4 2014. The planned works that will be undertaken as part of the Project will include, but are not limited to:

- Construction of signalling pads and associated infrastructure;
- Installation of communication and signalling equipment;
- Trenching, directional drilling and installation of new signal cables;
- Construction of permanent access roads and ramps;
- Construction of drains for storm water flow;
- Use of existing metered water supplies within the area for construction water;
- Installation of fencing and gates for passive level crossings;
- Topsoil and spoil stockpiling for use in construction and rehabilitation; and
- Rehabilitation of areas not required for permanent infrastructure.

The disturbance area shall not 4.3 hectares (**ha**) of which no more than 0.98 ha shall be permanent.

The Company is aware of its responsibilities under the *Aboriginal Heritage Act 1972* (WA). Company policy is for no clearing and/or earthworks to commence until heritage surveys have been undertaken. Should clearing and earthworks discover new heritage material / site(s) then suitable measures will be taken as per regulatory requirements and Company procedures.

Environmental aspects and factors considered relevant to this proposal have been identified, assessed and addressed within the Project Construction Environmental Management Plan (**CEMP**) and are summarised in this MP.

No Threatened Ecological Communities (**TEC**) or Declared Rare Flora (**DRF**) were recorded in the proposed work area. No Schedule listed fauna, or fauna species listed under the *Environmental Protection and Biodiversity Conservation Act 1999* (**EPBC Act**) were recorded.

This portion of the Project will be located adjacent to the existing railway corridor within the MCNP and will traverse the Harding Dam Catchment Area, a Priority 1 Public Drinking Water Source Area (**PDWSA**).

The location of the development, within an existing disturbance corridor, indicates a low risk of significant impact to any environmental receptor.

In summary, the Project is largely contained within an existing disturbed footprint and is unlikely to significantly impact on the existing natural and social environment.

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APPENDICES

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1.0 ABBREVIATIONS

AHA	*Aboriginal Heritage Act* 1972
Biota	Biota Environmental Sciences Pty Ltd
DEB	Deepdale Emu Bypass Chainages
CEMP	Construction Environmental Management Plan
Ch	Chainage
Company	Robe River Mining Co. Pty Ltd
DAFWA	Department of Agriculture and Food, Western Australia
DEC	Department of Environment and Conservation
DMP	Department of Mines and Petroleum
DoW	Department of Water
DRF	Declared Rare Flora
EPA	Environmental Protection Authority
EP Act	*Environmental Protection Act 1986*
EPBC Act	*Environmental Protection and Biodiversity Conservation Act* 1999
FOC	Fibre Optic Cable
GPS	Global Positioning System
ha	Hectare
HSEQ	Health, Safety, Environmental and Quality Management System
IBRA	Interim Biogeographic Regionalisation of Australia
kp	Kilometre Point
LAA	*Land Administration Act* 1997
Landgate	Western Australian Land Information Authority
MCNP	Millstream Chichester National Park
MP	Mining Proposal

mtpa	million tonnes per annum
NATA	National Association of Testing Authorities
NVCP	Native Vegetation Clearing Permit
OC	Operations Centre
PDWSA	Public Drinking Water Source Area
PEC	Priority Ecological Community
RCE	Rail Capacity Enhancement
RIWI Act	*Rights in Water and Irrigation Act 1914*
RMT	Rail Maintenance Track
RRIA	Robe River Iron Associates
RTIO	Rio Tinto Iron Ore
RTIOEP	Rio Tinto Iron Ore Expansion Projects
RTRD	Rio Tinto Railway Division
TEC	Threatened Ecological Community
WPWSS	West Pilbara Water Supply Scheme

2.0 INTRODUCTION

2.1 Proponent

Robe River Mining Co. Pty Ltd (**the Company**), a wholly owned subsidiary of Rio Tinto, proposes to automate mainline locomotive operations (AutoHaul™) from the ports to mines (excluding the Deepdale line west of Emu siding).

This Mining Proposal (**MP**) has been prepared to request authority to use Miscellaneous Licences (**L**) L47/47 and L47/228 to undertake the works associated with AutoHaul™ (the **Project**) located adjacent to the Tom Price Rail Line within the Millstream Chichester National Park (**MCNP**). This MP outlines these supporting works. Due to the construction and commissioning schedule of the overall project, the works are proposed to commence in Q1 2013 and be completed during Q4 2014.

Miscellaneous Licences L47/47 and L47/228 are held by Robe River Iron Associates joint venture (**RRIA**), which is managed by the Company and comprises:

- Mitsui Iron Ore Development Pty Ltd

  C/- John Smith, General Manager Administration GPO Box 2585, Perth, WA 6001

- North Mining Ltd

  C/- Rio Tinto Iron Ore, Land Assets Department GPO Box A42, Perth, WA 6837

- Robe River Mining Co. Pty Ltd

  C/- Rio Tinto Iron Ore, Land Assets Department GPO Box A42, Perth, WA 6837

- Cape Lambert Iron Associates (Registered Business Name)

  C/- John Smith, General Manager Administration GPO Box 2585, Perth, WA 6001

- Pannawonica Iron Associates (Registered Business Name)

  C/- John Smith, General Manager Administration GPO Box 2585, Perth, WA 6001

2.2 Location and Tenure

The proposed activities will be located on L47/47 and L47/228 the purpose of which are '*to conduct all necessary activities for the design, planning, construction, operation and maintenance of a railway and all associated infrastructure in connection with mining operations pursuant to the provisions of Iron Ore (Robe River) Agreement Act 1964*'. The purpose of these tenements is considered suitable for the proposed activities.

The Project description is provided in Section 3.0 of this MP and the proposed activities relating to this portion of the Project are detailed in Figures 1 to 3. A majority of the proposed works associated with the Project will take place on the *Land Administration Act* 1997 (**LAA**) tenure (LGE I195323). The works on the LAA tenure are not subject to approval via the MP process and will only be undertaken once all of the necessary approvals from the relevant regulatory authorities have been obtained.

This portion of the Project will be located adjacent to the existing railway corridor within the MCNP and will traverse the Harding Dam Catchment Area, a Priority 1 Public Drinking Water Source Area (**PDWSA**).

A completed MP Checklist is included in Appendix 1.

2.3 Project Overview

Rio Tinto Iron Ore (**RTIO**) is currently expanding its iron ore operations in the Pilbara from 220 million tonnes per annum (**mtpa**) to 353 mtpa as part of the Rail Capacity Enhancement (**RCE**) Project. This involves the establishment of new and expansion of existing mines, expansion of its existing port facilities at Dampier and Cape Lambert and the upgrade of its existing rail network and associated infrastructure to accommodate this additional capacity.

RTIO's AutoHaul™ Autonomous Train Project forms part of the RCE initiative. The AutoHaul™ system will automate mainline locomotive operation from the ports to mines (excluding the Deepdale line west of Emu siding). This shall be achieved by integrating additional technology with existing proven train management systems of the RTIO rail network.

This MP covers a portion of the works (adjacent to the Tom Price Rail Line within the MCNP) and have been prepared to allow for the upgrade of the existing Rio Tinto Rail Division (**RTRD**) network with the installation of upgraded passive level crossings, a narrow band shunt and signal pads. Appendix 2 includes detailed descriptions of the proposed equipment and infrastructure to be located within the Miscellaneous Licences.

The proposed activities within the Miscellaneous Licences include:

- Installation of Passive Level Crossing at 72.5kp, 78.7kp, 81kp and 114.8kp.
- Installation of Fibre Optic Cable (**FOC**) at 76.4 to 77.4kp.
- Construction of a Signal Pad and associated equipment at 77.5km.
- Installation of a Narrow Band Shunt at 108.5kp.

This MP details all proposed works and associated infrastructure and is intended to provide information to the Director, Environment, Department of Mines and Petroleum (**DMP**) to meet the requirements of the tenure conditions.

MP REG ID 36900 was submitted to DMP on 04 September 2012 and covered the remainder of the AutoHaul™ sites outside MCNP from Ibis to Rosella sidings.

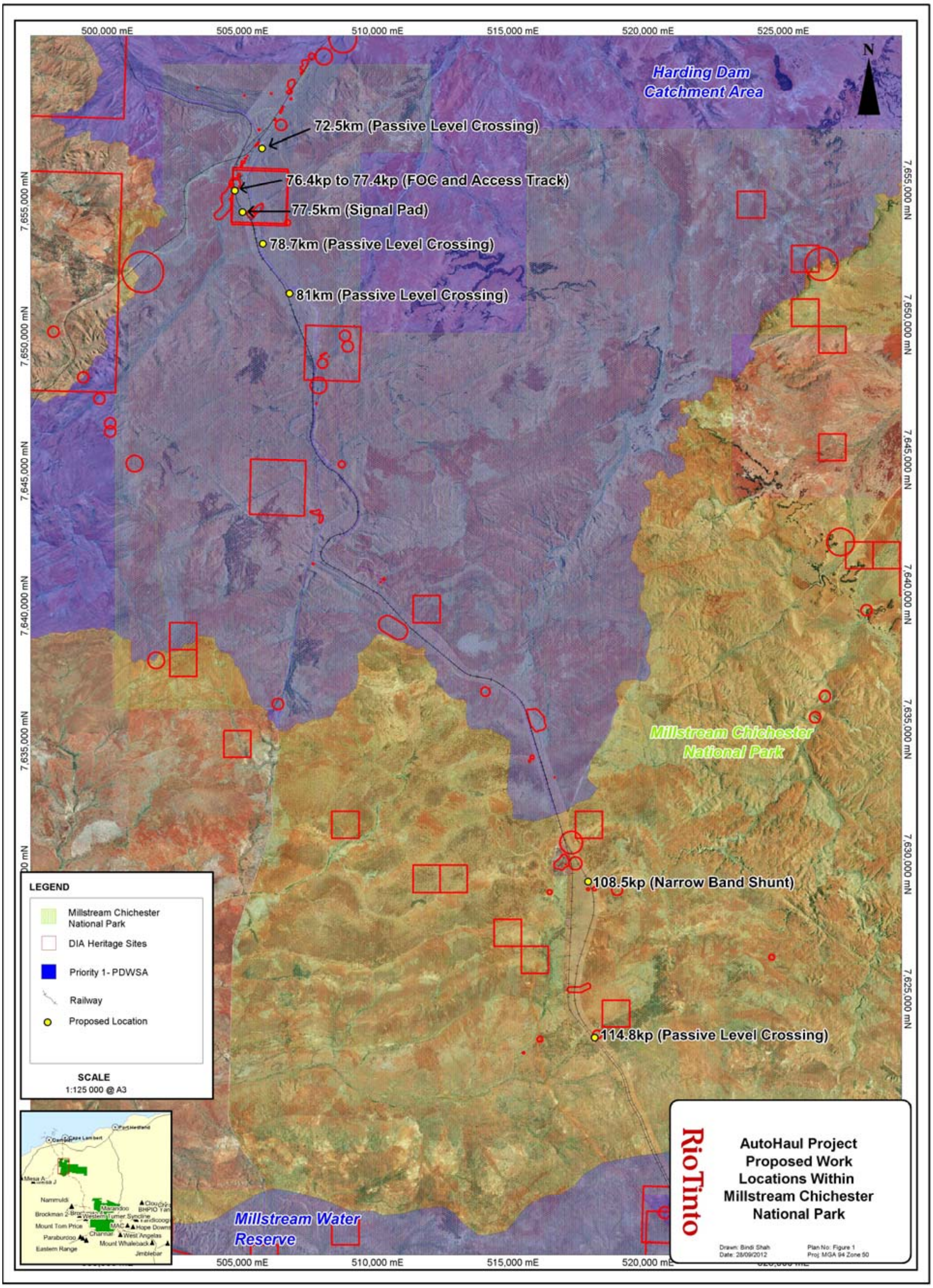

Figure 1 Proposed Work Locations in MCNP

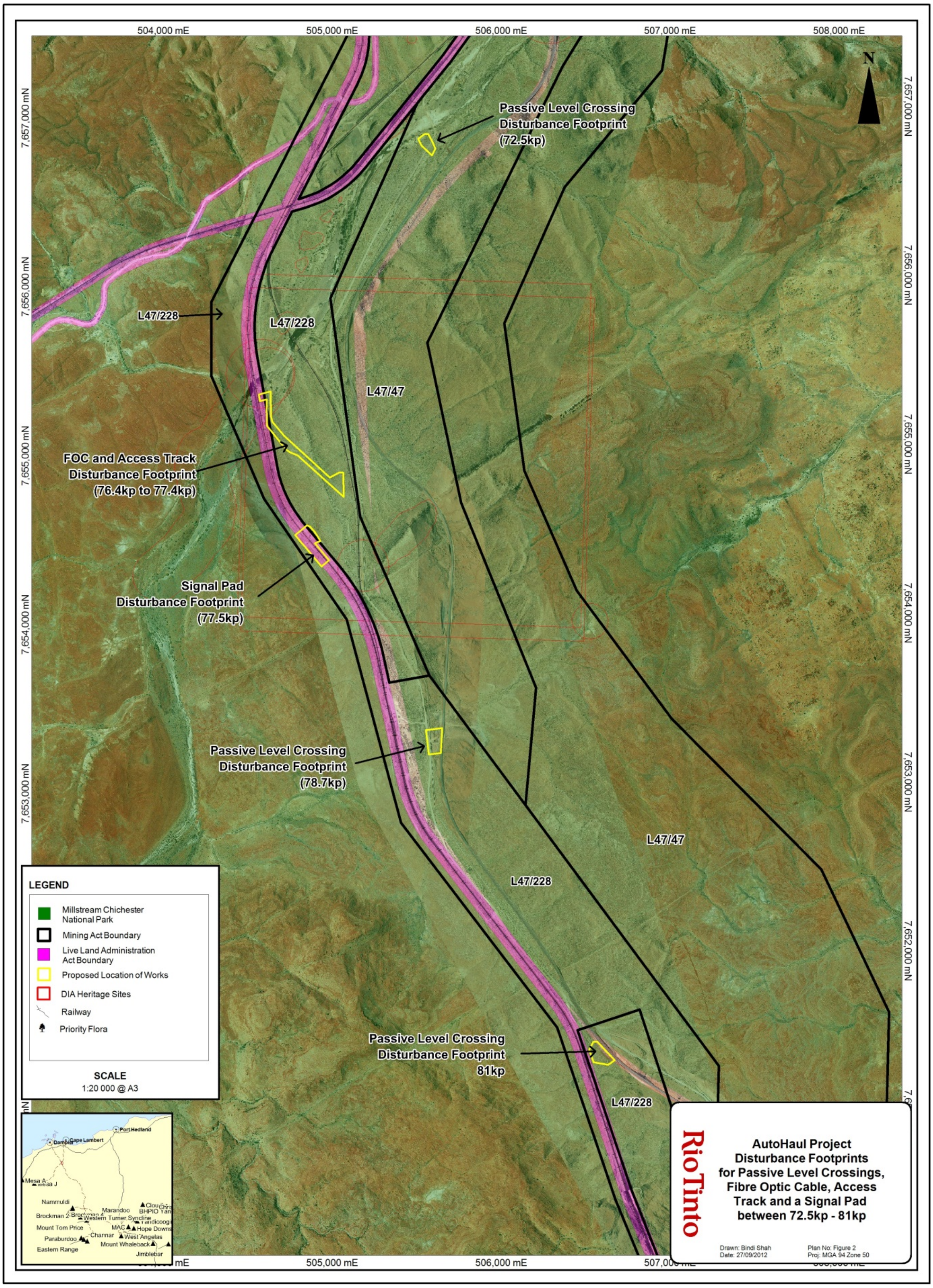

Figure 2 Disturbance Footprint for Passive Level Crossings and Signal Pad between 72kp and 82kp

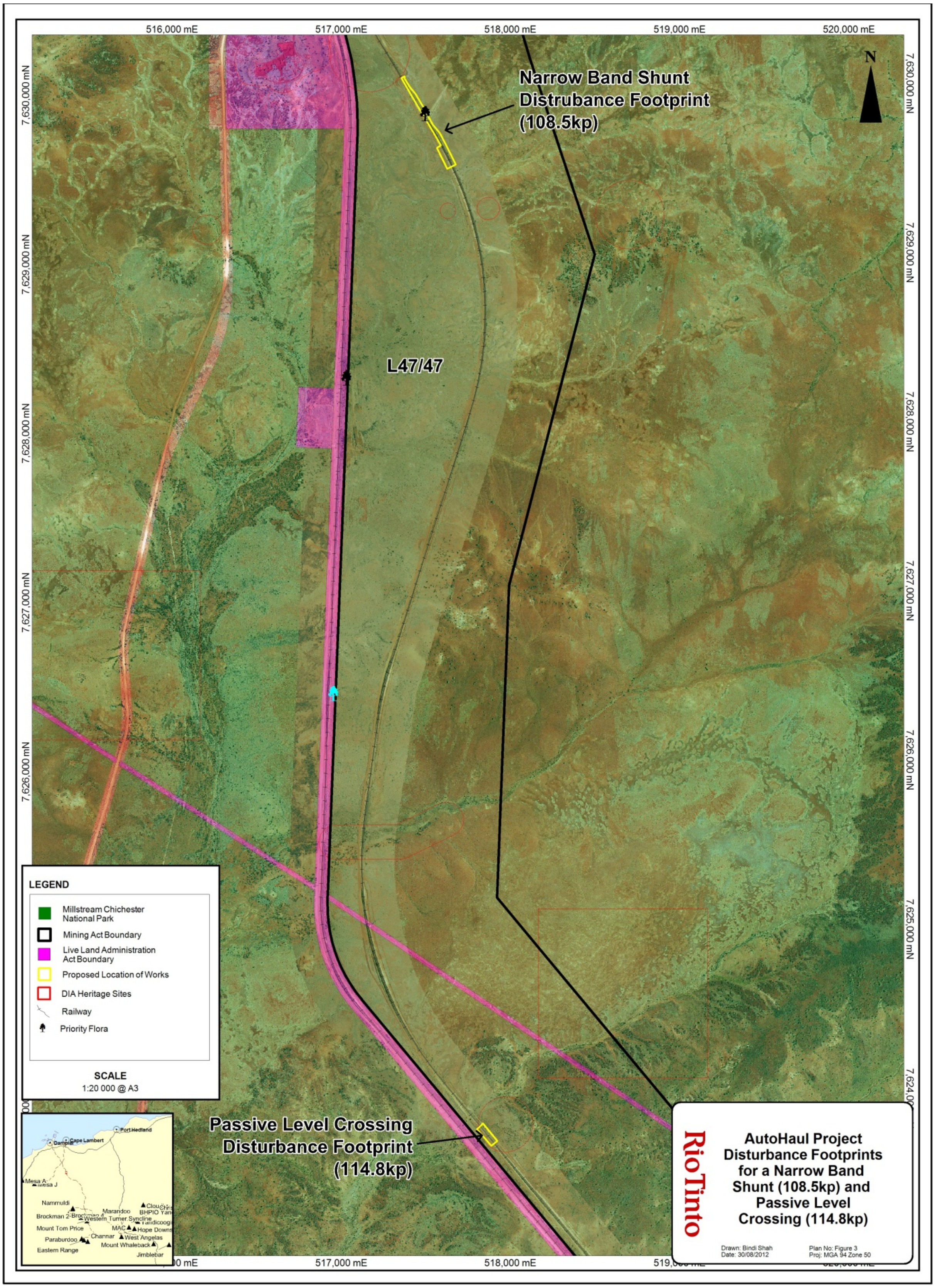

Figure 3 Disturbance Footprint for Passive Level Crossings and Narrow Band Stunt between 105kp and 115kp

3.0 PROJECT DESCRIPTION

3.1 Project Objectives

The AutoHaul™ system will automate mainline locomotive operation from the ports to mines, excluding the Deepdale line west of Emu siding. This shall be achieved by integrating additional technology with existing proven train management systems of the RTIO rail network. A completed AutoHaul™ system shall comprise the following major elements:

- An automated train driving system that computes the required power and brake settings to control trains within predetermined operational and safety parameters;
- The computer control of all the major operational scenarios needed to safely move trains between mine sites and the coastal yards;
- Operations Centre (**OC**) equipment to allow their staff to configure trains for automatic operation, monitor train and locomotive performance, and intervene in the event of abnormal system behaviours;
- A data network to allow communication between the OC and the on-board systems of automated trains;
- Additional wayside systems to support driverless operation to, from and in mine loops; and
- A substantial increase in level crossing protection at public and private access crossings.

The works in this MP associated with the Project includes, but is not limited to:

- Construction of signalling pads and associated infrastructure;
- Installation of communication and signalling equipment;
- Trenching, directional drilling and installation of new signal cables;
- Construction of permanent access roads and ramps;
- Construction of drains for storm water flow;
- Use of existing metered water supplies within the area for construction water;
- Installation of fencing and gates for passive level crossings;
- Topsoil and spoil stockpiling for use in construction and rehabilitation; and
- Rehabilitation of areas not required for permanent infrastructure.

3.2 History

Pastoralism has been active in the region for the past 100 years with grazing of sheep until about 1970 and cattle thereafter. The RTRD network was established in the late 1960s as a means of transporting iron ore for export from the Tom Price Mine in the Pilbara to port facilities at Parker Point.

The railway has been expanded over time to service newly established mines and increased production at existing operations such that the current capacity of the rail network is 220 mtpa. The company proposes to further expand the rail network capacity to 353 mtpa.

Duplication of the track and other modifications to the current rail network are required to cater for the increase in rail traffic associated with the railing of ore from the mines to the port in a safe and efficient manner. Modification of the existing rail network, as outlined, shall ensure that this is achievable.

3.3 Area of Disturbance

Table 1 to 3 detail the proposed disturbance footprints required for construction activities within L47/47 and L47/228. The disturbance area shall not exceed 4.3 hectares (**ha**) of which no more than 0.98 ha shall be permanent (Table 3).

Note: Location of temporary facilities are subject to minor changes based on site conditions but will meet all the environmental commitments and requirements of this MP and the AutoHaul™ Construction Environmental Management Plan (**CEMP**) (Appendix 3). Existing tracks shall be used where practicable and may require additional clearing and grading for safety purposes.

Table 1 Disturbance Areas for L47/47

Tenement	Description of Disturbance	Temporary Disturbance (ha)	Permanent Disturbance (ha)
L47/47	Passive Level Crossing (72.5kp)	0.1	0.005
L47/47	Narrow Band Shunt (108.5kp)	0.5	0.015
L47/47	Passive Level Crossing (114.8kp)	0.1	0.005
Total (ha)		**0.7**	**0.03**

Table 2 Disturbance Areas for L47/228

Tenement	Description of Disturbance	Temporary Disturbance (ha)	Permanent Disturbance (ha)
L47/228	Passive Level Crossing (78.7kp)	0.3	0.005
L47/228	Passive Level Crossing (81kp)	0.2	0.005
L47/228	FOC (76.4kp – 77.4kp)	1.4	0.53
L47/228	Access Track for FOC (76.4kp – 77.4kp)	1.4	0.39
L47/228	Signal Pad including FOC connection (77.5kp)	0.3	0.02
Total (ha)		**3.6**	**0.95**

Table 3 Summary of Disturbances

Tenement	Temporary Disturbance (ha)	Permanent Disturbance (ha)
L47/47	0.7	0.03
L47/228	3.6	0.95
Total (ha)	**4.3**	**0.98**

3.4 Operations

The proposed construction works will commence in Quarter 1 2013 with commissioning works set to finish in Quarter 4 2014. The Company shall manage construction works and oversee maintenance of the new infrastructure during the operating life of the rail network.

3.5 Workforce

Construction personnel associated with this MP for works will be accommodated at Northern Link Camp (65kp on the Tom Price – Dampier rail line). The construction workforce is expected to be approximately 30 persons at peak time. Personnel shall be sourced locally or from fly in fly out contractors.

3.1 Fibre Optic Cable Installation

A portion of the fibre optic cable is required to be relocated as part of the signal pad works.

Route Preparation includes, but is not limited to:

- Clearing of the route approximately 10-15 metres wide to allow machinery and any support vehicles to operate safely;

- Exposure and relocation if the existing underground services which conflict with the proposed cable depth and alignment;
- Directional drilling and/or under boring of roads, railways and creeks where required;
- Preparation of creek crossings to ensure the depth of cover is adequate under all conditions; and
- FOC installation will be pre-ripped to a depth that provides a sufficient bed of fines for the FOC.

The FOC will be laid in a rift created by the dozer tyne (or similar), and will be backfilled as soon as practicable, thus preventing opportunities for fauna entrapment.

Reinstatement works involve areas to be cleared of all excess materials, road base, stones, clay and any other foreign materials so the route is left clean and tidy. Minimisation of environmental impacts will be achieved through control techniques such as reducing vegetation disturbance particularly riparian vegetation. The areas cleared for the installation of FOC will be rehabilitated by respreading topsoil and vegetation cleared.

3.2 Signalling and Communications

Signalling, asset protection and communications infrastructure will be constructed and installed along the length of the rail works. These works comprise of location cases and equipment rooms, housing electronic equipment that is used to ensure the safe and reliable operation of the railway. Depending on the type of installation, the work varies, but in general comprises of:

- Clearing of appropriate areas for access, equipment, cabling and earthing;
- Installation of concrete foundations, concrete pits and pipe (conduit);
- Installation of earthing rods and connections (utilising vertical drill rigs);
- Installation of location cases or equipment rooms;
- Installation of appropriate power systems, either mains powered or solar;
- Installation of battery backup power systems;
- Installation of communication masts;
- Trenching of signalling, power, earthing and communication cables;
- Horizontal boring to allow cable connections under existing infrastructure or geographical features; and
- Installation of specialised signalling and communications equipment as required.

3.3 Civil Works

Minor civil works will be required to support the installation of the signalling and communications equipment. These works comprise of infrastructure pads, widening/grading of existing access roads and ramps and the installation of gates and fencing. All civil works have been designed to ensure that existing drainage lines are maintained with the installation of floodways where required.

3.4 Laydown Areas

Laydown areas are required for the storage of the construction material, vehicle parking, crib, toilet facilities, equipment laydown and material conditioning. Vehicle traffic is anticipated in these areas and therefore sheeting of some form of road base material will be required in some areas.

Laydown areas shall be located on pre-disturbed areas within the proposed disturbance areas to minimise environmental impacts. In addition topsoil and spoil removed at the site of construction will be stockpiled in the laydown areas.

3.5 Construction Water

Water for construction purposes will be sourced from existing and new metered take-off points from the Water Corporation West Pilbara Water Supply - Millstream Supply Main pipeline along the proposed alignment.

In addition, if required, construction water may also be sourced from licenced water bores along the proposed rail alignment to reduce the water demand required from the Water Corporation West Pilbara Water Supply - Millstream Supply Main pipeline.

3.6 Resource Requirements

The required resources for the Project include:

- A construction workforce of approximately 30 persons at peak;
- Accommodation at Northern Link Camp;
- Water for construction will be sourced from the Water Corporation West Pilbara Water Supply - Millstream Supply Main pipeline or existing water bores; and
- Hydrocarbons will be stored off site in purpose built storage and distribution facilities and will be stored and handled as outlined in Section 6.8 of this Mining Proposal. Further details can also be found in the CEMP Section 6.11 (Appendix 3).

3.7 Existing Facilities

The infrastructure within the vicinity of the Project is part of the existing RTRD Rail Network which is also referred to as the Tom Price rail line. The nearest town is Karratha.

Access to the works will be via the North West Coastal Highway along the Rail Maintenance Track (**RMT**) or the Karratha to Tom Price Road depending on the location of the works. From these roads, personnel will enter the existing rail maintenance tracks which form part of RTRD network or purpose built new access roads.

4.0 EXISTING ENVIRONMENT

4.1 Regional Setting

The Pilbara forms part of the Great Plateau of Western Australia with the Hamersley plateau forming the most extensive elevated land area in Western Australia. Landform in the Project area is dissected so that it consists of residual uplands and ridges, standing above drainage basins.

MCNP contains a diverse and unique terrestrial and aquatic natural environment, and supports significant Indigenous cultural values. The major landforms in the MCNP lie on an east-west axis and include the Fortescue Valley, the Chichester Range, the Chichester tablelands between the ranges, and the Roebourne Plains. The Roebourne Plains run inland until they strike the Chichester escarpment, which rises to about 350 m above sea level. Behind the Chichester escarpment is the Chichester tablelands. The Project detailed within this MP is located within the Chichester Range landform.

4.2 Geology and Soils

The Project lies wholly within the Chichester Pilbara Interim Biogeographic Regionalisation of Australia (**IBRA**) subregion.

The Chichester subregion is described as undulating Archean granite and basalt plains, including significant areas of basaltic ranges (CALM, 2002). The soils of the Chichester Ranges are predominantly shallow, porous, loamy soils.

4.3 Hydrology

Rivers in the Pilbara region are ephemeral and flow only occasionally. Sheet flow areas characteristically occur where there is poorly defined drainage and low gradients, or in flat or gently undulating areas where ponding occurs.

The Project will traverse the Harding Dam Catchment Area, a Priority 1 PDWSA. The Harding River is formed by the junction of three main tributaries: Western Creek, Harding River and Harding River East. This catchment covers an area of approximately 1,100km^2 and runoff is high due to intense rainfall events and the relatively impervious nature of the catchment.

The groundwater occurrences are limited to shallow unconfined alluvial or fractured rock aquifers and are generally hydraulically connected. There is also significant surface water to groundwater interaction. Groundwater flow in the alluvial aquifers is generally in the same direction as stream flow. Drainage channels provide recharge to the underlying fractured rock aquifers during periods of river flow, in between rainfall events the subsurface flow and groundwater levels will likely follow a recession pattern. The groundwater quality in fractured rock and alluvial aquifers ranges from fresh to brackish.

Along all creeks and rivers in the Pilbara, where riverine vegetation has established, it is likely that this vegetation is supported by groundwater in the river alluvium.

4.4 Climate

The Project lies within a semi-arid area of Western Australia and is characterised by high temperatures, low and variable rainfall and high evaporation. Between October and April, average temperatures reach or exceed 32°C every day. Temperatures can range up to 47°C (BOM, 2012). In the winter months, the average temperature is 25°C.

The average rainfall for the area is 200 to 350 mm, although this is highly intermittent and is generally associated with cyclones and thunderstorms which occur mainly in the summer months. January and February are generally the wettest months of the year while September and October are the driest.

4.5 Vegetation and Flora

The Project falls within the Chichester subregion of the Pilbara Interim Biogeographic Regionalisation for Australia (**IBRA**) region. The Chichester subregion comprises undulating Archaean granite and basalt plains including significant areas of basaltic ranges (Kendrick and McKenzie 2001). Plains support a shrub steppe characterised by *Acacia inaequilatera* over *Triodia wiseana* (formerly *Triodia pungens*) hummock grasslands, while *Eucalyptus leucophloia* tree steppes occur on ranges.

The quality of vegetation varies across the Project area due to varying degrees of grazing and construction related impact and previously cleared areas that have been rehabilitated (Biota, 2008a; Biota, 2010; Rio Tinto, 2012). The majority of the areas are considered to be in Good condition.

Three priority flora species were identified within 2km of the Project area:

- Priority 3: *Themeda* sp. Hamersley Station (M.E. Trudgen 11431);
- Priority 3: Rostellularia adscendens var. latifolia; and
- Priority 4: *Goodenia nud*a.

Themeda sp. Hamersley Station (M.E. Trudgen 11431) was identified within the proposed NBS at 108.5kp at one location. Priority flora will be managed in accordance with Section 6.2 and Management Procedure 6.4 in the CEMP (Appendix 3).

4.5.1 Declared Rare Flora

No Declared Rare Flora (**DRF**) species or flora species listed under the *Environmental Protection and Biodiversity Conservation Act 1999* (**EPBC Act**) were recorded and none are expected to occur within the Project footprint (Rio Tinto 2012).

Copies of these reports can be provided on request.

4.5.2 Introduced Flora

Five species of weeds occur within the Project area, with no Declared Plant (declared weeds) according to the *Agriculture and Related Resources Protection Act 1976*. However, a number of species are considered to be serious environmental weeds, particularly Kapok Bush (*Aerva javanica*), two Cenchrus species (*Buffel and Birdwood grass*) and Mimosa Bush (*Vachellia farnesiana*).

4.6 Fauna

No Schedule 1 fauna or other protected fauna as listed under the federal *EPBC Act* or by the DEC were recorded within the project area.

4.7 Threatened Ecological Communities / Priority Ecological Communities

There are no Threatened Ecological Communities (**TEC**) listed by the DEC within the Project area (Biota, 2010).

4.8 Millstream Chichester National Park

This portion of the Project is located within the Class A Nature Reserve – MCNP. The MCNP contains significant conservation, recreation and heritage values. Management of the construction workforce adjacent to the MCNP is outlined in Section 6.14 of this MP and CEMP Section 6.21 (Appendix 3).

A Communications Protocol has been developed in collaboration with the DEC and the rangers of the MCNP. The Communications Protocol is included in Appendix 5. The Protocol aims to provide transparency to the works in MCNP and ongoing communications between Rio Tinto and the DEC.

4.9 Social Environment

4.9.1 Aboriginal Heritage

The Project lies within two Native Title claim areas of the Ngarluma and Yindjibarndi people.

A search of the Department of Indigenous Affairs (**DIA**) Aboriginal Enquiry System indicates that there are four registered sites that intersect with the proposed footprints between 75 kp and 77 kp. (see Figure 1).

The Project area has been surveyed for heritage sites and none were identified. In the event new heritage material / site(s) are found during construction, then suitable measures will be taken as per regulatory requirements (*Aboriginal Heritage Act 1972* (**AHA**)) and Company procedures. All construction activities will be managed as per the CEMP Section 6.6 (Appendix 3) to avoid and/or minimise impacts to Cultural Heritage aspects. Heritage sites in the vicinity will be fenced according to Rio Tinto Iron Ore Expansion Projects (**RTIOEP**) Heritage Fencing Guidance note (RTIO-HSE-0121790).

4.9.2 Adjacent Land Use and Tenure

The Project area traverses a broad cross-section of land ownership and tenure including various tenements under the Western Australia *Mining Act* 1978, leases under the LAA and freehold land as administered by the Western Australian Land Information Authority (**Landgate**).

The Project area is inside environmentally sensitive areas such as MCNP and Harding Dam Catchment.

Copies of the MP have been forwarded to the Department of Environment and Conservation (Michelle Corbellini – Environmental Impact Assessment for the Pilbara; Stephen Breedveld – Acting Senior Ranger MCNP) and the Department of Water (Natalie Leach – Natural Resource Management Officer, Pilbara Region).

5.0 COMPLIANCE WITH LEGISLATION

5.1 Environmental Protection Act 1986 (WA)

This project is not regarded as warranting referral to the Environmental Protection Authority (**EPA**) under Section 38 of the *Environmental Protection Act 1986* (**EP Act**) by virtue of its minimal impact on the environment.

The undertaking of the works associated with the Project where native vegetation is being clearing is be managed through Part V of the EP Act through the Native Vegetation Clearing Permit (NVCP) process and other necessary approvals. RTIO has obtained an NVCP for the proposed works in MCNP (CPS 5117/1) and have recently requested an amendment to include an additional site for the signal pad, FOC and access track between 75 kp and 77 kp.

5.2 State Agreement

State Agreements approval associated with the Work as outlined in the Mining Proposal were approved by the Minister for State Development on the 7 May 2012 under the Wildflower Camp and Integrated Use – Detailed Proposal.

5.3 Mining Act 1978 (Tenure Conditions)

Installation and upgrade of infrastructure associated with the Project occurs on tenure which is suitable for this purpose. The Miscellaneous Licences L47/47 and L47/228 have been granted under the *Mining Act 1978*. All tenement conditions will be complied with as indicated in Appendix 6.

5.4 Rights in Water and Irrigation Act 1914

Water required to support construction of the Project will be preferentially sourced from the West Pilbara Water Supply Scheme (**WPWSS**) existing metered take off points. Should additional water be required, licenced bores will be accessed to obtain further water for construction.

5.5 Local Government

The Company shall obtain all of the necessary local government approvals from the Local Councils within the Project area in relation to the construction of the proposed works. This shall include Building Permits for all associated facilities, if required.

6.0 ENVIRONMENTAL IMPACTS AND MANAGEMENT

The company is ISO 14001:2004 certified for its Health, Safety, Environmental and Quality Management System (**HSEQ**). The primary purpose of HSEQ is to demonstrate sound health, safety, environmental and quality performance consistent with the business Policy and Objectives. Under HSEQ, environmental aspects and impacts have been identified and subsequently detailed in the CEMP that was developed for the Project (Appendix 3). Environmental management of key environmental issues and rehabilitation shall be in accordance with the Project CEMP (Appendix 3) and associated HSEQ documents. The HSEQ documents referenced in the CEMP, or the sections below, can be provided on request.

Associated management actions for potential impacts identified include, but are not limited to, the following:

- Ground disturbance and clearing management;
- Topsoil management;
- Flora and fauna management;
- Weed management;
- Heritage management;
- Dust management;
- Surface water, groundwater and drainage management;
- Hydrocarbon management;
- Hazardous materials management;
- Noise and vibration management;
- Fire management;
- Greenhouse gas management;
- Demobilisation, Rehabilitation and Handover; and
- Workforce Management and Public Access within MCNP.

Land clearing activities are controlled by the Rio Tinto Approvals Coordination process which ensures that all heritage and biological surveys are undertaken, tenure obtained (and appropriate for the activity) and that other necessary approvals are obtained prior to clearing activities. In addition to this process other levels of permit controls are required for individual clearing activities.

The CEMP (Appendix 3) will be used to manage all known and potential environmental risks and impacts associated with the construction of infrastructure associated with the Project. The key management actions for each identified environmental impact (as detailed in the CEMP) are summarised in the relevant sections below.

6.1 Land Clearing

Land clearing will be managed under Management Procedure 6.2 in the CEMP (Appendix 3). Key environmental management actions for land clearing are summarised below:

- All personnel shall be informed of the importance of ground disturbance and clearing management and the requirement to keep to existing tracks and roadways.

- The boundary of the area to be cleared will be surveyed and suitably defined with survey tape or equivalent to minimise the risk of disturbance outside the designated area.
- Ground disturbance activities will be planned to ensure minimal disturbance is achieved through the use of appropriate ground engaging plant, use of designated tracks, roadways and use of pre-existing disturbed areas.
- Mobile plant operators will receive clear instructions from the Site Construction Manager or delegate on ground disturbance activities; including maps and GPS coordinates of disturbance area.
- Topsoil will be managed in accordance with HSEQ procedure – Soil Resource Management (RTIO-HSE-0011596). Key points include demarcate areas to be cleared, recover topsoils and appropriate subsoil, inspect for sufficient soil recovery, stockpile as per site requirements, stockpile topsoil (maximum 2m), install signage at stockpiles, complete documentation and undertake stockpile monitor and management.
- Areas undergoing ground disturbance will be monitored (as appropriate) to ensure they meet the guidelines. A supervisor will be present during all boundary clearing activities or clearing adjacent to exclusion areas.
- Verification of compliance with the Clearing Permit will be undertaken at completion of the clearing works. Any ground disturbance outside of permit boundaries shall be recorded as an incident and shall be rehabilitated in consultation with Rio Tinto Iron Ore Expansion Projects (**RTIOEP**) requirements.
- All ground disturbance activities and area of disturbance shall be reported in the Monthly Environmental Report.
- Emergency clearing activities (e.g. for fire, flood protection) shall follow the approvals coordination emergency procedure.

6.2 Topsoil Management

Topsoil shall be managed under Management Procedure 6.3 of the CEMP (Appendix 3). Key environmental management actions for the removal of topsoil is summarised below:

- All vegetation and topsoil shall be removed during the clearing associated with construction and shall be appropriately stockpiled to maintain viability of topsoil or used for progressive rehabilitation of other areas.
- Topsoil shall be managed in accordance with HSEQ procedure – Soil Resource Management (RTIO-HSE-0011596).
- Procedures for ground disturbance shall be followed prior to removing topsoil and include dust suppression procedures.
- Inspection of topsoil removal practices and stockpiles shall be conducted (as appropriate) to ensure they meet the guidelines.
- Topsoil stockpiles shall be limited to 2 m in height and with side slopes of no more than 20 degrees to reduce the risk of erosion. The stockpiles shall not be located within drainage lines. Stockpiles shall be sited appropriately to ensure erosion and run off are minimised.
- Burning of vegetation stockpiles shall not be permitted.

6.3 Flora and Fauna Management

Measures to minimise impacts to significant vegetation communities or habitat features has been incorporated into the design. Flora and fauna shall be managed under Management Procedure 6.4 in the CEMP (Appendix 3). Key management actions for the protection of flora and fauna are summarised below:

- The requirements of the Wildlife Interaction Policy and the Rio Tinto Biodiversity Guidance shall be communicated and implemented on site.
- Maps and photographs/images of priority flora and fauna shall be supplied to personnel to facilitate identification and on-ground management.
- Fencing and/or signage shall be used to protect areas containing significant flora and fauna habitat where there is risk of damage. Access to these areas shall be restricted or not permitted.
- Safe relocation of fauna and management of injuries will be managed in accordance with HSEQ Procedure – Wildlife Interaction Guidelines (RTIO-HSE-0013116). Key commitments include the reporting of unusual animal sightings and contacting the site Emergency Management Officer or delegate for the removal of fauna, for assistance in the event of a fauna strike by a vehicle and for the euthanasia of seriously injured wildlife.
- All drill/bore holes will be capped.
- Feeding and/or capture of native and fauna or feral animals shall not be permitted. Any deaths of native fauna shall be reported as per incident report procedures.
- All trenches and excavations will be constructed in accordance with DMP Environmental Note – Fauna Egress Matting and Ramps (DMP, 2012). Trained fauna handlers will be available and routine inspections of any open trenches will be made twice daily, as stated in the CEMP.

6.4 Weed Management

The construction of the Project has the potential to introduce and spread weeds species into the area as a result of vehicle movements and equipment mobilisation. To prevent the introduction and/or spread of weed species, weed hygiene and control measures shall be managed as outlined in Management Procedure 6.5 of the CEMP (Appendix 3).

Key environmental management actions for weed hygiene and control is summarised below:

- Weed hygiene and control measures shall be managed as outlined in Iron Ore (WA) Weed Management Work Practice (RTIO-HSE-0013504). The most effective means of weed control is through the prevention of their introduction to a site or location. This will primarily be achieved through the implementation of the Weed Hygiene Procedure (RTIO-HSE-0014273) that documents the process for the cleaning of heavy vehicles and equipment prior to transportation to and between Rio Tinto sites to prevent the introduction and spread of weed seeds. All employees and contractors will comply with this procedure.
- All earthmoving and mobile construction equipment shall be inspected for weeds prior to mobilisation to site as per the Vehicle Hygiene and Weed Inspection Form (RTIO-HSE-0044261).
- Posters and other educational material relating to weed identification and control shall be displayed in the workplace.

- Signage shall be used where necessary to communicate the weed status of an area and hygiene requirements.
- Sites which are likely to facilitate germination of weeds shall be inspected after rainfall events.
- Sites shall be inspected at the completion of works to verify effectiveness of weed hygiene controls.
- Where weeds are identified, Global Positioning System (**GPS**) coordinates of the site shall be taken and locations maintained on a register or database.
- Site inductions shall educate all construction personnel in the identification of serious environmental weeds. A list and means of identifying Declared Plants and other weeds on site shall be made available by the Site Environmental Advisor to all personnel involved in or managing clearing activities.
- Where an area is infested with weeds, wash-down of ground engaging equipment such as; front end loaders, graders, tracked equipment and out riggers shall be undertaken at the site prior to leaving.
- Wash down facilities shall be designed to prevent run off to the surrounding environment.
- Failures in weed hygiene processes shall be reported as an incident through the incident reporting process.

6.5 Surface and Groundwater Management

Water requirements shall be sourced from the Water Corporation West Pilbara Water Supply - Millstream Supply Main pipeline from existing and new metered off take points and licensed water bores in the local construction area. The Company is currently liaising with the Water Corporation to ensure sufficient water is available from the pipeline, and the approvals obtained as required.

Water use shall be managed under Management Procedure 6.10 as outlined in the CEMP (Appendix 3) and the following DoW Water Quality Protection Guideline Series. DoW will be notified of spills in PDWSA areas. Key environmental management actions for the use of water are summarised below:

- A water inventory shall be maintained and will include:
  a) Abstraction monitoring from each source including quantity and quality; and
  b) Quantity of water that is recycled and reused.
- Water shall only be taken from points authorised by RTIOEP and licensed by DoW. Copies of all water licences shall be held on site.
- Meters, Calibration and Reporting:
  a) Bores shall be fitted with a sample tap / valve and probe access point in addition to a flow meter that complies with the Rights in Water and Irrigation (Approved Meters) Order 2009;
  b) Calibration of meters shall be carried out prior to usage / installation and certificates for meters will be retained on site;
  c) The amount of groundwater extracted shall be reconciled regularly against the Licence limit; and

d) Water usage data shall be recorded and forwarded monthly to RTIOEP Environmental Advisor.

- Water Monitoring:

  a) Water quality monitoring shall be undertaken as defined by water licences or as directed by RTIOEP;

  b) Equipment used to monitor groundwater quality shall be calibrated and associated records maintained;

  c) A monitoring programme and schedule shall be established for groundwater, surface water and wastewater where applicable. This shall be approved by RTIOEP;

  d) The monitoring programme shall outline where a National Association of Testing Authorities (**NATA**) registered laboratory shall be used for analysis of water samples; and

  e) Any works within beds and banks of watercourses will be managed in accordance with the conditions of the permits issued by DoW for those activities

- Stormwater/Surface Water Management:

  a) All potentially contaminated stormwater (sediment and hydrocarbons) shall be treated prior to discharge to the environment;

6.6 Waste Management

Waste shall be managed under Management Procedure 6.12 of the CEMP (Appendix 3). Key waste management actions are summarised below:

- All non-hazardous waste shall be transported offsite to nearest local waste management facility.
- All waste streams and their collection and disposal shall be identified. This includes but is not limited to: waste oil, grease, solvent, inhibitors, oily-water mixtures, empty hydrocarbon drums, oil filters, oily rags, absorbent materials.
- Sufficient bins, recycling and general waste collection areas, shall be established to facilitate the management of waste. Putrescibles waste shall be stored in covered receptacles at all times, prior to disposal.
- Waste bins are colour coded and labelled in accordance with RTIOEP procedures to facilitate segregation of waste types.
- The following materials shall be recycled unless otherwise approved: scrap metal, paper/cardboard, wood pallets, aluminium, glass, and waste oil.
- Receipts shall be maintained of waste type, amount (mass/volume), and disposal method (recycled, re-used, disposal location). Records shall be reported in monthly report.
- Controlled waste (including hydrocarbons) shall be handled and disposed of in accordance with the *Environmental Protection (Controlled Waste) Regulations* 2004.
- Cut material generated from construction activities shall be used as fill and for rehabilitation wherever possible and practicable.
- Littering shall be avoided at all times and work and office sites shall be kept clean and tidy.

6.7 Spoils Management

Spoils removed from the construction of the Project shall be used to backfill borrow pits where practicable. Where spoil cannot be used for backfill it will be stored and landformed to surrounding landscape and in accordance with the Rio Tinto Rehabilitation Handbook (RTIO-HSE-11608).

6.8 Hydrocarbon Management

Hydrocarbon storage and use shall be managed under Management Procedure 6.11 of the CEMP (Appendix 3). Some parts of the Project fall within the Harding Dam Catchment Area P1 PDWSA. DoW guidelines, as referenced in Section 6.11 of the CEMP, will be complied with during all construction activities.

Key environmental management actions associated with the storage and use of hydrocarbons are summarised below:

6.8.1 General

- Regular inspections of construction sites, storage areas, washdown bays shall be undertaken for adherence to hydrocarbon management procedures.

6.8.2 Storage And Bunding:

- Hydrocarbon storage, including temporary storage, shall comply with the applicable minimum standard set out in the HSEQ Environmental Design Principles for Temporary facilities (DC-N002) (Appendix 4).
- Monthly inspections of construction sites shall be carried out to check for adherence to good hydrocarbon management procedures.

6.8.3 Refuelling:

- Refuelling of service trucks shall occur offsite in accordance with the CEMP.
- Where mobile refuelling is conducted, dry-break hoses shall be used.
- Refuelling shall not occur within 30 m of a watercourse.
- Operator to be present during refuelling at all times.
- Mobile refuelling trucks must carry spill kits and drip trays.

6.9 Spill Response and Reporting

Spill response and reporting will be in accordance with Management Procedure 6.11 of the CEMP (Appendix 3). Key environmental management actions are summarised below:

- Spill response must be in accordance with the requirements of the HSEQ procedure – spill response (RTIO-HSE-0010867).
- Spill response equipment, including absorbent booms/socks/pillows and matting, should be readily available and used in work environments including mobile plant. Equipment should be clearly marked and housed in a manner that facilitates quick response to spills, including mobile plant.
- An incident report should also be completed where there has been a non-compliance with internal procedures that may or may not have resulted in an environmental impact or harm.
- Any spills in the PDWSA will be reported in accordance with DoW guidelines, CEMP and tenement conditions.

6.10 Dangerous Goods and Hazardous Substances

Hazardous Materials shall be managed under Management Procedure 6.14 the CEMP (Appendix 3). Key environmental management actions are summarised below:

- Application must be made to RTIO for approval of any chemical or hazardous material prior to its mobilisation to site. Chemicals not approved shall be prohibited from site.
- Where required, Dangerous Goods Licences shall be obtained and copies held on site.
- The RTIO ChemAlert hazardous material register shall be maintained.
- Design of hazardous material storage areas shall comply with legislative guidelines, RTIOEP standards and RTIOEP design guidelines.
- Safety Data Sheets shall be kept on site for all hazardous materials in their area of use. A duplicate copy must be provided to emergency response personnel.
- All containers shall be labelled.
- Spill response shall be in accordance with HSEQ procedure – Spill Response (RTIO-HSE-0010867). Key points include notifying the site emergency response team immediately if a spill could potentially cause a fire or explosion, using appropriate PPE, following the Control, Contain and Clean up approach, and to undertake appropriate reporting.
- Any spills in the PDWSA will be reported in accordance with the DoW guidelines, CEMP and tenement conditions.
- Appropriate fire detection and protection equipment shall be installed in all fire hazardous work and storage areas.
- All hazardous materials storage areas shall be inspected on a weekly basis as part of a monitoring programme.

6.11 Dust

Dust may be produced from the earthworks undertaken during the construction of the Project. In order to minimise dust, atmospheric pollution shall be managed under Management Procedures 6.9 of the CEMP (Appendix 3). Key environmental management actions are summarised below:

- Vegetation clearing shall be minimised where practical.
- Vehicle speeds on access roads, work areas and camp sites shall be restricted.
- Dust suppression such as water trucks shall be used on access roads, operating borrow pits, earthwork fronts and campsites.
- If dust levels are excessive, work activities shall be restricted until dust suppression is implemented.
- Excessive dust generation shall be reported as per incident reporting procedures.
- Visual dust assessments shall be included in the site inspection programme and formal dust monitoring shall be carried out as required.

6.12 Noise and Vibration

Noise and vibration are not expected to be significant whilst the proposed works are being undertaken. Construction activities shall increase local noise levels in the short term with earthworks expected to contribute the greatest increase in local noise. Construction activities shall be conducted in daylight hours where possible. However, there are no sensitive receptors in the vicinity of the works.

All works and equipment shall comply with the *Environmental Protection (Noise) Regulations 1997*. Noise and vibration shall be managed under Management Procedures 6.15 of the CEMP (Appendix 3). Key environmental management actions are summarised below:

- Construction vehicles and equipment shall be fitted with exhaust mufflers to suppress noise.
- Equipment with faulty or inefficient mufflers/noise dampers shall not be used.
- Construction work shall be carried out in accordance with Section 6 of Australian Standard 2436-1981 "*Guide to Noise Control on Construction, Maintenance and Demolition Sites*".
- A risk assessment shall be performed to identify potential environmental risks and controls associated with blasting.

6.13 Fire Management

Management Procedure 6.16 of the CEMP (Appendix 3) and the Project Safety Management Plan outlines the management measures that will be implemented to prevent construction activities in resulting in fires.

Measures that will be taken to prevent fires include:

- Obtaining a Hot Work Permit where required.
- Proper disposal of cigarette butts.
- Maintaining vehicles and equipment to be free of vegetative material.
- Reviewing work activities and weather conditions daily; and delaying works if the conditions are unsuitable / create a high fire risk.

6.14 Workforce Management in the Millstream-Chichester National Park

Management Procedure 6.22 of the CEMP (Appendix 3) outlines the aims of the Project to avoid and/or minimise impacts to the MCNP and to its visitors that may result from personnel working in the area.

Measures that will be taken to avoid and/or minimise impacts include:

- Personnel induction emphasising on the rules and values of the MCNP.
- Organised and controlled trips / tours will be scheduled with the presence of an appropriate Supervisor and consumption of alcohol will not be permitted during these trips.
- Compliance with the Communications Protocol agreed with, and approved by, the DEC (Appendix 5).
- Complying with the Fire Management requirements as outline in Section 6.13.

7.0 REHABILITATION AND CLOSURE

Due to the nature of this Mining Proposal, requirements of the Guidelines for Preparing Mine Closure Plans have not been addressed in full. However, relevant aspects have been incorporated into this section of the Proposal.

7.1 State Agreement Closure Obligations

Rio Tinto Iron Ore (RTIO) operates numerous iron ore mining operations across the Pilbara, with most of its activity covered under State Agreements. These agreements impose unique closure obligations. The facilities addressed in this MP are subject to State Agreement (see Section 5.2 of this MP), they do need to be considered in the context of a broader mining network.

All State Agreements under which RTIO operates in the Pilbara contain clauses in relation to ownership of infrastructure, including both fixed and mobile infrastructure, following closure. Whilst the various State Agreements differ in specifics, the relevant clauses are similar in general terms:

- Infrastructure remaining after closure becomes property of the State; and
- If the Company intends to remove infrastructure during closure, it must provide the State prior opportunity to purchase it an agreed price.

The proposed works subject to this MP will occur in association with the Company's rail construction and operations under the *Iron Ore (Robe River) Agreement Act 1964*. Whilst the specific facilities covered under this Proposal are not formally captured under such an obligation, it is logical to assume that they will be included in the negotiations that will occur under the relevant State Agreement. Until this occurs, no detailed commitments can be made with regard to decommissioning and rehabilitation.

7.2 Closure Strategies

Since the operational life of the facilities covered under this proposal is expected to exceed 30 years, detailed decommissioning, rehabilitation strategies have not been developed for this specific proposal. However, RTIO applies general decommissioning and rehabilitation strategies across the breadth of its operations. RTIO will prepare project specific decommissioning and rehabilitation plans as closure approaches; this is nominally undertaken when the site is five years from scheduled closure.

7.3 Decommissioning

General principles for decommissioning of infrastructure include:

- Rehabilitation of all non-permanent areas will be conducted progressively throughout the construction period.
- Subject to stakeholder negotiation, above ground installations are removed and materials recycled or reused where practicable or if agreed in writing by the appropriate regulatory authority, retention of plant and infrastructure agreed in consultation with relevant stakeholders.
- Infrastructure that is located more than 1 metre below the final surface level (e.g. footings, cables) remain in-situ following closure provided they are non-polluting.
- Any infrastructure with the potential to be a source of contamination after closure is managed to prevent contamination from occurring.

7.4 Rehabilitation

General principles for rehabilitation include:

- Where applicable and appropriate, contaminated soils are removed and treated in a landfarm facility.
- Rehabilitation is undertaken so as to make the land compatible with the agreed final land use.
- Consideration is given to the surface water drainage impacts, and steps taken to ensure appropriate flows following closure.
- Compacted soils are ripped in order to improve rehabilitation outcomes.
- Rehabilitation is conducted with seeds collected from a provenance zone appropriate for each species.
- Completion criteria are agreed with stakeholders as closure approaches.
- Post closure monitoring programmes are developed and implemented to demonstrate progress towards, and compliance with completion criteria.

Rehabilitation of disturbed areas that are not required for ongoing operation of the Project will be conducted progressively or after the completion of construction activities in accordance with the RTIO HSEQ Rehabilitation Handbook. This will include borrow pits, temporary construction disturbance and any other disturbed areas that are no longer required. The natural landform and drainage patterns will be re-established, stockpiled topsoil and vegetation will be returned, and the area ripped or scarified (dependent on degree of compaction) on the contour.

The final Rehabilitation and Closure Plan for the Rio Tinto rail network will address options for closure of this Project, including decommissioning and rehabilitation if no longer required for other projects or land users in the area. If decommissioning is required, all infrastructure including the siding and other associated infrastructure will be removed. Bituminised surfaces will be broken up and removed to an appropriate disposal site. Disturbed areas will be shaped to re-establish the natural landform and drainage patterns, stockpiled topsoil and vegetation will be returned, and then areas will be ripped or scarified (dependent on degree of compaction) on the contour. Rehabilitated areas will be monitored for erosion and vegetative regrowth.

7.5 Financial Provision for Closure

RTIO maintains financial provisions for closure of its extensive global mining network in accordance with its corporate obligations. Cost estimates are prepared for each operation using standard unit-rate methodologies, and reviewed annually to ensure currency.

There is no risk that the infrastructure and facilities covered under this proposal will pose an ongoing financial liability to the State. Even in the unlikely event of unexpected closure, the size and diversity of Rio Tinto’s mining operations, along with financial obligations arising from its State Agreements, provide adequate mitigation.

8.0 REFERENCES

Aquaterra (2011) Pre Feasibility Study – 320MT Expansion Project Rail Upgrade Water Supply; unpublished report prepared for RTIO.

Bureau of Meteorology (BOM) (2012) Climate summary, http://www.bom.gov.au.

Biota Environmental Services (2008a) A Vegetation and Flora Survey of the Rio Tinto Rail Duplication Project – Cape Lambert to Emu Siding. Unpublished report prepared for Rio Tinto.

Biota Environmental Sciences, (2008b) Pilbara Iron Rail Duplication Fauna Assessment: Emu Siding to Rosella Siding. Unpublished report prepared for Rio Tinto.

Biota Environmental Services (2010) A Vegetation and Flora Survey of the Rio Tinto Rail Duplication – Emu Siding to Rosella Siding Development Areas. Unpublished report prepared for Rio Tinto.

Department of Conservation and Land Management (CALM) (2002) Bioregional summary of the 2002 Biodiversity Audit for Western Australia.

Department of Mining and Petroleum (2012) Environmental Note – Fauna Egress Matting and Ramps.

Ecologia (2004) Tunkawanna Cockatoo Rail Duplication Flora & Fauna Assessment. Unpublished report prepared for Rio Tinto.

Kendrick, P. and Mckenzie, N. (2001) *Pilbara 1 (PIL 1 – Chichester subregion)* in Biodiversity Audit of Western Australia 2002.

Rio Tinto (2012) Statement Addressing the 10 Clearing Principles, AutoHaul Emu to Rosella. Unpublished internal report.

APPENDIX 1

MINING PROPOSAL CHECKLIST (as per Appendix 6 in Mining Proposal Guidelines)

This checklist must be completed and be included in the Mining Proposal after the title page.

The Mining Proposal checklist is designed to ensure that the proponent provides all required information and to enable fast and accurate assessment without the need for the assessing officer to seek further information or clarification. If critical issues have not been addressed in the Mining Proposal or the checklist information is found to be incorrect then assessment may be discontinued and the Mining Proposal returned.

Please cross reference page numbers from the Mining Proposal where appropriate.

Q No	Mining Proposal checklist	Y/N NA	Page No	Comments
	Public availability			
1	Are you aware that this Mining Proposal is publicly availability?	Y		
2	Is there any information in this mining proposal that should not be publicly available?	N		
3	If 'No' to Q2, do you have any problems with the information contained within this Mining Proposal being publicly availability?	N		
4	If 'Yes' to Q2, has confidential information been submitted in a separate document / section?	N/A		
5	Has the Mining Proposal been endorsed? (See last page Checklist.)	Y		
	Mining Proposal Details			
6	Have you included the tenement number(s), site name, proposal overview and date in the title page?	Y		
7	Who authored the Mining Proposal?	Shreya Shah		
8	State who to contact for enquiries about the Mining Proposal?	Shreya Shah Shreya.Shah@riotinto.com 08 6332 8545		
9	How many copies were submitted to DoIR?	Online Submission (EARS)		
10	Is this Mining Proposal to support lease application?	N		
11	Has a Geological Resource Statement been included (refer section 4.3.2 of Mining Proposal Guidelines)	N		
12	Will more than 10 million tonnes of ore and waste be extracted per year? State total tonnage: __________	N		
13	Will more than 2 million tonnes or ore be processed per year? State total throughput. ____________	N		
14	Is the Mining Proposal located on pre-1899 Crown Grant Land? (not subject to the *Mining Act 1978*)	N		
15	Is the Mining Proposal located on Reserve Land? If 'Yes' state Reserve types in space below:	N		
16	Will the Mining Proposal occur within or affect a declared occupied townsite?	N		

17	Is the Mining Proposal within 2 km of the coastline or a Private Conservation Reserve?	N		
18	Is the Mining Proposal wholly or partially within a World Heritage Property, Biosphere Reserve, Heritage Site or Soil Reference Site.	N		
	Tenement Details			
19	Are all mining operations within granted or applied for tenement boundaries?	Y		Figures 1-3
20	Are you the tenement holder of all tenements?	Y		
21	If 'No' at 20, do you have written authorisation from the tenement holder(s) to undertake the Mining Proposal activities? (Refer to section 4.2.1 of the Mining Proposal Guidelines)	N/A		
22	Is 'Yes' at 21, then is a copy of the authorisation contained within the Mining Proposal?	N/A		
23	Have you checked for compliance against tenement conditions?	Y		Appendix 6
	Location and Site Layout Plans			
24	Have you included location plans showing tenement boundaries and mining operations?	Y		Figures 1-3
25	Have you included site layout plans showing all mining operations and infrastructure in relation to tenement boundaries?	Y		Figures 1-3
26	Have you included "Area of Disturbance Tables" for all tenements impacted by mining operations?	Y		Tables 1 - 4
	Environmental Protection Act			
27	Does the Mining Proposal require referral under Part IV or the MOU? If 'Yes' describe why in space below:	N		
28	Has the EPA set a level of assessment? If yes state:	N/A		
29	Is a clearing permit required? If 'No' then explain why in space below?	Y		
30	If 'Yes' at Q29 then has a permit been applied for?	Y		CPS No.5117/1
31	Is a Works Approval required by the DEC (formerly DoE)?	N		
32	Has a Works Approval application been submitted to the DEC (formerly DoE)?	N/A		
33	**Stakeholder Consultation -**Have the following stakeholders been consulted? (use N/A if not applicable)			
	Shire?	N/A		
	Pastoralist?	N/A		Works are to be undertaken immediately adjacent to the rail line.

	DEC (formerly CALM)?	N/A		
	Main Roads?	N/A		
	Others? (specify):	Y		DMP, Native Vegetation Assessment Branch, re granting of CPS 5117/1
	Environmental Assessment and Management			
34	Is the Mining Proposal wholly or partially within DEC (formerly CALM) managed areas?	N		
35	If 'yes' at Q34 has DEC (formally CALM) been consulted?	N/A		
36	Is the Mining Proposal wholly or partially within a Red Book Area or a Bush Forever Site?	N		
37	Will the Mining Proposal impact upon a Water Resource Area, Water Reserve, declared or proposed catchment, Groundwater Protection Area, significant lake or Wetland?	Y		Work will be conducted within the Millstream Water Reserve; a copy of the MP will be forwarded to Natalie Leach (DOW Karratha) for review and comment.
38	Is a water or de-watering licence required?	Y		
39	If 'Yes' at Q38 then has the licence(s) application been submitted?	Y		Refer to Section 5.4
40	Does the Mining Proposal include a new tailings storage facility or changes to existing tailings storage facility?	N		
41	Have waste characterisation assessments been undertaken (eg. AMD, dispersiveness, salinity)?	N/A		
42	Have flora and fauna surveys been undertaken?	Y		Refer to Sections 4.5, 4.6 and 4.7
43	Are any rare / endangered species present?	N		
44	Has a Preliminary Closure Plan has been included?	Y		Refer to Section 7

I hereby certify that to the best of my knowledge the above checklist accurately reflects the information contained within this Mining Proposal.

Name: Shreya Shah Position: Specialist Advisor Government Approvals

Signed: ______________________ Date: 26/10/2012

APPENDIX 2

AutoHaul™ Terminology Guide

Item	Description	Example
Passive Level Crossing	"Passive" level crossings require the use of signs to manage the road rail interface. The AutoHaul™ Project aims to restrict access to level crossings through installation of lockable gates, signage and fencing for public safety reasons. Proposed construction work activities at passive level crossing sites include the following: • surveying; • minor vegetation clearing and grubbing; • topsoil removal and stockpiling; • material handling, conditioning and stockpiling; • cut to fill earthworks operations; • installation of concrete footings; • road and access track realignment and/or widening (where applicable); • installation of fencing, signage and lockable gates (including that for survey, heritage and perimeter); and • rehabilitation of disturbed areas	

<table>
<tr>
<td>Narrow Band Shunts</td>
<td>A Narrow Band Shunt is associated with the active level crossing upgrades and is located within 1-2km from the crossing. Works include the installation of new equipment, pits and terminations to the track through bootleg sleepers within the existing rail formation. An example of a Narrow Band Shunt is provided in the image.

Construction activities include the following:

surveying;
minor vegetation clearing and grubbing for access and laydown only;
topsoil removal and stockpiling for access and laydown only;
laydown areas for vehicles and equipment;
installation of new equipment, pits and terminations to the track through bootleg sleepers within the existing rail formation; and
rehabilitation of disturbed areas.

</td>
<td>
</td>
</tr>
</table>

Signals Pad	A signals pad controls the wayside infrastructure that allows the safe passage of a train through the section of rail for which it is associated. This includes construction of an asset pad, and housing of interlocking equipment (switches and controls) which directly interfaces to all wayside equipment. Signals equipment, solar panelling and radio masts at this assets pad send information through the switch box points and along the control of the rail switches and the track circuits to permit the train authority to proceed further down the line. The infrastructure assets on the pad retain control of the points of communication from the pad, the disconnect boxes adjacent to the track, the locomotive/train to the remainder of the signalling assets along the line. Construction activities include: • clearing and earthworks embankment construction; • removal of existing signal equipment; • installation of new signal pad equipment; • trenching and installation of fibre optic cable; • widening of existing access roads/improvement of existing access road surface using earthworks; • widening and improving existing access road creek crossing; • construction of under-bore development pads and pits, under-boring activities and • rehabilitation of temporally cleared areas.	

APPENDIX 3

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calibre

AutoHaul™ Project
Construction Environmental Management Plan

RCEATO-PLN-T-003

MASTER COPY

1	Revisions to reference documents and Flora & Fauna section	B.Shah	P.Zwart	M.Forward	D.Blainey	October 2012
0	Issued for Use	B.Shah	P.Zwart	M.Forward	D.Blainey	June 2012
B	Issued for Client Review	B.Shah	P.Zwart	M.Forward		May 2012
A	Issued for Internal Review	V. Sekizovic				May 2012
Rev	**Description**	**Author**	**Checked**	**Approved**	**Authorised**	**Date**

CONTENTS

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1.0 GLOSSARY

API	Assessment on Proponent Information
ASS	Acid Sulphate Soils
BSA	Borrow Pit Search Area
Contractor	Person or Company to whom an award of contract or sub-contract has been made by the Client
CEMP	Construction Environmental Management Plan
CERR	Construction Environmental Risk Register
CLY	Cape Lambert Yard
DAFWA	Department of Food and Agriculture Western Australia
DEC	Department of Environment and Conservation
DMP	Department of Mines and Petroleum
DoH	Department of Health
DoW	Department of Water
DRF	Declared Rare Flora
EAR	Environmental Approvals Register
EMS	Environmental Management System
EoS	End of Siding
EPBC Act	*Environment Protection and Biodiversity Conservation Act* 1999 (Commonwealth)
EPCM	Engineering, Procurement and Construction Management
EPRP	Emergency Preparedness and Response Plan
GHG	Greenhouse Gases
HBD	Hot Box Detector
HDPE	High Density Polyethylene
HSE	Health, Safety and Environment
HSEQMS	Health Safety Environment and Quality Management System
IBRA	Interim Biogeographic Regional Assessment
JHA	Job Hazard Analysis
KNP	Karijini National Park

LAA	*Land Administration Act* 1997
Landgate	Western Australian Land Information Authority
LAORS	Legal and Other Requirements System
MCNP	Millstream Chichester National Park
MLK	Microlock
MSDS	Material Safety Data Sheets
Mtpa	Million Tonnes Per Annum
MTX	Microtrax
NATA	National Association of Testing Authorities
NGER	National Greenhouse and Energy Reporting
OC	Operations Centre
PDWSA	Public Drinking Water Source Area
PEC	Priority Ecological Community
PPE	Personal Protective Equipment
RBS	Radio Base Station
RCE	Rail Capacity Enhancement
RTC	Rosella to Coast
RTDN	Radio Based Data Network
RTIO	Rio Tinto Iron Ore
RTIOEP	Rio Tinto Iron Ore Expansion Projects
RTM	Rosella to Mines
RTRD	Rio Tinto Railway Division
Robe	Robe River Mining Co. Pty. Ltd.
The Engineer	Calibre Rail
The Project	Rail Capacity Enhancement (RCE) AutoHaul™ Project
TEC	Threatened Ecological Community

2.0 INTRODUCTION

2.1 Purpose and Structure of the Construction Environmental Management Plan

This Construction Environmental Management Plan (CEMP) shall be used to manage the known and potential environmental impacts of the Project and associated activities.

The CEMP has been prepared to cover elements of AS/NZ ISO14001 (2004) and the Rio Tinto Iron Ore (RTIO) Health, Safety, Environment and Quality Management System (HSEQMS). As such it will be subject to internal audit by RTIO, ISO14001 accreditation audits and external audits (as required) by relevant regulatory agencies including but not limited to, the Department of Water (DoW), Department of Mines and Petroleum (DMP) and Department of Environment and Conservation (DEC).

All personnel working on the Rail Capacity Enhancement (RCE) AutoHaul™ Autonomous Train Project must adhere to the CEMP to ensure any impacts on the environment are managed in accordance with proponent commitments and with legal requirements.

3.0 PROJECT OVERVIEW

The RCE project is part of RTIO's plan to expand its iron ore export capacity. RTIO's AutoHaul™ Autonomous Train Project forms part of this initiative.

The AutoHaul™ system will automate mainline locomotive operation from the ports to mines, excluding the Deepdale line west of Emu. This shall be achieved by integrating additional technology with existing proven train management systems of the RTIO rail network. A completed AutoHaul™ system shall comprise the following major elements:

- an automated train driving system that computes the required power and brake settings to control trains within predetermined operational and safety parameters;
- the computer control of all of the major operational scenarios needed to safely move trains between mine sites and the coastal yards;
- Operations Centre (OC) equipment to allow their staff to configure trains for automatic operation, monitor train and locomotive performance, and intervene in the event of abnormal system behaviours;
- a data network to allow communication between the OC and the on-board systems of automated trains;
- additional wayside systems to support driverless operation to, from and in mine loops; and
- a substantial increase in level crossing protection at public and private access crossings.

AutoHaul™ will not provide for automation of the Deepdale line west of Emu, banker movement within mines, maintenance, yard Work, ad-hoc journeys, and recovery of trains in the event of system failures.

Figure 3-1 shows a simplified layout diagram of the RTIO rail network with the AutoHaul™ network upgrades highlighted.

3.1 Background

RTIO in conjunction with its joint venture partners, owns and operates a rail network within the Pilbara region of Western Australia to transport iron ore from its mine sites to its ports at Dampier and Cape Lambert. The Cape Lambert to Pannawonica line was originally constructed pursuant to the *Iron Ore (Robe River) Agreement Act* 1964 in the 1970s.

Robe River Mining Co. Pty. Ltd. (Robe) solely operated the rail line between Cape Lambert and Pannawonica until sharing of infrastructure was approved by the Minister of State Development pursuant to the relevant State Agreements. Following this agreement, the rail line is managed by Pilbara Iron Pty Ltd which oversees the daily operation of the RTIO rail network.

Rio Tinto is currently expanding its iron ore operations in the Pilbara from 220 million tonnes per annum (Mtpa) to 353Mtpa as part of the Rail Capacity Enhancement Project. This involves the establishment of new and expansion of existing mines, expansion of its existing port facilities at Dampier and Cape Lambert and the upgrade of its existing rail network and associated infrastructure to accommodate this additional capacity.

3.2 Project Scope

The Project will be deployed in two phases which include Rosella to Coast (RTC) and Rosella to Mines (RTM). Operational support will include 7 dedicated crib rooms located across the network, installation of 18 Hot Box Detectors (HBDs), upgrades to End of Sidings (EoS) (which consists of MicroTrax (MTX) sites and Microlock (MLK) communication boxes), 49 Rio Tinto Data Network (RTDN) base stations, OC upgrades and on board systems. In addition, the works will include 40 active level crossings and 84 sites with access restrictions to minor level crossings (based on installation of new secure lockable gates and signage).

The RTC section will include work associated with:

- Cape Lambert Yard and 7 Mile Yard to Rosella, Bellbird, Wombat and Crest sidings;
- 14 active level crossings;
- 22 RTDN base stations;
- EoS (number of sites to be finalised); and
- Sites with access restrictions to minor level crossings (number of sites to be finalised)

The RTM section will include work associated with:

- Rosella / Bellbird to all mines;
- 26 active level crossings;
- 27 RTDN base stations;
- EoS (number of sites to be finalised); and
- Sites with access restrictions to minor level crossings (number of sites to be finalised)

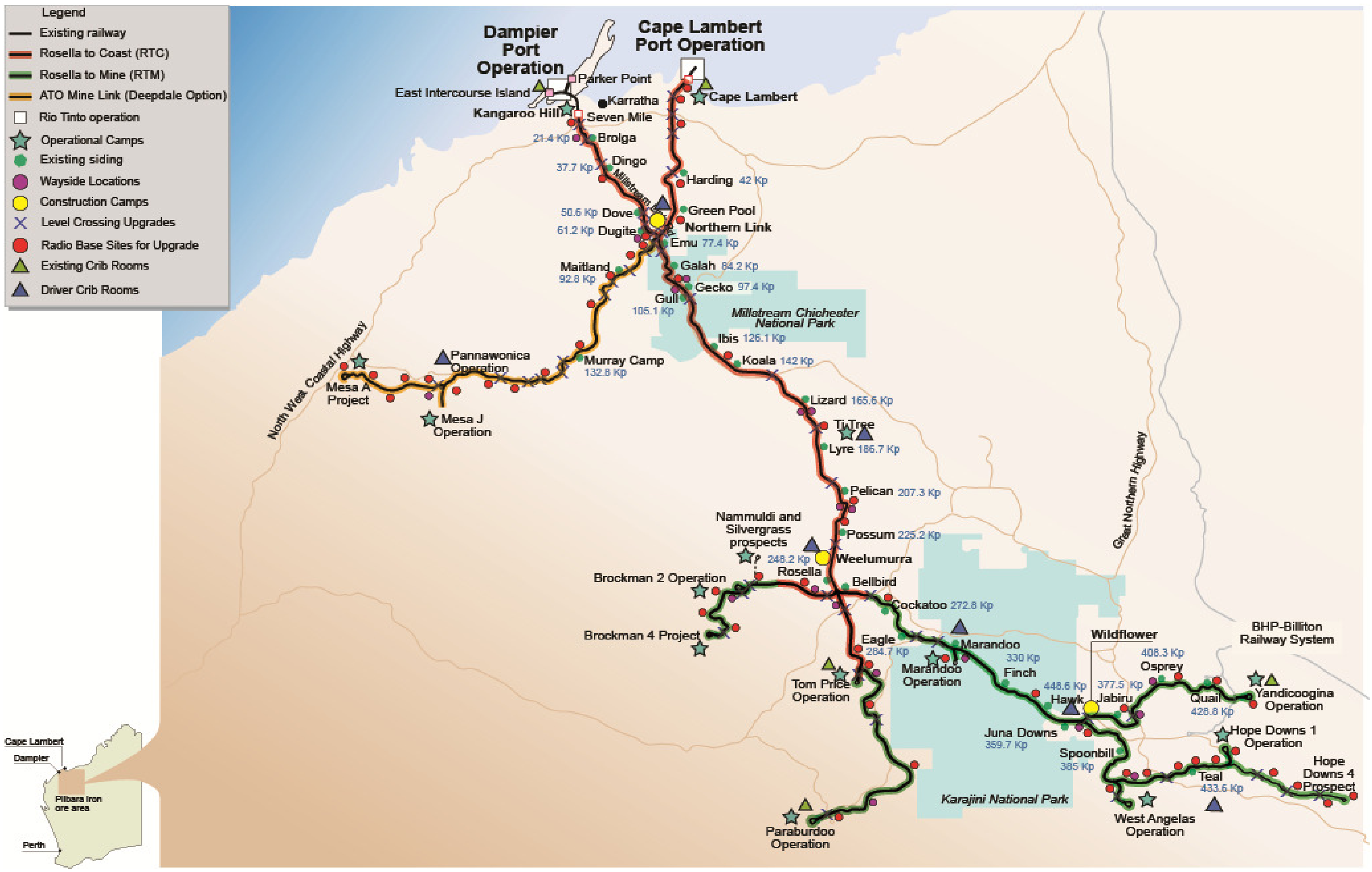


Figure 3-1: Simplified Layout of Existing Railway Network and Proposed Works

4.0 ENVIRONMENTAL SETTING

4.1 Physical Environment

4.1.1 Landform and Geology

The Pilbara forms part of The Great Plateau of Western Australia with the Hamersley plateau forming the most extensive elevated land area in Western Australia. The Project crosses numerous landforms, including residual uplands, ridges and drainage basins.

The Project lies wholly within the Pilbara Interim Biogeographic Regionalisation of Australia (IBRA) region, and crosses the four IBRA subregions, namely:

- Chichester;
- Fortescue Plains;
- Hamersley; and
- Roebourne.

The Chichester subregion is described as undulating Archean granite and basalt plains, including significant areas of basaltic ranges (CALM, 2002). The Fortescue Plains subregion is described as alluvial and has river frontages. An extensive calcrete aquifer (originating within a palaeodrainage valley) feeds numerous permanent springs in the central part of the Fortescue region, supporting large permanent wetlands. The Hamersley subregion is a mountainous area of Preterozoic sedimentary ranges and plateaux, dissected by basalt, shale and dolerite gorges. The Roebourne subregion comprises of Quaternary alluvial and older colluvial coastal and sub-coastal plains. Resistant linear ranges of basalts occur across the coastal plains (CALM, 2002).

Due to the geographic extent of the Project, it crosses numerous Department of Agriculture and Food (DAFWA) Rangelands Land Systems.

There are potential ‘moderate to low risk’ Acid Sulphate Soils (ASS) in the coastal area and in areas close to the estuarine, river and wetland systems (DEC Acid Sulphate Soil Risk Maps).

4.1.2 Climate

Across the Project area, climate can be characterised as arid tropical with two distinct seasons, summer and winter. Temperatures range from maximums of up to 47°C (36°C March mean daily maxima) in summer whilst winter temperatures are cooler (25°C July mean daily maxima) (BOM, 2012).

Rainfall is highly variable due to the influence of periodic summer cyclones but is approximately 300mm per annum. Peak rainfall occurs in the summer months (December – February), with smaller peaks occurring during May and June as the result of cold fronts moving across the south of the state which occasionally extend into the Pilbara.

4.1.3 Hydrogeology

Rivers in the Pilbara region are ephemeral and flow only occasionally. Sheet flow areas characteristically occur where there is poorly defined drainage and low gradients, or in flat or gently undulating areas where ponding occurs.

Due to the geographic extent of the Project, numerous aquifers are crossed, namely:

- Hamersley fractured rock;
- Hamersley-Fortescue fractured rock;
- Pilbara fractured rock; and
- Wittenoom combined fractured rock.

There is also significant surface water to groundwater interaction. Groundwater flow in the alluvial aquifers is generally in the same direction as stream flow. Drainage channels provide recharge to the underlying fractured rock aquifers during periods of river flow, in between rainfall events the subsurface flow and groundwater levels will likely follow a recession pattern (Aquaterra, 2011).

The Project will traverse two Priority 1 Public Drinking Water Source Areas (PDWSA), namely the Harding Dam Catchment Area, and the Millstream Water Reserve (DoW, 2010).

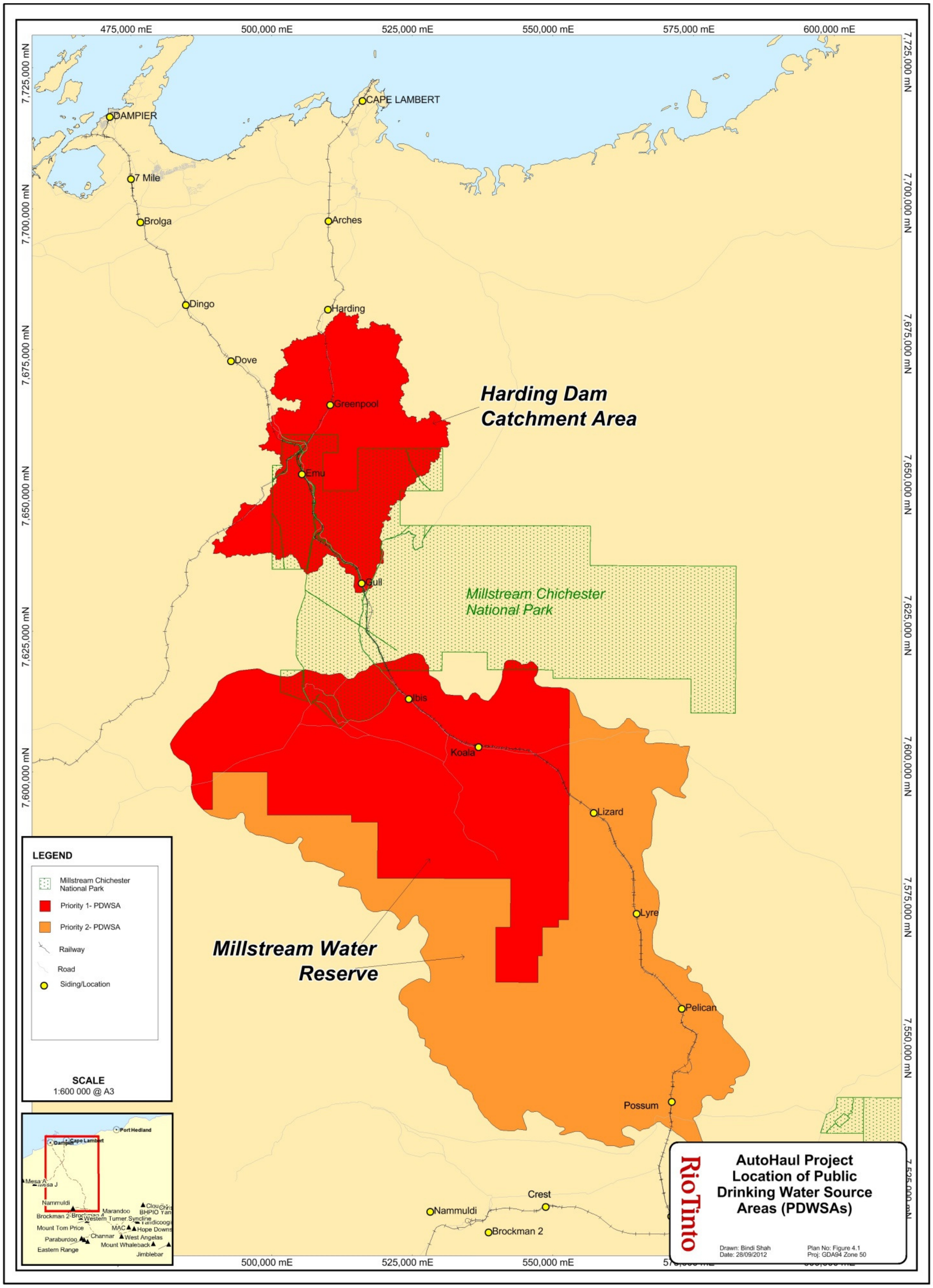


Figure 4-1 Location of PDWSA Areas in relation to AutoHaul™Project scope of works

4.2 Biological Environment

The Project falls within the four Pilbara IBRA subregions.

- The Chichester subregion is described as containing plants, which support a shrub steppe characterised by *Acacia pyrifolia* over *Triodia pungens* hummock grasslands, while Eucalyptus leucophloia tree steppes occur on the ranges.
- The Fortescue Plains subregion is described as having extensive salt marshes, mulga-bunch grass and short grass communities on the plains in the east. River gum woodlands fringe the drainage lines. In areas where permanent springs support permanent wetlands, the vegetation consists of extensive stands of river gum and cajuput.
- The Hamersley subregion contains valley floors which have low mulga woodland over bunch grasses on fine textured soils, while the ranges have *Eucalyptus leucophloia* over *Triodia brizoides* on skeletal soils. Roebourne subregion is an area dominated by grass savannas of mixed bunch and hummock grasses, and dwarf shrub steppe of *Acacia translucens* or *Acacia pyrifolia* and *Acacia inequilatera*.
- The uplands of Roebourne subregion are dominated by *Triodia* hummock grasslands. Ephemeral drainage lines support Eucalyptus woodlands. Samphire, Sporobolus grasslands and mangal occur on the marine alluvial flats and river deltas.

4.2.1 Classification of conservation significant flora and fauna

Protected species – Commonwealth

In 1974, Australia signed the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES). As a result, an official list of endangered, vulnerable or presumed extinct species was prepared (Schedule 1) and is regularly updated (*Endangered Species Protection Act 1992*). In July 2000 this Act was replaced by *The Environment Protection and Biodiversity Conservation Act 1999 (EPBC Act 1999)*, which retained the schedule of threatened species of the Act it replaced.

The vertebrate fauna listed in the current EPBC Schedule 1, differs from the *Wildlife Conservation Act 1950* and the DEC Priority Flora and Fauna list, although there are several species that appear on both,

- Extinct
- Extinct in the wild
- Critically endangered
- Endangered
- Vulnerable
- Conservation dependent.

Threatened Ecological Communities (TEC's) and Priority Ecological Communities (PEC's)

Vegetation Communities are described as TEC's if they have been defined by the Western Australian Threatened Ecological Communities Scientific Advisory Committee and found to be Presumed Totally Destroyed (PD), Critically Endangered (CR), Endangered (EN) or Vulnerable (VU).

Possible TEC's that do not meet survey criteria or that are not adequately defined are added to the Priority ecological communities list under Priority 1,2 and 3. These three categories are ranked in order of priority for survey and /or definition of the community, and evaluation of conservation status, so that consideration can be given to their declaration as threatened ecological communities. Ecological communities that are adequately known, and are rare but not threatened or meet criteria for Near threatened, or that have recently removed from the threatened list, are placed in Priority 4. These ecological communities require regular monitoring. Conservation Dependant ecological communities are placed in Priority 5.

Protected Species – Western Australia

Currently in Western Australia, rare or endangered species are protected by the *Wildlife Conservation Act 1950 (WC Act 1950)*. The relevant schedules defined under this *Act* are:

- Schedule 1: Fauna that is ranked as presumed extinct, critically endangered, endangered or vulnerable
- Schedule 4: Other Specially Protected Fauna.

The Act is periodically reviewed and the current list of protected flora and fauna is listed in Appendix A (2010 amendment).

Species of flora and fauna are defined as Declared Rare or Priority conservation status where their populations are restricted geographically or threatened by local processes. DEC recognises these threats of extinction and consequently applies regulations towards population and species protection.

DEC's Priority Flora and Fauna List classify species as:

Conservation Code	Category
R	**Declared Rare Flora – Extant Taxa** "Taxa which have been adequately searched for and are deemed to be in the wild either rare, in danger of extinction, or otherwise in need of special protection and have been gazetted as such."
P1	**Priority One – Poorly Known Taxa** "Taxa which are known from one or a few (generally <5) populations which are under threat, either due to small population size, or being on lands under immediate threat. Such taxa are under consideration for declaration as 'rare flora', but are in urgent need of further survey."
P2	**Priority Two – Poorly Known Taxa** "Taxa which are known from one or a few (generally <5) populations, at least some of which are not believed to be under immediate threat (not currently endangered). Such taxa are under consideration for declaration as 'rare flora', but urgently need further survey."
P3	**Priority Three – Poorly Known Taxa** "Taxa which are known from several populations, and the

	taxa are not believed to be under immediate threat (i.e. not currently endangered), either due to the number of known populations (generally >5), or known populations being large, and either widespread or protected. Such taxa are under consideration for declaration as 'rare flora' but need further survey."
P4	**Priority Four – Rare Taxa** "Taxa which are considered to have been adequately surveyed and which, whilst being rare (in Australia), are not currently threatened by any identifiable factors. These taxa require monitoring every 5-10 years."
P5	**Priority Five – Taxa in need of monitoring** "Taxa which are not considered threatened but are subject to a specific conservation program, the cessation of which would result in the species becoming threatened within five years."

4.2.2 RTC Flora and Vegetation

The region between RTC is dominated by Hummock and Tussock grasslands with isolated areas of open Eucalyptus woodlands and broader regions of low open Acacia shrublands in the north, moving into tall open acacia shrub land communities in the south (Biota, 2008a; Biota, 2010). The quality of vegetation varies across the RTC area varies in quality due to varying degrees of grazing and construction related impact and previously cleared areas that have been rehabilitated (Biota, 2008a; Biota, 2010).

Six ecological communities are considered to be of high conservation significance:

- Threatened Ecological Community (TEC): '*Themeda grasslands of the Pilbara bioregion*';
- Priority Ecological Community (PEC) 1: '*Cracking clay communities of the Chichester Range and Mungaroona Range'* comprising of mixed grassland and 'Roebourne Plains coastal grasslands'.
- PEC 1: '*Wona Land System*' has been identified around Gull siding. Four plant assembleys of the Wona Land System. This community is considered to be under threat of modification across much of its range from processes including heavy grazing by both native and feral herbivores, and high level erosion;
- PEC 3: *Hor*seflat land system of the Roeburn Plains that contains *Eragrostis xerophila* tussock grasslands on clay and encompasses the remainder of the Horseflat land system – not including the Roebourne Plains gilgai grasslands and the Chenopod association of the Roebourne Plains area. Extend from Cape Preston to Balla Balla (Whim Creek);
- and

Seven priority flora species have been recorded in the RTC area:

- Priority 1: *Nicotiana heterantha*. Annual herb species occurring on heavy clays on seasonally wet flats. A single specimen has been collected from the southern edge of the Harding Dam flood-out area. However, this identification is to be confirmed;
- Priority 1: *Josephinia* sp. Marandoo;

- Priority 1: *Rhagodia* sp. Hamersley;
- Priority 3: *Themeda* sp. Hamersley Station;
- Priority 3: *Oldenlandia* sp. Hamersley Station, recorded from clayey plain near Gull Siding;
- Priority 3: *Tephrosia Bidwillii*;
- Priority 3: *Terminalia supranitifolia;*
- Priority 3: *Rostellularia adscendens*;
- Priority 3: *Sid* asp Barlee Range
- Priority 3: *Triodia* sp. Mt Ella
- Priority 3: *Triodia* sp. Barlee Range
- Priority 3: Goodenia sp.East Pilbara
- Priority 3: *Iotasperma sessilifolium*
- Priority 3 *Glycine falcate*
- Priorty 3: *Astrebla lappacea*
- Priority 3: *Vigna* sp. rockpiles
- Priority 3: *Polymeria* species recorded at Hamersley Station, largely restricted to areas of heavy cracking clay and alluvium;
- Priority 4: *Rhynchosia bungarensis*
- Priority 4: *Eremophila magnifica;* and
- Priority 4: *Goodenia nuda* recorded in six locations throughout the RTC (Biota, 2008a; Biota Environmental Services, 2010).

Based on Biota Environmental Services studies, no Declared Rare Flora (DRF) were recorded in the RTC area and none are located within the Rio Tinto Geographical Information System Priority Flora layer.

Nineteen species of weeds occur within the RTC area, with no Declared Plant (declared weeds) according to the *Agriculture and Related Resources Protection Act 1976*. However, a number of species are considered to be serious environmental weeds, particularly Ruby Dock (*Acetosa vesicaria*), Kapok Bush (*Aerva javanica*) and two *Cenchrus* species (Buffel and Birdwood grass).

4.2.3 RTM Flora and Vegetation

The RTM region is dominated by limestone Spinifex, *Triodia melvillei* hummock grasslands, tall shrublands of Snakewood and open woodlands to forests of Coolibah over mixed shrubs and tussock grasslands in major creeks (Biota, 2008b).

Two ecological communities have been identified with high conservation significance, namely:

- PEC 1: Coolibah – Lignum Flats: sub type 3. Coolibah woodland over lignum over silky brown top (Mt Bruce flats)
- PEC 1: West Angelas Cracking Clays
- PEC 1: Weeli Wolli Spring Community

One DRF species *Lepidium catapycnon* was recorded within the RTM arealocated at approximately 296.8km on the Paraburdoo rail line and adjacent to the Yandi Mine Rail Loop. These specimens will not impacted from Project construction activities.

Five Priority flora species have been recorded from the RTM area, including:

Priority 1: *Calotis squamigera;*

- *Priority 1: Brunonia* sp. Longhairs*;* Priority 2: *Pilbara trudgenii;*
- Priority 2: *Stylidium weeliwolli*;
- Priority 2: *Euphorbia clementii*;
- Priority 3: *Acacia effuse*;
- Priority 3: *Glycine falcate*;
- Priority 3: *Indigofera gilesii*;
- Priority 3: *Themeda* sp. Hamersley Station;
- Priority 3: Geijera salicifolia;
- Priority 3: *Goodenia lyrata;*
- Priority 3: *Goodenia* sp. East Pilbara
- Priority 3: *Oldenlandia* sp. Hamersley Station
- Priority 3: *Rhagodia* sp. Hamersley;
- Priority 3: *Astrebla lappacea*;
- Priority 3: *Rostellularia adscendens* var. *latifolia.*
- Priority 3: *Swainsona* sp. Hamersley Station
- Priority 4: *Eremophila magnifica*
- Priority 4: *Goodenia nuda*

Eleven introduced flora species were recorded from the RTM area. None of the species are Declared Plants according to the *Agriculture and Related Resources Protection Act 1976*, however, **Acetosa vesic*aria, **Aerva javanica* and **Cenchrus* species are considered to be serious environmental weeds.

4.2.4 Fauna

There were fauna surveys completed which covered the Project area. The results of the separate surveys were as follows:

Tunkawanna to Rosella rail duplication flora and fauna report (Ecologia, 2004)
There were no conservation significant fauna species recorded in the study area.

Rio Tinto Rail Duplication Fauna Assessment: Bellbird Siding to Juna Downs (Biota Environmental Services, 2008d)
Five potential Short Range Endemic (SRE) Invertebrates were recorded in the study area. The conservation significance of these species is unknown, however, based on the broad distribution habitat types and vegetation units in which SRE's were recorded, it was concluded that that the taxa are likely to be more widely distributed beyond confines of the study corridor.

There were four conservation significant fauna species recorded in the study area:

- Schedule 4: The Peregrine Falcon (*Falco peregrinus*)
- Priority 4: The Australian Bustard (*Ardeotis australis*)

- Priority 4: Star Finch (*Neochimia ruficauda subclarescens*)
- Priority 4: Western Pebble-mound Mouse (*Pseudomys chapmani*)

The Rainbow Bee-eater (*Merops ornatus*) is listed under Federal legislation as Migratory and was also sighted in the study area. The habitat types in the project area are quite widespread and therefore the clearing of vegetation for this Project is unlikely to significantly impact the distribution of fauna species above.

Hope Downs Rail Corridor (Juna Downs Section) Fauna Survey (Biota Environmental Services, 2006)
Invertebrate short-range endemic taxa were recorded from the study site. This included five species of mygalomorph spider encompassing four families, two species of pulmonate snail encompassing two families, a species of pseudoscorpion, and one millipede species.

Two conservation significant fauna species were recorded in the project area including the Ghost Bat, *Macroderma gigas** (Priority 4) and the Western Pebble-mound Mouse, *Pseudomys chapmani* * (Priority 4). The Western Pebble-mound Mouse is noted as being common in the Pilbara. There were no abandoned mines or caves recorded in the project area confirming that the nesting habitat of this species will not be impacted from the AutoHaul project.

Pilbara Iron Rail Duplication Fauna Assessment: Emu Siding to Rosella Siding (Biota Environmental Services, 2008c)
There were five conservation significant fauna species recorded in the study area. These include:

- Schedule 4: The Peregrine Falcon (*Falco peregrinus*)
- Priority 4: The Australian Bustard (*Ardeotis australis*)
- Priority 4: Star Finch (*Neochimia ruficauda subclarescens*) -
- Priority 4: Western Pebble-mound Mouse (*Pseudomys chapmani*)
- Priority 4: The Rainbow Bee-eater (*Merops ornatus*)

Biota Environmental Services (2008c) have noted there is only a relatively small proportion of local habitat suitable for the taxa above that would be cleared for this Project in comparison to the wider distribution in the region and thus impacts would be minimal.

Hope Downs 4 (Ninox 2009) Infrastructure corridor option 6 Terrestrial Fauna Survey Report (Ninox Wildlife Consulting, 2009)

Two fauna species, of conservation significance, were recorded in the study area: the Western Pebble-mound Mouse (*Pseudomys chapmani*) (Priority 4) and Yellow-bellied Sheathtail-bat (*Saccolaimus flaviventris*) (IUCN List – Lower Risk). Both species are relatively common in the Pilbara and their distribution unlikely to be significantly impacted from the proposed works.

Turee Syncline Infrastructure Area Flora, Vegetation and Fauna Surveys - October 2009 (GHD, 2009)
A total of 14 conservation significant fauna taxa were recorded from the Study Area:

- Western Pebble Mound Mouse (*Pseudomys chapmani*, Priority Four [DEC]);
- Olive Python (*Liasis olivaceus barroni* Vulnerable [EPBC Act], Schedule 1 [*WC*

Act 1950]);

- Star Finch (*Neochmia ruficauda clares*cens; Near Threatened IUCN Red List; Priority Four [DEC]);
- Australian Bustard (*Ardeotis australis*, Priority Four [DEC]);
- Wedge tailed Eagle (*Aquila audax*) (Migratory);
- Whistling Kite (*Haliastur sphenurus*) (Migratory, Marine);
- Pacific Black Duck (*Anas superciliosa*) (Migratory);
- Blackfaced Cuckoo shrike (*Coracina novaehollandiae*) (Marine);
- Blackfronted Dotterel (*Charadrius melanops*) (Migratory);
- Pallid Cuckoo (*Cuculus pallidus*) (Marine);
- Sacred Kingfisher (*Todiramphus sanctus*) (Marine);
- Rainbow Bee eater (*Merops ornatus*) (Migratory, Marine);
- Glossy Ibis (*Plegadis falcinellus*) (Migratory, Marine); and
- Magpie lark (*Grallina cyanoleuca*) (Marine).

The continuance of analogous vegetation types and habitats is a result of the same ridgeline extending east, beyond the bounds of the study area, into Karijini National Park. Therefore such vegetation types and habitats are likely to be well represented and protected in the Park and surrounding areas. The majority of the species listed above are transitory in nature and unlikely to be significantly reliant on the Project area for habitat. Non transitory species are unlikely to be significantly impacted as their habitat is well represented in surrounding areas.

4.3 Social Environment

4.3.1 Local Government and Towns

The Project runs through the Shires of Roebourne, Ashburton and East Pilbara. Major towns in proximity to the Project include:

Table 4-1 Approximate Town Proximity to Works

TOWN	APPROXIMATE PROXIMITY TO WORKS
Tom Price	1km
Dampier	2km
Wickham	5km
Roebourne	6km
Karratha	10km

4.3.2 Aboriginal Heritage

The Project area lies within seven Native Title claims inclusive of the Ngarluma Group, Kuruma Marthudunera, Narlawonga Ngarlawangga, Yindjibarndi, Eastern Guruma, Yinhawangka and Innawonga Bunjima people.

There are a number of Aboriginal communities in close proximity to the Project Area. A number of Aboriginal heritage archaeological sites have been identified within the

AutoHaul™ Project area. All efforts will be made to avoid disturbance to these sites (refer to Section 6.6).

Those sites requiring disturbance will be managed in compliance with the *Aboriginal Heritage Act 1972*, including liaison with affected Aboriginal Groups. All heritage sites will be fenced according to Rio Tinto Iron Ore Expansion Projects (RTIOEP) Heritage Fencing Guidance note (RTIO-HSE-0121790).

4.3.3 Adjacent Land Use and Tenure

The Project area traverses a broad cross-section of land ownership and tenure including various tenements under the Western Australia *Mining Act* 1978, Leases under the LAA and freehold land as administered by the Western Australian Land Information Authority (Landgate).

The Project area falls within a number of pastoral stations, including Mt Welcome, Coolawanyah, Hamersley, Juna Downs and Marillana Stations.

The Project area is inside or passes within close proximity to environmentally sensitive areas such as Millstream Chichester National Park (MCNP), Karijini National Park (KNP), Harding Dam and Millstream Water Reserve Catchments.

5.0 ENVIRONMENTAL MANAGEMENT SYSTEM

5.1 Health Safety Environment and Quality Management System Overview

Impacts on the environment will be managed via the Rio Tinto Health, Safety Environmental and Quality Management System (HSEQMS). The primary purpose of the HSEQMS is to demonstrate sound environmental performance through control of impacts on the environment, consistent with the business Environmental Policy and Objectives.

Rio Tinto's Environmental Management System was certified to AS/NZS ISO1400:2004 in November 2010. The HSEQ Management System standard has been developed and is consistent with ISO 14001 (Environment); OHSAS 18001 and AS/NZS 4801 (Health and Safety); ISO 9001 (Quality); and equivalent international standards.

The HSEQMS standard is divided into 17 elements which this CEMP will implement. The 17 elements have been summarized in the diagram below:

Whilst all contractors are encouraged to establish and maintain their own Environmental Management System (EMS), compliance with the HSEQMS is required as a minimum.

The Engineer, also ISO14001 certified, has developed this CEMP for specific environmental risks based on Rio Tinto HSEQMS procedures. The Engineer will ensure that the relevant HSEQMS documentation will be available to Project personnel on advice from RTIO, including contractors.

5.2 Environmental Policy

The RTIO Health Safety, Environment and Quality Policy (RTIO-HSE-0070927) provides a framework for RTIO activities. The policy applies to all personnel and contractors undertaking work on behalf of RTIO. The content of these policies will be presented during site inductions and displayed in site common rooms and the Health, Safety and Environment (HSE) office. Contractors are also encouraged to have their own environmental policy. Calibre's HSEQ policy will be communicated on the Project, however, the RTIO HSEQ policy applies.

5.3 Environmental Risk Register

Hazard identification and risk management is a part of the RTIO HSEQMS (refer to RTIO-HSE-0062207). A part of the process is the qualitative risk assessment matrix (RTIO-HSE-0067793), which has been prepared for the AuoHaul™ project.

A project-specific Construction Environmental Risk Register (CERR), RCEATO-REG-T-001, has also been produced for the AutoHaul™ Project. A risk identification workshop was held on the 7 June 2012 and register was created in accordance with HSEQ Risk Register Creation, Review and Maintenance Work Practice (RTIO-HSE-0067977). The purpose of the workshop was to identify environmental aspects (in addition to health, safety and quality) applicable to the Project, and assess the significance of their impacts. This CEMP has been reviewed and updated as a result of the risks and management practices identified in the risk assessment.

The Qualitative Risk Assessment Matrix or equivalent will be reviewed at least annually and will also be reviewed in response to significant incidents, changes in legal requirements, changes in Project scope, and findings of inspections, audits and management reviews. The document will be managed as a controlled document and each revision will require the approval of RTIO. Risk registers are approved by the RTIO Project Manager. RTIO recognises that regular updates of risk registers are an indicator of a healthy environmental management system.

At a task level, Job Hazard Analyses, Take 5s, safety interactions or equivalent will be used by personnel to identify potential environmental risk and appropriate control measures.

5.4 Legal and Other Requirements

5.4.1 Legal Requirements

The Project will comply with all relevant Federal, State and Local Government legal requirements. Copies of all Licences/Approvals/Permits will be held on the Legal and Other Requirements System (LAORS), as per Section 4.3.1 of the Legal and Other Requirements Work Practice (RTIO-HSE-0072772) and Environment Legal and Other Requirements Procedure (RTIO-HSE-0130122). The Engineer’s Legal and Other Requirements Register (RCEATO-REG-T-003) will be utilised on the AutoHaul™ Project.

5.4.2 Legal and Other Requirements Register

The AutoHaul™ Environmental, Legal and Other Requirements Register (RCEATO-REG-T-003) will identify legislation, policies, systems and standards relevant to the Project. The register will be reviewed on a quarterly basis with new legislation and standards added as required. An AutoHaul™ Project specific online LAORS has been created and shall be updated as required to consolidate all the legal and other requirements relevant to the Project. The Engineer will ensure that communication of legal responsibilities between accountable persons is duly undertaken.

5.5 Demonstrating Legal Compliance

The Project will be subject to annual internal audits for compliance against LAORS. The Senior Environmental Advisor or Site Environmental Advisor will undertake inspections and audits against legal compliance and communicate the results of these inspections/audits to relevant persons. Results of audits will be included in reports to RTIO.

Internal and external audit programs will include, but not limited to:

- The Expansion Projects Assessment Schedule 2012 (RTIO-HSE-0015765);
- Regulatory inspections/audits of the site (e.g. DEC, DoW, DMP);
- ISO14001 surveillance process (also see section 5.17)
- AutoHaul™ Project Environmental Audit Schedule (RCEATO-SCH-T-003)
- Audits against Rio Tinto HS&E Standards;
- Contractor audits;
- Internal Calibre audits; and
- Weekly inspections.

5.6 Environmental Objectives and Targets

Project specific Environmental Objectives and Targets (RCEATO-PLN-T-004) have been developed for the construction phase that will be consistent with RTIOEP Objectives and Targets (as outlined in HSEQ Management Improvement Planning Work Practice – RTIO-HSE-0072773) and which will be reflective of identified risks. Appropriate actions will be selected to support these. RCEATO-PLN-T-004 shall be reviewed on an annual basis and will be a controlled Calibre document. Performance will be measured by individually reporting on Targets set via Compliance Monitoring, Monitoring Reports and Incident Report Tracking at regular intervals.

5.7 Project Responsibilities and Authorities

All personnel managing, preparing documentation or working on construction of the Project shall have specific responsibilities for environmental management. Table 5-1 Summary of Key Roles and Responsibilities, details the roles and responsibilities of key personnel in relation to environmental management of the Project. The Rail Capacity Enhancement ECP/AutoHaul ™/DA/OSS Organisation Chart which is attached as Appendix C details specific roles and responsibilities for the Project.

Each of the roles listed is filled by an individual that has been approved and deemed competent by RTIO. In particular, environmental personnel have been selected on past demonstrated competence in environmental management.

Table 5-1 Summary of Key Roles and Responsibilities

ROLE	KEY RESPONSIBILITIES
RTIO	
RTIOEP Environmental Manager (Off-site)	• Providing specialist environmental management support to RTIOEP; • Providing management advice on implementation of the HSEQMS; • Ensuring effective application of HSEQMS for RTIOEP; • Authorising and communicating with external agencies as required; • Approving the Risk Register and the CEMP in coordination with the RTIOEP Site Representative; • Managing environmental audits of the project to ensure compliance with legal requirements and the HSEQMS; and • Providing environmental advice in the event of an incident or emergency event.
RTIOEP Specialist Environmental Advisor (Off-site)	• Assisting RTIOEP Environmental Manager with tasks as above.
RTIOEP Project Manager (Off-site)	• Demonstrating effective leadership to all project personnel with respect to matters concerning the environment; • Ensuring the project complies with all applicable requirements set out in the Rio Tinto Environment Standards and HSEQMS; and • Ensure legal compliance.
RTIOEP Site Representative (On-site)	• Demonstrating effective leadership to all project personnel with respect to matters concerning the environment and implementation of the HSEQMS at a project level; and • Reporting significant incidents and emergency events to the RTIOEP Environmental Manager or RTIOEP Environmental Advisor as soon as reasonably practicable.
The Engineers	
The Engineers Project Managers (Off-site)	• Coordination of project deliverables with RTIOEP Project Manager; • Ensuring effective delivery and implementation of HSEQMS within project scope of works; and • Ensuring effective communication of project requirements to the Engineers personnel and contractors.

ROLE	KEY RESPONSIBILITIES
The Engineers Site Construction Manager (On-site)	• Ultimate responsibility in relation to compliance with the requirements of the contract including the environmental specifications; • Facilitating review and approval of the Engineers Risk Registers and CEMP in consultation with the Engineers Senior Environmental Advisor; • Reviewing environmental management performance on a monthly basis; • Ensuring appropriate management actions are put into place to facilitate compliance with all legal requirements; • Ensuring they are aware of Project legal obligations and requirements; and • Ensuring the requirements of the Construction Environmental Management System, including the CEMP, are met.
The Engineers Environmental Manager	• Overall responsibility for the Engineer's EMS; • Liaison with RTIOEP Environment Manager and other project personnel as required; and • Providing environmental support to the Engineers Project Manager and the Engineers Environmental Advisors.
The Engineers Senior Environmental Advisor (Off-site)	• Providing assistance to the Engineers Environmental and Approvals Manager; • Providing the Engineer Management advice in implementation of the HSEQMS; • Coordination of project approvals in conjunction with the Rio Tinto Approvals Team; • Development, management and review of HSEQMS component of works; • Ensuring effective application of HSEQMS for RTIOEP; • Conducting internal audits of site HSEQMS and legal compliance; • Liaison with RTIO Environmental Manager and other project personnel as required; and • Providing environmental support to the Engineers Construction Manager in the absence of the on-site The Engineers Environmental Advisor.

ROLE	KEY RESPONSIBILITIES
The Engineers Environmental Advisor (On-Site)	• Assisting the Engineers Senior Environmental Advisor with development of CEMP and HSEQMS requirements; • Implementation of the CEMP as necessary; • Specialist on-site environmental advice and guidance to contractors as necessary; • Monitoring of environmental parameters and submission of monthly environmental reports; • Delivery of site environmental inductions; and • On-site liaison with RTIO Specialist Environmental Advisor and/or RTIO Environmental Manager as required.
The Engineers Construction Superintendent	• Ensuring the requirements of the Construction Environmental Management System including the CEMP are met for their area of responsibility; • Ensure compliance with RTIO clearing approvals process; topsoil and subsoil stockpiling requirements; borrow pit management plans; and spill control measures on site; and • Ensure that employees and sub contractors under their direction have appropriate training, inductions, qualifications, permits and understand the environmental restrictions associated with their work activities.
The Engineers Occupational Health and Safety Manager	• Implement components of the CEMP that relate to Health and Safety as required; • Provide inspection and monitoring information to the Engineers Site Environmental Advisor as appropriate; and • Provide advice and assistance in the coordination of Health, Safety and Environmental aspects of the project.

ROLE	KEY RESPONSIBILITIES
CONTRACTORS	
Successful Tender Applicants for the provision of construction materials and services	• Committing to provide services and materials that minimise impact to the environment; • Complying with relevant environmental legislation; • Supervise and train their personnel and subcontractors in order to meet the requirements of this CEMP. This includes delivery of at least 1 environmental awareness session per month; • Implementing the requirements of this CEMP that relate to their work scope; • Conducting routine environmental inspections of worksites both informally and formally with the Engineers Site Environmental Advisor; • Provide personnel and maintained equipment for emergency response to environmental incidents; • Reporting any incidents as soon as practicable to the Engineers Site Environmental Advisor or other authorised personnel as appropriate; • Undertaking corrective/preventive actions arising from inspections, audits and management reviews in a timely manner; • Liaising with the Engineers Site Environmental Advisor to manage their environmental impacts; and • Managing any sub contractors environmental performance.

5.8 Environmental Awareness and Training

5.8.1 Inductions

All Project employees and contractors are required to undertake an Expansion Projects HSEQ Induction before they commence work on any RTIO site.

The Engineer is developing Project specific induction training package to supplement the Northern or Southern inductions, which will communicate the environmental impacts introduced by Project activities and specify the control measures to be implemented by all personnel. The induction will also cover specific management actions for working in environmentally sensitive areas such as the National Parks. This induction will be reviewed on a quarterly basis, or as significant Project changes arise.

5.8.2 Training Needs Analysis and Matrix

The Engineer has conducted a training needs analysis for all personnel, taking into account the employees/contractor's position, responsibility and duties as presented in Training Needs Analysis Plan (RCEATO-PLN-T-005). A Training Matrix and Training Records will be maintained, and kept for a minimum of seven years and will be a controlled document.

Environmental awareness training will be held at monthly intervals, in accordance with the Training Needs Analysis Plan (RCEATO-PLN-T-005). Contractors are required to ensure that environmental training, on topics relevant to their scope of work, is provided to its employees and sub-contractors throughout the construction period. Environmental concerns and issues will be discussed daily at morning pre-starts. Environmental toolbox meetings will be held on a monthly basis during construction, to discuss site-specific environmental issues and management. The Senior Environmental Advisor will coordinate the preparation and distribution of Environmental themes to the Engineers employees, in conjunction with the Site Environmental Advisor. These toolbox meetings will also be available to subcontractors at their request. However, it is the responsibility of individual subcontractors to present monthly environmental toolbox themes to employees relevant to their scope of works.

5.8.3 Training Materials

Existing training materials are available from RTIO which will be used where appropriate. When required, the Engineer will update these materials or develop Project specific training and supply copies of this training material to the RTIOEP Environmental Manager. Additional training materials will be available from the Engineers Corporate Environmental Team.

5.9 Communication

5.9.1 Internal

Environmental notice boards will be established on-site to inform personnel of relevant environmental information such as the RTIO HSEQ Policy, minutes of meetings, results of monitoring, performance standards, environmental incident alerts and company environmental notices. The notice board will be refreshed monthly with up-to-date information, or as it becomes available.

Communication of environmental issues that will require actioning will occur via:

- Progress Meetings;
- Audit Report Findings;
- Environmental Site Inspections;
- Incident Reports;
- Corrective Action Register; and
- Meetings between RTIO EP Specialist Environmental Advisor (Off-site), Calibre Project Managers (Off-site) and the Engineers Senior Environmental Advisor (Off-site).

Communications with rail operations and other existing operations will be conducted via the interface meetings that occur between construction and operations. These meetings will be minuted and distributed to relevant staff.

The result of these communications will be recorded in minutes of meetings or will be contained within reports.

5.9.2 External

The RTIOEP Environmental Manager or as delegated, is the sole person authorised to communicate externally in relation to matters concerning the environment. This includes but is not limited to communications with the media and government agencies and particularly in relation to reporting of incidents that may have occurred.

A communication protocol has been developed to inform the DEC of all construction activities in the National Parks including the management of weeds and fire issues (see Appendix D).

All community complaints received by Project personnel will be directed to the RTIOEP Site Representative immediately for action. Complaint information will be recorded using the RTIO incident report forms.

All communication will occur as per Table 5-2: Communication Mechanisms on the AutoHaul™ Project.

Table 5-2: Communication Mechanisms on the AutoHaul™ Project

COMMUNICATION MECHANISM	ACTIVITY	TYPE OF INFORMATION	RECORDS	FREQUENCY	WHO
Induction	Training	Presentation	Assessment	As Required	The Engineers Trainer (HS or E Advisor)
Job Hazard Analysis (JHA)	Reporting	Document	Filed	As Required	All
Environmental Alerts	Awareness	Poster	Noticeboard	As Required	The Engineers Environmental Advisor (Site)
LAORS Updates	Awareness	Report	Database	As Required	The Engineers Environmental Advisor (Site)
Monthly RTIOEP Environmental Reporting	Reporting	Report	Published	Monthly	The Engineers Senior Environmental Advisor
EPCM Monthly Project Report	Reporting	Report	Published	Monthly	The Engineers Project Manager

COMMUNICATION MECHANISM	ACTIVITY	TYPE OF INFORMATION	RECORDS	FREQUENCY	WHO
EPCM Quarterly Environmental Report	Reporting	Report	Circulated	Quarterly	The Engineers Senior Environmental Advisor
Monthly Contractor Environmental Report	Reporting	Report	Circulated	Month	Contractor
HSEQMS Awareness Posters	Awareness	Poster	Noticeboard	As Required	The Engineers Site Environmental Advisor
Environmental Inspections	Reporting	Check Sheet	Filed	As Required	The Engineers Site Environmental Advisor
Audit Report Findings	Reporting	Report	Circulated	As Required	The Engineers Site Environmental Advisor
Incident Reports	Awareness	Report	Circulated	As Required	HSE Administrator
Corrective Action Register	Reporting	Report	Published	As Required	The Engineers Site Environmental Advisor
Environmental Training Programs	Training	Presentation	Assessment	As Required	The Engineers Senior Environmental Advisor
Pre-start Meetings	Awareness	Meeting	Attendance	Daily	All
Toolbox Meetings	Awareness	Meeting	Minuted	Weekly	All
Weekly Progress Meetings	Awareness	Meeting	Minuted	Weekly	The Engineers Senior Environmental Advisor
Weekly HSE Meetings	Awareness	Meeting	Minuted	Weekly	All
Environmental Performance Review	Reporting	Meeting	Attendance	Annual	The Engineers Project Manager
Project Information Centre (Notice Board)	Reporting	Poster	Noticeboard	On-going	HSE Administrator
Works in National Parks	Reporting	Report	Published	As Required	RTIO Specialist Environmental Advisor

COMMUNICATION MECHANISM	ACTIVITY	TYPE OF INFORMATION	RECORDS	FREQUENCY	WHO
Perth Notice Board	Reporting	Poster	Noticeboard	On-going	The Engineers Senior Environmental Advisor
Pre-tender Meetings	Awareness	Presentation	Minuted	As Required	All
Tender Award Kick-off Meeting	Awareness	Presentation	Minuted	As Required	All

5.10 Document Control

The following documents will be managed as controlled documents and revisions will be re-submitted to RTIOEP for approval:

- Construction Environmental Management Plan (this document) RCEATO-PLN-T-003;
- Project Environmental Objectives and Targets RCEATO-PLN-T-004;
- Training Needs Analysis RCEATO-PLN-T-005; and
- Calibre Legal and Other Requirements Register for RTIOEP CP-PLN-HSEQ-004.

These documents are intended to be of a live nature for reference by project personnel and will be updated annually or as required by designated personnel.

5.11 Management Procedures

Section 6 of this CEMP details the management principles and objectives for particular tasks or environmental risk. These incorporate the Rio Tinto Construction Environmental Management Guidance Notes and Specification (RTIO-HSE-0015533) and specific items listed in the Risk Register or equivalent document.

5.12 EMS Documentation

HSEQ documentation is available to the Engineers personnel via the HSEQ intranet site and from RTIO. The Engineer will forward relevant HSEQ documentation to Calibre staff and contractors as necessary, and under RTIO’s direction.

The Engineer will also use controlled documentation available to it via the the Engineers EMS certified to AS/NZS ISO 14001:2004 where required.

5.13 Emergency Response

A Safety Management Plan (SMP) Addendum (RCEATO-PLN-K-001) and Emergency Preparedness and Response Plan (RCEATO-PLN-K-002) have been developed for the AutoHaul ™ project. Both of these documents will be used in conjunction with the

Safety Management Plan for RTIOEP (CP-PLN-HSEQ-001). These documents will detail:

- actions and procedures required to mitigate environmental harm associated with incidents and emergencies;
- procedure on communications;
- responsibility for notifying the RTIOEP Site Representative of the incident or emergency;
- Memorandum of Understanding with existing operations in relation to response from site;
- contact details of all personnel who are required to be notified;
- other site emergency contact numbers and details, including external site emergency services and service providers;
- schedule for emergency response drills; and
- training requirements.

The Engineer and the projects Contractors will maintain equipment and personnel for response to potential environmental incidents and identify likely emergency scenarios. Equipment and personnel requirements will also be outlined in the safety documents.

5.13.1 Emergency Drills

At least one emergency response drill will be conducted annually, which addresses potential environmental emergency issues. This will be managed under the Emergency Preparedness and Response Plan (RCEATO-PLN-K-002).

5.14 Monitoring and Measurement

The environmental performance of construction activities and the identification of environmental monitoring requirements will be assessed throughout the construction period. Regular monitoring of all activities, contractors and personnel will be conducted, including daily, weekly and monthly checklists, calibration record checks, mobilisation and demobilisation audits and audits for compliance with HSEQ. Monitoring procedures to be developed will include (but are not limited to):

- Sampling Methods;
- Sampling Locations;
- Quality Assurance and Quality Control;
- Responsibilities;
- Storage;
- Disposal;
- Dispatch; and
- Calibration.

The Engineers Site Environmental Advisors will be responsible for coordinating the monitoring of construction activities and compilation of data for reports. Monitoring of environmental performance is not limited to environmental personnel. Project supervisors are responsible for all works and personnel under their supervision to ensure that work activities adhere to the requirements of this CEMP.

Any monitoring required to meet legislative, licensing, work approval or monthly environmental reporting requirements will be undertaken in accordance with Measuring, Monitoring and Data Analysis Work Practice (RTIO-HSE-0068753).

Monthly Environmental Reporting will include, but not be limited to:

- ground disturbance and clearing locations (including area cleared) [monthly};
- amounts of topsoil and subsoil stockpiled [quarterly];
- borrow pit activities (area disturbed and volumes);
- water usage data [quarterly];
- waste type, amount (mass/volume), and disposal method (recycled, re-used, disposal location) [quarterly];
- hazardous materials data [quarterly];
- greenhouse gas (energy consumption) / National Greenhouse and Energy Reporting (NGER) data; [monthly}and
- environmental incidents [monthly}.

The Engineers will submit a Monthly and Quarterly Environmental Report to RTIOEP and provide other Project reporting as required.

5.15 Incidents and Corrective Actions

5.15.1 Incident reporting

Incidents include events that cause environmental impacts or harm, near misses and complaints. All environmental hazards and incidents shall be reported promptly to ensure identification of corrective actions, appropriate cleanup to reduce the likelihood of reoccurrence. Environmental incidents that are considered significant may legally be required to be reported to Government by RTIO. Any incidents located within the National Parks will be treated as a significant incident and will be informed as per the DEC protocol.

An incident report should also be completed where there has been a non-compliance with internal procedures that may or may not have resulted in an environmental impact or harm. Notification to RTIO is required for those incidents within 24 hours, with a written report to be submitted within 7 days, as per the Incident Management Verbal Notification Reporting Matrix (RCEATO-LST-T-001). Environmental hazard reporting aids in the identification of environmental risks and the implementation of preventative management procedures and controls.

Significant incidents require reporting to RTIO immediately. This will involve the completion and submission of the RTIO Incident Report Form (RTIO-HSE- 0037074) using the Environmental Incident Classification Guide (RTIO-HSE-0013610) within 24 hours of the incident. The Engineers Environmental Manager must be notified immediately upon occurrence of significant environmental incidents.

All Project personnel (RTIO, Calibre, contractors and subcontractors) have the responsibility to report incidents either directly or indirectly through their supervisors.

Investigation of environmental incidents is conducted to determine contributing factors to the event and to accurately identify and initiate appropriate preventative and management actions. An incident register will be kept by the Engineers and be used to analyse incident trends and actions. JIRA shall be used to record all incidents and ensure that corrective actions are identified, implemented and closed out.

5.15.2 Corrective Action

Corrective actions generated from incident and hazard reporting will be recorded on the Engineers Site Corrective Action Register to ensure items raised are recorded, rectified and closed. Where necessary, the project will retain evidence (documents, photographs, personnel statements) on file to demonstrate completion of actions. The Site Corrective Action Register, recording all corrective actions raised and closeout details will be maintained on site.

5.16 Managing Records

All documentation (such as training records, incident reports etc.) will be managed through the Engineers Document Control Procedure (CPJ-02-PRO-PM-07) and HSEQMS requirements (in accordance with HSEQ Document, Record and Data Management and Control Work Practice – RTIO-HSE-0072775) to ensure a thorough archiving system and handover of environmental management records to Rio Tinto within the operations and maintenance manual at handover. Records will be retained for the required period.

5.17 Internal Audits and Inspections

Monitoring and internal audits are required to ensure RTIOEP management are advised of environmental performance and environmental non-conformances. In addition, construction project audits form part of RTIOEP's self-assessment program (required under Rio Tinto's corporate assurance program) and covered under HSEQMS.

Inspections and audits will be conducted when applicable to ensure compliance with this CEMP, HSEQMS, RTIOEP Standards, Ministerial conditions and licensing requirements (DEC/DoW/DMP). Works undertaken in Millstream Chichester National Park, within the MS 880 footprint, are subject to inspections by Park Rangers and DEC as per commitments under RTIO Offset package. Types and frequencies of audits and inspections are outlined in Table 5-3 Inspection and Audit Types. An Audit Schedule will be produced by the Engineer and is a controlled document. Actions and management requirements resulting from audits and inspections remain the ultimate responsibility of the Engineer and RTIO Project Director, with site-specific management being administered by the Engineers Site Construction Manager.

Table 5-3 Inspection and Audit Types

TYPE INSPECTIONS	REQUIREMENT	PURPOSE	REFERENCE DOCUMENTS
THE ENGINEERS CONDUCTED INSPECTIONS			
Daily Site Inspections	CEMP Requirement	Informal site inspection to be conducted by all project supervisory staff. Day to day work activities are evaluated for compliance with CEMP. Provides informal setting for contractor to seek advice to ensure compliance with CEMP.	RCEATO-SCH-T-001 AutoHaul™ Environmental Audit Schedule.

TYPE INSPECTIONS	REQUIREMENT	PURPOSE	REFERENCE DOCUMENTS
Calibre Environmental Site Inspections	CEMP Requirement	Formal site inspections conducted by Calibre in conjunction with contractor supervisory staff to evaluate contractor understanding and maintenance of CEMP procedures and controls. It also provides an opportunity for Calibre and Contractors to discuss effectiveness of environmental procedures and controls at a particular job site so that, if necessary, adjustments can be made to ensure continued compliance with CEMP objectives and targets. Environmental Inspection Checklist to be completed by the Engineers Site Environmental Advisor. the Engineers Site Environmental Advisor to log Corrective Actions into Incidents & Corrective Action Register. Form 163 will be kept on file by the Engineers Site Environmental Advisor.	Form 163 – Environmental Inspection Checklist. RCEATO-SCH-T-001 AutoHaul™ Environmental Audit Schedule.
THE ENGINEERS CONDUCTED AUDITS			
Mobilisation Audit	CEMP Requirement	To assess the Engineers and Contractor compliance with CEMP. To ensure that design criteria have been met and that environmental procedures are being implemented.	Formal, Form 162 - RCEATO-SCH-T-001 AutoHaul™ Environmental Audit Schedule.
Demobilisation Audit	CEMP Requirement	To assess the Engineers and Contractor compliance with CEMP and to ensure that all project handover documents have been completed.	Formal, Form 128 RCEATO-SCH-T-001 AutoHaul™ Environmental Audit Schedule.
Internal Legal Compliance Audit	CEMP Requirement	To review compliance with legal requirements.	RCEATO-SCH-T-001 AutoHaul™ Environmental Audit Schedule..
Internal CEMP Audit	CEMP Requirement	To review compliance with the CEMP and assess its' effectiveness.	RCEATO-SCH-T-001 AutoHaul™ Environmental Audit Schedule.
Internal CEMP Audit – Contractors	CEMP Requirement	To review Contractor compliance with CEMP.	RCEATO-SCH-T-001 AutoHaul™ Environmental Audit Schedule.
Internal Environmental Management System Audit	RTIOEP Requirement	A formal review of the EMS conducted annually during construction.	Form 162 ISO 14001.
THIRD PARTY CONDUCTED AUDITS			
RTIOEP Audits	ISO 14001	HSEQMS compliance performance, compliance against RTIO Standards, general performance audits, legal compliance audits.	External procedure which will be adhered to as required.
3rd Party Surveillance Audits (DEC, DMP, EPA)	License and permits, ISO 14001	Assess compliance with licenses, works approval and permits.	External procedure which will be adhered to as required.

TYPE INSPECTIONS	REQUIREMENT	PURPOSE	REFERENCE DOCUMENTS
Calibre EMS Audits	ISO 14001	Assess compliance with Calibre's EMS.	Formal, a report of the audit findings will be compiled by the certifying body. Actions will be reported internally to Calibre.

5.18 Management Review

Review and revision of this CEMP will be undertaken on an annual basis (or at least once if project life less than one year) based on statutory approvals and conditions, results of monitoring, inspections, recurring incidents, auditing and improved industry practices. In order to provide effective feedback, this CEMP ensures that:

- regular inspections of key areas highlighted in the CEMP are undertaken;
- actions identified in the CEMP are implemented and reported; and
- environmental incidents are reported and rectified.

Any revision of the CEMP will be forwarded to RTIOEP for approval prior to adoption. Any corrective actions arising from such review will be entered into an actions database (JIRA) to ensure appropriate closeout.

The Project will be reviewed against this CEMP approximately 12 weeks after construction commencement as per Construction Environmental Management Guidance Note and Specification RTIO-HSE-0015533.

6.0 MANAGEMENT PROCEDURES

The management actions that follow in this section form the basis for control of construction activities. Where reference is made to procedures or guidelines the latest approved version (as located on the HSEQMS website) is to be used.

The person named under the "responsibility" column within each management action refers to the person whose ultimate responsibility is the implementation of that action. Responsibility of particular management procedures and actions can be delegated, though overall responsibility will remain with the responsible person.

Where the Engineers Construction Manager is named and the work is being conducted by a Sub Contractor, the Sub Contractor is responsible for ensuring the effective implementation of the management action. The Engineers Construction Manager has ultimate responsibility as the manager of the Sub Contractor and the sub contract.

6.1 Use of Existing Infrastructure and Facilities

The construction of new facilities and infrastructure in previously unused areas may cause unnecessary impacts to the natural environment. The environmental footprint of infrastructure and facilities can be reduced by making use of those facilities already in place. Examples include the use of existing landfills or borrow pits rather than construction of new landfills. Any liaison work with Operations or other RTIOEP projects must occur with the RTIOEP site rep or Project Managers sanctioning.

Objectives:

- To effectively communicate with Operations management on the use and maintenance of existing facilities in accordance with agreed management requirements; and
- Management actions for ensuring the use of existing infrastructure wherever possible, is detailed in Table 6-1 Management Actions for Use of Existing Infrastructure and Facilities.

Table 6-1 Management Actions for Use of Existing Infrastructure and Facilities

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.1.1	The Engineers shall liaise with a suitable Operations Representative to determine the existing facilities and infrastructure available for use prior to mobilisation.	The Engineer Construction Manager	Pre-construction/ Construction	
6.1.2	The Engineers shall provide relevant details to the Operations Representative regarding usage and consumption demands. For example, this would include estimated landfill waste volumes, potable water bore extraction rates, waste water treatment inflow volumes, etc.	The Engineers Construction Manager	Construction	
6.1.3	The Engineers may at its discretion conduct a due diligence audit to confirm the current condition of the facilities and infrastructure prior to use. The current condition must be agreed by the Operations Representative.	The Engineers Construction Manager	Construction	
6.1.4	The Engineers shall comply with the current operating procedures of the existing facilities/infrastructure.	The Engineers Construction Manager	Construction	
6.1.5	The Engineers shall establish communication systems such that it can appropriately respond to operational changes including but not limited to procedural changes associated with existing facilities. This must address communication requirements associated with emergency response.	The Engineers Construction Manager	Construction	
6.1.6	The Engineers shall meet the reasonable requests of the Operations Representative in relation to compliance with catchment-related management issues. For example, this may include weed management	The Engineers Construction Manager	Construction	

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
	(spraying) programmes, feral animal management, and firebreak management. The Engineers may also be subject to audits conducted by the relevant Operation in relation to these catchment-related management issues.			
6.1.7	The Engineers shall be subject to inspections and audits performed by the Operations Representative to facilitate overall environmental management of the existing infrastructure/facilities. The Engineers shall respond with details of corrective actions taken to resolve any issues identified by the Operations Representative.	The Engineers Construction Manager	Construction	
6.1.8	The Engineers shall notify the Operations Representative of any defects or maintenance required on existing infrastructure and facilities that if not fixed, may impact the environment.	The Engineers Construction Manager	Construction	

6.2 Ground Disturbance and Clearing

Ground disturbance includes any activities that disturb, alter or damage the Project area in any way and can lead to a loss of flora, fauna habitat and disturbance of heritage sites. This is not limited to previously undisturbed ground and procedures must be followed to reduce rehabilitation costs, soil rehandling, ensure compliance with appropriate legislation and limit damage to the natural environment and heritage values.

Clearing and ground disturbance breaches have significant penalties and compliance with the projects clearing procedure is mandatory.

Breaches may impact on the Millstream Chichester National Park (MCNP) and the Karijini National Park (KNP). Additional controls to manage this risk is described in section 6.22 – A Class Reserve Management.

Ground disturbance activities include:

- disturbance or removal of vegetation;
- change in ground compaction from vehicle and plant movements and parking;
- drilling, grading, trenching and excavation activities;
- borrow pits and quarries;
- changes to natural drainage;
- establishment of office and laydown areas;
- topsoil stockpiling; and
- importation of fill material.

Objectives:

- To ensure sustainable stewardship of the land by undertaking construction with minimum ground disturbance;
- To ensure that ground disturbance does not occur outside of approved areas;
- Ground disturbance complies with relevant legal provisions; and
- Ground disturbance complies with areas specified in RTIO clearing permits and Approval Request areas.

Table 6-2 Management Actions for Ground Disturbance and Clearing

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.2.1	All personnel to be informed of the importance of ground disturbance and clearing management and the requirement to keep to existing tracks and roadways.	The Engineers Construction Manager / Contractor	Pre-Construction/ Construction	Induction
6.2.2	Prior to ground disturbance activities, an Approvals Request permit must be obtained by The Engineers. A copy must be held on site.	The Engineers Construction Manager	Construction	RTIO-HSE-0069098 Approvals Request Guidance Note
6.2.3	Prior to ground disturbance activities contractors must obtain a Calibre Clearing Permit and complete a clearing 'Take 5'. The Calibre Clearing Permit will be smaller than the area covered by the Approvals Request permit unless specific approval by RTIOEP Environment Manager and RTIOEP Project Manager is given and include conditions from RTIOEP Approvals Request Permit.	Contractors	Construction	RTIO-HSE- 0043075 Clearing Permit Procedure RTIO-HSE-0043306 Clearing Permit Form No 188 - Clearing Permit - Take 5
6.2.4	The boundary of the area to be cleared must be surveyed and suitably defined with survey tape or equivalent to minimise the risk of disturbance outside the designated area.	The Engineers Construction Manager	Construction	RTIO-HSE- 0043075 Clearing Permit Procedure

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.2.5	Any areas containing significant flora, fauna, heritage or service and utility corridors shall be fenced or suitably marked to prevent damage. All heritage sites to be fenced according to RTIOEP Heritage Fencing Procedure (RTIO-HSE-0121790) Buffer zones will be employed where applicable. Exclusion areas will be demarcated as per the requirements specified in the approvals permit. In addition to these requirements sites that are inside or within 10m of approvals permit boundaries shall also be demarcated. Deviations can only be approved by the RTIOEP Environment Manager and endorsed by the RTIOEP Project Manager.	The Engineers Construction Manager	Construction	RTIOEP-HSE-0121790 RTIOEP Guidance Note Heritage Sites Fencing
6.2.6	Plan ground disturbance activities to ensure that minimal disturbance is achieved through the use of appropriate ground engaging plant, use of designated tracks, roadways and use of pre-existing disturbed areas.	The Engineers Construction Manager	Construction	RTIO-HSE-0043075 Clearing Permit Procedure
6.2.7	Prior to clearing permit approval a site inspection of the area to be disturbed will be conducted with the contractor, The Engineers site supervisor and The Engineers environmental advisor prior to application for a Clearing Permit. At time of site inspection the following will be explained and detailed: a) extent and purpose of ground disturbance activities, b) Verification with GPS, of boundary pegging c) Where cleared vegetation and topsoil will be stockpiled for later use and management of stockpiles d) Method and depth of topsoil removed e) Method and depth of subsoil removed f) Rehabilitation activities	The Engineers Construction Manager	Construction	RTIO-HSE- 0043075 Clearing Permit Procedure

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.2.8	Mobile plant operators must receive clear instructions from the Site Construction Manager or delegate on ground disturbance activities; including maps and GPS coordinates of disturbance area. Take 5's must be conducted prior to any shift involving clearing by the operator(s), the contractor supervisor and the Engineers supervisor. The Engineers Clearing Permit must be attached to the Job Hazard Analysis (JHA) and kept with the operator at all times.	Contractors	Construction	RTIO-HSE- 0043075 Clearing Permit Procedure RTIO-HSE-0043306 Clearing Permit Form Form No 188 - Clearing Permit - Take 5
6.2.9	Topsoil must be managed in accordance with HSEQMS procedure –Soil Management	The Engineers Construction Manager	Construction	RTIO-HSE-0011596 Soil Resource Management
6.2.10	Areas undergoing ground disturbance will be inspected (as appropriate) to ensure they meet the guidelines. The Engineers supervisor must be present during all boundary clearing activities or clearing adjacent to exclusion area.	The Engineers Construction Manager	Construction	RTIO-HSE- 0043075 Clearing Permit Procedure RTIO-HSE- 0043306 Clearing Permit Form
6.2.11	Verification of compliance with the Engineers Clearing Permit must be undertaken at completion of the clearing works. Ground disturbance outside of permit boundaries will be recorded as an incident and will be rehabilitated in consultation with RTIOEP requirements.	The Engineers Construction Manager	Construction	RTIO-HSE-0037074 Incident Report Form
6.2.12	All ground disturbance activities, areas of disturbance and rehabilitation will be reported in the Monthly Environmental Report	The Engineers Environmental Advisor	Construction	RTIO-HSE-0015107 Monthly Environmental Report
6.2.13	Emergency clearing activities (e.g. for fire, flood protection) must follow the approvals coordination emergency procedure.	The Engineers Construction Manager	Construction	RTIO-CR-0001029 Approvals Coordination Emergency procedure

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.2.14	Ground disturbance activities in National Parks shall adhere to the management plan for these areas. No borrow pits or quarries will be established in these areas without specific formal approval. Please refer to section 6.22 for additional management actions with respect to ground disturbance in the National Parks.	The Engineers Construction Manager	Construction	DEC 2011. Millstream-Chichester National Park and Mungaroona Range Nature Reserve Management Plan

6.3 Topsoil Management

Successful rehabilitation efforts are dependant on the availability of good quality, viable topsoil. Successful topsoil management will result in successful, minimum cost rehabilitation which will meet RTIO criteria and legislative obligations.

Objectives:

- Successfully remove, store, manage and reuse topsoil resources; and
- Compliance with approved licenses, permits and legal provisions.

Table 6-3 Management Actions for Topsoil Management

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.3.1	Topsoil must be managed in accordance with HSEQMS procedure – Soil Resource Management.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0011596 Soil Resource Management Earthworks Specification SS-C101
6.3.2	Ground disturbance permit procedures shall be followed prior to removing topsoil and include dust suppression procedures.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0011596 Soil Resource Management

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.3.3	Depth of topsoil to be removed is to be in accordance with the Soil Resource Management Procedure and as such; a) 100mm of topsoil will be stripped and stored, where available. If less than 200 mm is available, all available topsoil will be stripped within the clearing area. b) Vegetation and topsoil shall be stripped and stored in windrows or stockpiles as deemed appropriate c) Stockpiles location to be identified and recorded by the site surveyors.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0011596 Soil Resource Management
6.3.4	Inspection of topsoil removal practices and stockpiles will be conducted by the Engineers Site Environmental Advisor (as appropriate) to ensure they meet the guidelines.	The Engineers Site Environmental Advisor	Construction	RTIO-HSE-0011596 Soil Resource Management
6.3.5	Total area cleared for pre-strip, and amounts of topsoil and subsoil stockpiled will be reported in the monthly environmental report.	The Engineers Site Environmental Advisor	Construction	RTIO-HSE-0015107 Monthly Environmental Report
6.3.9	Topsoil stockpiles will be limited to 2m in height and with side slopes of no more than 20 degrees to reduce the risk of erosion The stockpiles shall not be located within drainage lines. Stockpiles will be sited appropriately to ensure erosion and run off are minimised.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0011596 Soil Resource Management
6.3.10	Burning of vegetation stockpiles is not permitted.	The Engineers Site Construction Manager	Construction	

6.4 Flora and Fauna Management

Construction activities have the potential to impact native flora and fauna. Protection of flora and fauna is controlled by the *Wildlife Conservation Act 1950.* Objectives:

- Minimise the impact of construction activities on existing native flora and fauna;
- Protect declared rare flora and significant priority flora / fauna through avoidance of construction related impacts, wherever practicable; and
- No introduction and/or spread of weeds as a result of construction works..

Table 6-4 Management Actions for Flora and Fauna Management

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.4.1	The requirements of the HSEQMS Wildlife Interaction Policy and the Rio Tinto Biodiversity Guidance shall be communicated and implemented on site.	The Engineers Site Environmental Advisor	Pre-Construction / Construction	Induction RTIO-HSE-0013116 Wildlife Interaction Policy
6.4.2	Maps and photographs/ images of Priority flora and fauna shall be supplied to personnel and displayed on office notice boards to facilitate identification and on-ground management.	The Engineers Site Environmental Advisor	Construction	
6.4.3	Fencing and/or signage shall be used to protect areas containing significant flora and fauna habitat where there is risk of damage. Access to these areas will be restricted or not permitted. The Engineer Site Environmental Advisor will inspect these areas prior to work commencing.	The Engineers Site Construction Manager	Construction	RTIOEP-HSE-0121790 Guidance Note Heritage Sites Fencing
6.4.4	A procedure shall be established that outlines the requirements for safe relocation of fauna (such as snake relocation) and management of injuries to fauna. The procedure shall include an internal and external contacts list for the purposes of fauna management.	The Engineers Site Environmental Advisor	Construction	To be completed
6.4.5	Temporary lighting, if required, shall take into consideration impacts on fauna and opportunities for minimising fauna impacts.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0049623 Lighting Natural and Artificial
6.4.6	Any suspected occurrences of significant flora and fauna will be reported to the Engineers Site Environmental Advisor.	The Engineers/ Site Construction Manager	Construction	Induction RTIO-HSE-0013116 Wildlife Interaction Policy
6.4.7	Any trenches will be inspected regularly and designed with suitable egress to prevent stock or fauna entrapment. All turkey nests, dams and other pit with potential to trap fauna shall be constructed with fauna egress ramps. Drill holes, pipes and conduits will be capped or blocked to prevent fauna getting trapped.	The Engineers Site Environmental Advisor	Construction	Induction DC-N002 for turkeys nests
6.4.8	Open trenches shall be ramped at the ends or egress structures	Contractor	Construction	N/A

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
	provided at an incline no greater than 45 degrees to allow fauna egress. Trenches shall be closed as soon as practicable.			
6.4.9	Turkeys nests must be fenced off to prevent fauna access and have egress devices suitable for fauna to egress the turkeys nest.	Contractor	Construction	DC-N002 Environmental Design Criteria for Temporary Facilities
6.5.0	Any sites of potential entrapment including trenches must be inspected for fauna at the beginning and end of each shift.	Contractor / Calibre Site Environmental Advisor	Construction	N/A
6.5.1	Feeding and/ or capture of native fauna or feral animals is not permitted at any construction sites.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0013116 Wildlife Interaction Policy
6.5.2	Any deaths of native fauna shall be reported as per incident report procedures.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0037074 Incident Report Form
6.5.3	Any fauna injury or mortality within MCNP shall be reported to the RTIOEP Specialist Environmental Advisor within 24hrs.	Calibre Site Environmental Advisor	Construction	

6.5 Weed Management

The spread or introduction of weeds is an identified risk for the Project. Physical disturbance, import of fill, and additional movement of vehicles and equipment within the Project area may result in the introduction of weed species or the spread of existing weed populations.

Objectives:

- Prevent the introduction and spread of weeds into the Project area;
- Prevent the introduction and spread of weeds into the Millstream Chichester and Karijini National Parks; and
- Vehicles and equipment shall be inspected, cleaned and issued with a weed hygiene certificate prior to site entry.
- To reduce weed species distribution and translocation during the project.

Table 6-5 Management Actions for Weed

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.5.1	All earthmoving and mobile construction equipment shall be inspected for weeds prior to	Contractors	Construction	Form 135 Mobilisation/Demobili-sation Weed Hygiene

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
	mobilisation to site. An application to bring vehicles to site, accompanied by certificate stating that the equipment is free from earth, weeds and weed propagules, is submitted to the Engineers and filed onsite. Additional weed hygiene measures will be developed and implemented where required. The weed hygiene inspection is conducted to ensure that: No dirt or debris is on vehicles and mobile plant; and No visible plant or organic material is attached.			Certificate Weed Washdown Procedure for Rail Construction Projects RTIO-HSE-0144759
6.5.2	Borrow pits will be inspected for weeds prior to disturbance, following rainfall events and periodically during the weed season. Where weeds are found to exist in a quarry/borrow pit, a specific management program will be established to minimise the risk of weed propagules leaving the area and to control weeds.	Contractors / The Engineers Site Environmental Advisor	Construction	Borrow Pit RTIO-HSE-0015216 Specification and Management
6.5.3	Posters and other educational material relating to weed identification and control shall be displayed in the workplace.	The Engineers Site Environmental Advisor	Construction	HSEQMS posters
6.5.4	Signage shall be used where necessary to communicate the weed status of an area and hygiene requirements.	The Engineers Site Environmental Advisor	Construction	
6.5.5	Sites will be inspected to determine weed status; - at the commencement of works - following rainfall events (timing of post rainfall inspections to be determined in consultation with RTIOEP Approvals Specialist) and; - at decommissioning / rehabilitation weed species and locations which have been identified during surveys will be documented within clearing permit information.	The Engineers Site Environmental Advisor	Construction	
6.5.6	Sites which are likely to facilitate germination of weeds will be	The Engineers Site Environmental	Construction	

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
	inspected after rainfall events.	Advisor		
6.5.7	Sites will be inspected at the completion of works to verify effectiveness with weed hygiene controls.	The Engineers Site Environmental Advisor	Construction	
6.5.8	A weedwash down procedure for managing equipment that contains or potentially contains weed propagules shall be established.	The Engineers Site Environmental Advisor	Construction	RTIO-HSE-0036005 Equipment Hygiene Inspection Work Practice
6.5.9	Where weeds are identified, GPS coordinates of the site shall be taken and locations maintained on a register or database.	The Engineers Site Construction Manager /the Engineers Site Environmental Advisor	Construction	
6.5.10	Site specific inductions will educate all construction personnel in the identification of serious environmental weeds such as Ruby Dock. A list and means of identifying Declared Plants and other weeds on site shall be made available by the Engineers Site Environmental Advisor to all personnel involved in or managing clearing activities.	The Engineers Site Environmental Advisor	Construction	Induction
6.5.11	Introduce a weed monitoring and control program, in consultation with the RTIOEP Approvals Specialist for existing weed sites, and in the event new sites are identified or introduced.	The Engineers Site Environmental Advisor	Construction	
6.5.12	Wash down of ground engaging equipment such as; front end loaders, graders, tracked equipment and out riggers is required at the site prior to leaving.	The Engineers Site Construction Manager	Construction	
6.5.13	Wash down facilities will be designed to prevent run off to the surrounding environment.	The Engineers Site Construction Manager	Construction	
6.5.14	Failures in weed hygiene processes will be reported as an incident through the incident reporting process.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0037074 Incident Report Form

6.6 Heritage Management

The Project area lies within several Native Title claims inclusive of the Ngarluma Group, Kuruma Marthudunera, Narlawonga Ngarlawangga, Yindjibarndi, Eastern

Guruma, Yinhawangka and Innawonga Bunjima people. Heritage surveys will be conducted prior to construction activities to identify any heritage sites.

Aboriginal heritage sites include cultural and ceremonial sites, rock engravings, scarred trees, stone arrangements, rock shelters, artefact scatters, water sources, burial sites, shell middens, stone quarries and campsites. Aboriginal heritage sites are protected by the *Aboriginal Heritage Act 1972.*

Several areas of non-indigenous heritage are also contained within the project area, most of these are protected under the Register of National Estate.

Objectives:

- Report and protect heritage sites that may be identified during construction;
- Maintain access to sites for traditional landowners; and
- Avoid unauthorised disturbance to heritage sites, as per the requirements of the *Aboriginal Heritage Act* 1972.

Table 6-6 Management Actions for Heritage

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.6.1	Report any possible newly identified Aboriginal heritage sites to the Engineers Site Environmental Advisor or the Engineers Construction Manager and stop work until further instruction.	All Personnel	Construction	Induction
6.6.2	Ensure that identified Aboriginal heritage sites within 100m of planned construction activities or where deemed of being a high risk of being disturbed are demarcated with signage, flagging and fencing as appropriate.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0121790 Expansion Projects Heritage Site Protection Fencing
6.6.3	Heritage sites that have a potential to be affected by construction activities will be monitored to ensure no unauthorised disturbance.	The Engineers Site Environmental Advisor/ The Engineers Site Construction Manager	Construction	
6.6.4	Any unauthorised disturbance of a heritage site will be reported immediately as an incident.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0037074 Incident Report Form
6.6.5	Requirements of the *Aboriginal Heritage Act 1972* and the management of heritage sites in the project area will be covered in the site inductions	The Engineers Site Environmental Advisor	Construction	Induction
6.6.6	The Engineers Supervisor or approved delegate will be present during all clearing	The Engineers Site Construction Manager	Construction	RTIO-HSE-0043075 Clearing Permit Procedure

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
	activities adjacent to known heritage sites and ensure appropriate demarcation.			

6.7 Borrow Pit Management

Borrow pits are an area where gravel, sand, rock or fill material is removed and used as construction material. They have the potential to affect site drainage, spread weeds, create dust and negatively impact visual amenity if not managed correctly.

Objectives:

- Develop and rehabilitate borrow pits to minimise their impact and comply with RTIO standards; and
- Borrow pits comply with approved borrow pit management plans.

Table 6-7 Management Actions for Borrow Pit Management

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.7.1	Borrow pits must be developed and managed in accordance with HSEQMS procedure – Borrow Pit Specification and Management	The Engineers Site Construction Manager	Construction	RTIO-HSE-0015216 Borrow Pit Specification and Management
6.7.2	A borrow pit management plan will be developed for each borrow pit which follows the Borrow Pit Specification and Management guidelines and includes: • material testing methodology; • extraction methods and dust suppression; • drainage management; • weed management; and • rehabilitation details. Any variations to the borrow pit management plan must approved by the RTIOEP Project Manager.	Contractors	Construction	RTIO-HSE-0015216 Borrow Pit Specification And Management
6.7.3	Any disturbance requires authorisation.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0015216 Borrow Pit Specification and Management
6.74	Inspection of borrow pits will be conducted (as appropriate) to ensure they meet the guidelines. Photographs of borrow pits will be taken during inspections to provide a photographic record of	The Engineers Site Environmental Advisor	Construction	RTIO-HSE-0015216 Borrow Pit Specification and Management

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
	borrow pit development.			
6.7.5	Borrow pit activities (area disturbed and volumes removed) will be reported in the monthly environmental report.	The Engineers Site Environmental Advisor	Construction	RTIO-HSE-0015107 Monthly Environmental Report
6.7.6	Weeds and hygiene at borrow pits will be managed in accordance with Section 6.5 Weed Management.	The Engineers Site Construction Manager/ The Engineers Site Environmental Advisor	Construction	Section 6.5 Weed Management.
6.8.6	No borrow pits will be constructed within National Parks	The Engineers Site Construction Manager/ The Engineers Site Environmental Advisor	Construction	

6.8 Acid Sulphate Soil Management

Acid Sulphate Soils may occur in the Project area, where the soil may contain iron sulphides. When these sulphides are oxidised through disturbance, they have the potential to form ASS. They have the potential to affect vegetation, groundwater and infrastructure assets by acidification.

Objectives:

- Determine location of ASS through geotechnical investigations along the alignment to ascertain the accuracy of the ASS maps; and
- Manage ASS during construction so that the environment and assets are not adversely impacted.

Table 6-8 Management Actions for Acid Sulphate Soil Managment

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.8.1	Areas of ASS to be determined in areas impacted by construction activities.	The Engineers Project Manager	Pre-Construction	RTIO-HSE-0113191 Acid Sulphate Soils Management Plan
6.8.2	Groundwater Bores in ASS areas to be tested for acidity and managed and monitored as required.	The Engineers Project Manager	Pre-Construction	RTIO-HSE-0113191 Acid Sulphate Soils Management Plan
6.8.3	Topsoil management practices to be followed and ASS areas and soil movement/usage to be documented.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0011596 Soil Resource Management

6.8.4	Lime-dosing (if required) to be used per determined in the Management plan based on the level of acidity.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0113191 Acid Sulphate Soils Management Plan

6.9 Dust Management

The Pilbara is a naturally dusty environment and strong winds can increase local dust levels. Dust can be generated during construction activities such as site preparation, materials movement and general equipment movements on unsealed construction areas. Excessive dust levels can have adverse effects on human health and adjacent vegetation and create an uncomfortable and potentially unsafe working environment.

Objectives:

- To prevent or minimise dust levels at construction areas so that a safe working environment is maintained.

Table 6-9 Management Actions for Dust

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.9.1	Vegetation clearing will be kept to a minimal area as practicable.	The Engineers Site Construction Manager	Construction	
6.9.2	Vehicle speeds on haul roads, access roads, and work areas will be restricted	All Personnel	Construction	Induction
6.9.3	Dust suppression such as water trucks will be used on haul roads, access roads, work and.	The Engineers Site Construction Manager	Construction	
6.9.4	Any excessive dust generation shall be reported as per incident reporting procedures	The Engineers Site Construction Manager	Construction	RTIO Incident Report Form RTIO-HSE 0037074
6.9.5	Visual dust assessments at construction sites will be included in the site inspection programme and formal dust monitoring shall be carried out if deemed required post earlier recommendations.	The Engineers Site Environmental Advisor	Construction	

6.10 Surface Water, Groundwater and Drainage Management

Fresh water is a valuable resource in the Pilbara and works associated with the Project traverse through a Public Drinking Water Source Areas (PDWSA). There is a potential for poorly managed construction operations to cause pollution of surface and ground water through spills or discharges of contaminated water and hazardous substances. Poor design and placement of drainage structures and poor design of excavations can result in ponding of water, drainage shadows leading to vegetation loss, scouring, sediment loading and bank erosion.

Groundwater drawdown has the potential to adversely impact the surrounding environment. Pollution of groundwater can occur from chemicals, hydrocarbon materials and wastewater streams and can impact on water catchments and drinking water supplies.

Objectives:

- Ensure efficient, safe and sustainable use and protection of water resources and ecosystems;
- Water use shall be minimised wherever possible;
- Water shall only be taken from licensed sources;
- Compliance with approved licenses, permits and legal provisions;
- Groundwater extraction and reuse volumes to be clearly defined, monitored and adhered to;
- Disruption and pollution to local drainage patterns shall be minimised;
- Where catch drains are located or required, clean surface runoff is to be kept separate from contaminated site runoff where practicable. All discharge points shall be designed to appropriate treatment quality and discharge velocity to match receiving catchment conditions and water corporation requirements.
- No significant erosion or sedimentation; and
- Protection of water quality and catchment area to ensure a safe drinking water supply.

Table 6-10 Management Actions for Surface Water, Ground Water and Drainage

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.10.1	A water inventory shall be maintained and shall include: • Abstraction monitoring from each source including quantity and quality • Discharge monitoring including quantity, quality and receptor • Quantity of water that is recycled and re-used • Aquifer monitoring including bore levels and water quality	The Engineers Site Construction Manager	Construction	DoW / DEC Licence (where applicable) RTIO-HSE-0040672 Legal and Other Requirements Work Practice
6.10.2	Water storages (e.g. Turkeys Nests) should be designed and constructed in accordance with the Environmental Design Criteria DC-N002 for Temporary Facilities.	The Engineers Site Construction Manager	Construction	Environmental Design Criteria for Temporary Facilities DC-N002
Groundwater and Surface Water Licences				
6.10.3	Water shall only be taken from points authorised by RTIOEP and licensed by DoW.	The Engineers Site Construction Manager	Construction	
6.10.4	Copies of all water licenses shall be held on site.	The Engineers Site Environmental Advisor	Construction	
Meters, Calibration and Reporting				

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.10.5	Bores shall be fitted with a sample tap / valve and probe access point in addition to a flow meter that complies with the Rights in Water and Irrigation (Approved Meters) Order 2009.	The Engineers Site Construction Manager	Construction	
6.10.6	Calibration of meters to be carried out prior to usage / installation and certificates for meters are to be retained on site.	The Engineers Site Construction Manager	Construction	
6.10.7	The amount of groundwater extracted shall be reconciled regularly against the licence limit. The frequency of reconciliation must be commensurate with usage and risk of approaching licence limit.	The Engineers Site Construction Manager	Construction	DoW licence (where applicable)
6.10.8	Water usage data must be recorded and forwarded monthly to the Engineers Environmental Advisor. All water data to be recorded in EnviroSys.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0015107 Quarterly Environmental Monthly Report Template RTIO-HSE-0062948 Monthly Environmental Report RTIO-HSE-0105335 Groundwater Tracking Register
Water Monitoring				
6.10.9	Water quality monitoring shall be undertaken as defined by water licenses or as directed by the Engineers/RTIOEP.	The Engineers Site Construction Manager	Construction	DoW licence (where applicable)
6.10.10	Equipment used to monitor groundwater quality shall be calibrated and associated records maintained.	The Engineers Site Construction Manager	Construction	
6.10.11	A monitoring program and schedule shall be established for groundwater, surface water and wastewater where applicable. This shall be approved by RTIO expansion projects.	The Engineers Site Construction Manager	Construction	Where applicable.
6.10.12	The monitoring program shall outline where a National Association of Testing Authorities (NATA) registered laboratory will be used for analysis of water samples.	The Engineers Site Construction Manager	Construction	
6.10.13	Monitoring Data to be entered into EnviroSys.	The Engineers Site Construction Manager	Monthly during Construction	
Stormwater/Surface Water Management				

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.10.14	All potentially contaminated stormwater or surface water (sediment and hydrocarbons) is treated prior to discharge to the environment in accordance with the Design Criteria.	The Engineers Site Construction Manager	Construction	Environmental Design Criteria for Temporary Facilities DC-N002
6.10.15	Sediment basins shall be constructed where discharges from the premises are likely to occur. These basins are to be designed to allow sufficient retention time to reduce suspended solids prior to discharge.	The Engineers Site Construction Manager	Construction	Environmental Design Criteria for Temporary Facilities DC-N002
6.10.16	Sediment basins must be regularly inspected and maintained.	The Engineers Site Construction Manager	Construction	Form 163 Weekly Environmental Inspection Checklist
6.10.17	Where required, clean water diversion systems and contaminated water collection systems shall be established as soon as practicable after commencement of site works.	The Engineers Site Construction Manager	Construction	
Works within Catchments - Public Drinking Water Source Areas (PDWSA)				

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.10.18	All works within public drinking water catchments will be in accordance with the relevant DoW Water Resource Protection Series (WRPS).	The Engineers Site Construction Manager	Construction	WRPS 60 – Tanks for Mobile Fuel Storage in PDWSA WRPS 65 - Toxic and Hazardous Substances – storage and use WRPS 36 – Protecting PDWSA WRPS – 83 Infrastructure corridors near sensitive water resources WRPS 84– Rehabilitation of Disturbed Land in PDWSA WRPS 25 – Land Use Compatibility WRPS 10 – Contaminant Spills: Emergency Response WRPS 28 – Mechanical servicing and workshops Appendix B
6.10.19	Any hydrocarbon or chemical spill incidents and near misses will be reported as per incident reporting procedures and reported to the site environmental advisor	The Engineers Site Construction Manager		RTIO Incident Report Form RTIO-HSE-0037074
6.10.20	All vehicles are required to trailed and tested outside of the PDWSA prior to mobilisation. Comprehensive pre-start checks Including a visual inspection of all hoses should be performed and documented by a suitably qualified person prior to mobilisation. Vehicles that have maintenance issues or oil leaks will not be deployed to PDWSA	The Engineers Site Construction Manager		

6.11 Hydrocarbon Management

Hydrocarbons (fuels, oils, lubricants) can contaminate soil, surface water and groundwater, and have adverse impacts on flora and fauna if released to the environment. During construction, there is a risk of hydrocarbon release due to:

- refuelling mobile plant, vehicles and equipment;
- hydraulic hose failures and other mechanical failure;
- hydrocarbon transport and storage;
- washdown and maintenance of mechanical equipment;
- hydrocarbon waste containment and disposal; or
- incorrect spill response.

Objectives:

- Ensure all hydrocarbons and hydrocarbon contaminated wastes are transported, handled, stored and disposed of correctly in order to prevent contamination of soils, surface water and groundwater.
- Compliance with approved licenses, permits and legal provisions.

Table 6-11 Management Actions for Hydrocarbons

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.11.1	Where dangerous goods licenses are required copies shall be held on site in the dangerous goods register.	The Engineers Site Environmental Advisor	Construction	
6.11.2	Regular inspections and audits of construction sites, workshops, storage areas, maintenance areas, washdown bays will be undertaken for adherence to hydrocarbon management procedures.	The Engineers Site Environmental Advisor / The Engineers Area Supervisors	Construction	
Storage And Bunding				
6.11.3	Hydrocarbon storage, including temporary storage, shall comply with the applicable minimum standard set out in the HSEQMS environmental design principles – permanent facilities (DC-N001) or temporary facilities (DC-N002). All works within public drinking water catchments/PDWSA will be in accordance with the relevant DoW Water Resource Protection Series (WRPS).	The Engineers Project Manager / The Engineers Site Construction Manager	Design/ Construction	Environmental Design Criteria for Permanent Facilities DC-N001 Environmental Design Criteria for Temporary Facilities DC-N002

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.11.4	All storage tanks and associated piping must be above ground and be secondarily contained unless otherwise approved by RTIOEP (Managing Director) and supported by a risk assessment. For exemption refer to 'Underground Hazardous Pipeline or Tank Authorisation Template'(RTIO-HSE-0033842).	The Engineers Project Manager / The Engineers Site Construction Manager	Design/ Construction	Underground Pipelines or Tanks Containing Hazardous Materials Authorisation Template (RTIO-HSE-0033842). Environment Standard E5 Hazardous Materials and Contamination Control.
6.11.5	Regular inspections of construction sites, workshops, storage and maintenance areas, and washdown bays are to be carried out to check for adherence to good hydrocarbon management procedures.	The Engineers Site Construction Manager	Construction	
Refuelling				
6.11.6	Refuelling shall be attended at all times.	All Personnel	Construction	
6.11.7	Refuelling shall take place inside a bunded area in accordance with the design criteria – temporary facilities	The Engineers Site Construction Manager	Construction	DC-N002 Environmental Design Principles – Temporary Facilities
6.11.8	Where mobile refuelling is conducted: • Dry-break hoses should be used; • Refuelling shall not occur within 30m of a watercourse; and • Mobile refuelling trucks must carry spill kits. • Drip trays will be used whilst mobile refuelling is conducted in PDWSA.	The Engineers Site Construction Manager	Construction	WQPN 65 - Toxic and Hazardous Substances – storage and use

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
Temporary Workshops				
6.11.9	Temporary workshops shall be constructed in accordance with DC-N002 Environmental Design Principles – Temporary Facilities and operated such that: • A High Density Polyethylene (HDPE) liner of at least 1mm thickness shall be installed approximately 300mm beneath surface level. • Transportable coalescing separator plates are to be used on discharge water. • The facility shall be adequately protected from stormwater flows. • No washing down is to be conducted within the workshop. • The site shall be subject to soil assessment at the completion of the works and appropriate remediation and validation shall be undertaken to restore the area to its pre-existing condition. • The HDPE liner shall be removed on completion of the works.	The Engineers Project Manager / The Engineers Site Construction Manager	Design / Construction	DC-N002 Environmental Design Principles – Temporary Facilities
Temporary Washdown Areas				
6.11.10	Equipment washdown must occur in designated areas only. Facilities to be in accordance with the Environmental Design Criteria – Temporary Facilities.	The Engineers Site Construction Manager	Construction	DC-N002 Environmental Design Criteria – Temporary
6.11.11	Water use should be minimised at all times, and the use of solvent-based degreasers is not permitted.	The Engineers Site Construction Manager	Construction	
6.11.12	All mechanical equipment must occur on an impervious pad. The area should be graded and bunded to divert stormwater runoff from entering the washdown area and becoming contaminated.	The Engineers Site Construction Manager	Construction	DC-N002 Environmental Design Criteria – Temporary

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.11.13	The hierarchy for management of waters collected from washdown facilities is as follows: • Treatment for sediments and hydrocarbons (via use of transportable coalescing separator plates) then re-use of treated waters (e.g. dust suppression). • Disposal (in order of preference) via: a) Evaporation and sludge removal using appropriately licensed providers as required b) Removal of waters offsite using appropriately licences providers or c) Treatment for sediments and hydrocarbons then discharge in a manner that minimises erosion impacts.	The Engineers Site Construction Manager	Construction	
6.11.14	For treatment of hydrocarbon contaminated waters referred to in clause 6.11.13 above, output water quality must not exceed the more stringent of: • 30mg/L oil/grease content for inland and 10mg/L for coastal; or • The site's specific licence/approval conditions (if applicable). Hydrocarbons recovered must be collected and securely stored for recycling, destruction or disposal at an approved site. Treatment facilities must be designed to cope with the peak wastewater flows. Sampling to be conducted to determine residual hydrocarbons	The Engineers Site Construction Manager	Construction	

Spill Response				
6.11.15	Spill response must be in accordance with the requirements of HSEQMS procedure – spill response	The Engineers Site Construction Manager	Construction	RTIO-HSE-0010867 Spill Response Work Practice
6.11.16	Spill response equipment, including absorbent booms/socks/pillows and matting, should be readily available and used in work environments including mobile plant. Equipment should be clearly marked and housed in a manner that facilitates quick response to spills. This includes mobile plant.	The Engineers Construction Manager	Construction	RTIO-HSE-0010867 Spill Response Work Practice
Waste Oil and Hydrocarbon Product Disposal				
6.11.17	Waste oil, oily rags, oil filters, and any other hydrocarbon related wastes shall be segregated and disposed through an appropriately licensed waste service provider.	The Engineers Site Construction Manager	Construction	
6.11.18	Hydrocarbon products shall not be incinerated on site as a disposal option.	The Engineers Site Construction Manager	Construction	
Waste Oil And Hydrocarbon – Product Disposal				
6.11.19	In the event that hydrocarbon or other contaminated areas are discovered during the course of construction, Rio Tinto Iron Ore Expansion Projects (RTIOEP) shall be notified immediately. Where practicable the impacted area shall be made safe (i.e. physical hazards associated with an excavation shall be controlled) and the areas cordoned off for further inspection and assessment. Impacted sites are to be added to the Contaminated Sites Register where applicable.	The Engineers Site Environmental Advisor	Construction	RTIO Incident Report Form RTIO-HSE-0109617 Contaminated Sites Register Work Practice

6.12 Waste Management

Various types of waste from construction activities will be generated including:

- general office waste – paper, cardboard, printer cartridges, plastic, cans, glass, packaging materials;
- building and demolition waste – packaging material, scrap steel, concrete, pipe/PVC cut-offs, wood scraps, welding rods/crucibles, plastic, tyres;
- putrescible waste – food scraps, kitchen waste; and
- hydrocarbon waste – engine oil, lubricants, solvents, oily rags.

Incorrect waste disposal can result in groundwater, surface water or soil contamination, vegetation or fauna impacts, poor visual amenity and health and safety issues.

Hazardous materials and waste will be covered under Management Procedure 6.14.

Objectives:

- To minimise the impact of waste disposal on the environment and prevent pollution of the air, land and water;
- To reduce, reuse or recycle the amount of waste;
- To collect, segregate, transport and dispose of wastes in an environmentally acceptable manner and in accordance with all relevant legislation;
- Compliance with approved licenses, permits and legal provisions; and
- Volumes of wastes recorded in reports match landfill records.

Table 6-12 Management Actions for Waste

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.12.1	All waste streams, their collection and disposal must be identified. This includes but is not limited to: waste oil, grease, solvent, inhibitors, oily-water mixtures, empty hydrocarbon drums, oil filters, oily rags and absorbent materials.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0010849 RTIO-HSE-0040347
6.12.2	Sufficient bins, recycling and general waste collection areas, will be established to facilitate the management of waste. Putrescibles waste shall be stored in covered receptacles at all times, prior to disposal	The Engineers Site Construction Manager	Construction	RTIO-HSE-0010849 Non Mineral Waste Management Work Practice
6.12.3	Waste bins are colour coded and labelled in accordance with RTIO procedures to facilitate segregation of waste types.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0010849 Non Mineral Waste Management Work Practice
6.12.4	The following materials shall be recycled unless otherwise approved: scrap metal, paper/cardboard, wood pallets, aluminium, glass, office waste (including printer cartridges) and waste oil.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0012439 Controlled Waste Guidelines
6.12.5	Burning of waste is prohibited.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0012439 Controlled Waste Guidelines
6.12.6	Receipts to be maintained of waste type, amount (mass/volume), and disposal method (recycled, re-used, disposal location). Records reported in monthly report.	The Engineers Construction Manager /Site Environmental Advisor	Construction (Monthly)	RTIO-HSE-0015107 Environmental Monthly Report Template
6.12.7	Controlled waste (including hydrocarbons and contaminated soil) will be handled and disposed of in	The Engineers Site Construction Manager	Construction	RTIO-HSE-0012439 Controlled Waste

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
	accordance with the *Environmental Protection (Controlled Waste) Regulations 2004.*			Guidelines
6.12.8	An inspection and maintenance program is established for all waste facilities including landfills, waste storage areas. Housekeeping inspections shall confirm segregation of waste and recyclables and that adequate recycling is occurring where practicable.	The Engineers Site Construction Manager /Site Environmental Advisor	Construction	Form 163 – Weekly Environmental Inspection Checklist
6.12.9	An appropriate landfill site to be identified for use, otherwise RTIO approval is required to establish a landfill site in accordance with RTIO Landfill Management Plan.	The Engineers Site Construction Manager	Design/ Construction	
6.12.10	Where it is intended to use existing Landfills, permission to use the facility must be obtained from the facility owner. This is also to include discussions with operations and their current waste management requirements for such landfills. All existing requirements of the facility must be complied with unless specified otherwise. Management of the facility including access and responsibilities must be negotiated with the facility owner.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0012439 Controlled Waste Guidelines
6.12.11	Cut material may also be generated from construction activities and may be viewed as waste material. This material should be used as fill and for rehabilitation wherever possible and practicable.	The Engineers Site Construction Manager	Construction	
6.12.12	Mineral wastes will be handled and disposed of inline with the RTIO Mineral Waste Management Plan	The Engineers Site Construction Manager	Construction	Mineral Waste Management Plan RTIO-HSE-0040347
6.12.13	Littering shall be avoided at all times and work and office sites will be kept clean and tidy. Non Mineral wastes will be managed in accordance with the non mineral waste management procedure	The Engineers Site Construction Manager		Non Mineral Waste Management Plan RTIO-HSE-0010849
6.12.14	All food wastes to be disposed of accordingly so as to not encourage fauna into camp or cribs areas.	The Engineers Site Construction Manager		

6.13 Sewage and Wastewater Management

Sewage and wastewater will be generated from construction crib rooms and ablutions within the project area. Sewage and wastewater has the potential to infiltrate surface water and groundwater, cause odours and pollute soil if not adequately controlled.

Objectives:

- To minimise the impact of sewage and wastewater disposal on the environment and prevent pollution of the air, land and water; and
- Compliance with approved licenses, permits and legal provisions.

Table 6-13 Management Actions for Sewage and Wastewater

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.13.1	Portable toilets will be designed in accordance with the design criteria – temporary facilities	The Engineers Project Manager / the Engineers Site Construction Manager	Design / Construction	DC-N001 Design Criteria – Permanent Facilities DC-N002 Design Criteria – Temporary Facilities
6.13.2	Contractors will be required to produce a Temporary Toilets Management Plan	Contractor	Construction	
6.13.3	The Engineers shall develop and ablution block management plan to decrease the likelihood of sewerage releases to the environment.	The Engineers Site Environmental Advisor	Construction	To Be Develop

6.14 Hazardous Material Management

Hazardous materials can present a significant risk to the environment if allowed to enter watercourses, leach into the soil or if particles become airborne. The health and safety of personnel can also be put at risk.

Hydrocarbons have been covered under Management Procedure 6.12.

Objectives:

- Ensure the safe and responsible use of all hazardous materials in ways commensurate with their risk to the environment; and
- Compliance with approved licenses, permits and legal provisions.

Table 6-14 Management for Hazardous Materials

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.14.1	Application must be made to the Engineers for approval of any	The Engineers Site Construction	Construction	The Engineers Form 038 –

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
	chemical or hazardous material prior to its mobilisation to site. Chemicals not approved shall be prohibited from site.	Manager		Application to bring Hazardous Material to site
6.14.2	Design of hazardous material storage areas will comply with legislative guidelines, RTIOEP standards and RTIOEP design guidelines.	The Engineers Project Manager / the Engineers Site Construction Manager	Design / Construction	DC-N001 Permanent Facilities DC-N002 Design Criteria – Temporary Facilities
6.14.3	Material Safety Data Sheets (MSDS) shall be kept on site for all hazardous materials in their area of use. A duplicate copy must be provided to emergency response personnel.	The Engineers Site Construction Manager	Construction	
6.14.4	All containers shall be labelled.	The Engineers Site Construction Manager	Construction	
6.14.5	Spill response shall be in accordance with HSEQMS procedure – Spill Response.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0010867 Spill Response Procedure
6.14.6	Hazardous material storage and usage shall be in approved areas for approved purposes only. A hazardous material register shall be maintained of all approved hazardous materials (Chemalert).	The Engineers Site Construction Manager	Construction	Form 038 – Application to bring Hazardous Material to site Chemalert
6.14.7	Appropriate fire detection and protection equipment shall be installed in all fire hazardous work and storage areas.	The Engineers Site Construction Manager	Construction	
6.14.8	All hazardous materials storage areas will be inspected on a weekly basis as part of a monitoring program.	The Engineers Site Environmental Advisor	Construction	Form 163 – Weekly Environmental Inspection Checklist
Emulsion Storage				
6.14.12	Emulsion tanks must be mounted on hardstand compacted earth or concrete pads. Concrete pads must be placed under transfer points for easy clean up of spillage.	The Engineers Site Construction Manager	Construction	
Asbestiform and Fibrous Materials				
6.14.13	Where there is a risk that asbestiform or fibrous materials may be encountered, a process must be established to manage associated risks. This must include: disposal requirements for personal protective equipment (**PPE**) and other	The Engineers Site Construction Manager	Construction	CP-PRO-HSEQ-010 Fibrous Material Management Procedure

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
	disposable equipment used, details of disposal points and collection areas and authorised waste service providers for removal of waste from site. This is also to include the procedures outlined in 6.20.14 and 6.20.15			
6.14.14	Machine air filters that may be affected by asbestiform or fibrous materials shall not be reconditioned for re-use.	The Engineers Site Construction Manager	Construction	CP-PRO-HSEQ-010 Fibrous Material Management Procedure
6.14.15	Material containing asbestos must be labelled with "caution asbestos" in letters not less than 50mm high (where practicable).	The Engineers Site Construction Manager	Construction	CP-PRO-HSEQ-010 Fibrous Material Management Procedure
Contaminated Sites Register				
6.14.16	The Engineers shall maintain a register of known contaminated sites as well as all sites with the potential to become contaminated sites. The register will be provided to Operations so that the Site register is up to date	The Engineers Site Construction Manager	Construction	(RTIO-HSE-0019412) Contaminated Sites Register
6.14.17	The register must be completed using the HSEQMS template – contaminated sites register	The Engineers Site Construction Manager	Construction	(RTIO-HSE-0019412) Contaminated Sites Register

6.15 Noise and Vibration Management

Noise and vibration will result from construction due to the operation of earthmoving and construction equipment and blasting activities. The largest towns within proximity to the Project area are Wickham, Roebourne and Tom Price.

Objectives:

- Ensure a safe working environment by minimising noise and vibration levels; and
- No significant impacts on the natural environment or local community from excessive noise or vibration.

Table 6-15 Management Actions for Noise and Vibration

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.15.1	Where modelling or monitoring results indicate that noise and/or vibration is a significant risk, a management plan shall be established to ensure that regulatory	The Engineers Site Construction Manager	Construction	RTIO-HSE-0133367 Noise Management Work Practice

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
	requirements and community expectations are met. This must address any out of hours work if it is to occur.			
6.15.2	Construction vehicles and equipment shall be fitted with exhaust mufflers to suppress noise	The Engineers Site Construction Manager	Construction	
6.15.3	Equipment with faulty or inefficient mufflers/noise dampers must not be used	The Engineers Site Construction Manager	Construction	
6.15.4	Construction work must be carried out in accordance with the Environmental Protection (Noise) Regulations 1997 (WA) and Section 6 of Australian standard 2436-1981 "Guide To Noise Control on Construction, Maintenance and Demolition Sites".	The Engineers Site Construction Manager	Construction	AS 2436-1981
6.15.5	Screens, enclosures and other noise mitigating devices shall be used where there is a risk of unacceptable noise levels to the community and employees.	The Engineers Site Construction Manager	Construction	
6.15.6	A risk assessment shall be performed to identify potential environmental risks and controls for blasting.	The Engineers Site Construction Manager	Construction	
6.15.7	Blast management plans must include an assessment of proximity to sensitive receptors and must duly consider; • Advanced notification to the community (exploration camp, construction personnel. • Requirements for additional environmental emergency procedures.	The Engineers Site Construction Manager	Construction	

6.16 Fire Management

The use of machinery and equipment, particularly hot work equipment increases the risk of fire in Project areas.

Prevention of fire is a key priority in all RTIO Operations, especially field operations in the highly flammable Spinifex vegetation. The Emergency Preparedness & Response Plan is developed and includes communications with key stakeholders including DEC.

Objective:

- To minimise the potential for construction activities to result in fires; and
- Contain all naturally occurring fires (e.g. lightning strikes).

Table 6-16 Management Actions for Fire

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.16.1	Lighting of fires is strictly prohibited.	All personnel, the Engineers Site Construction Manager	Construction	Induction
6.16.2	Ensure hot work permits are obtained and hot work only conducted in designated areas	Contractor	Construction	CP-PRO-HSEQ-007 Hot Works Procedure for RTIOEP RTIO-HSE-0049936 Hot Work Standard
6.16.3	Regular fire safety inspections and routine maintenance of fire suppression equipment.	The Engineers Site Construction Manager	Construction	
6.16.4	Notify relevant personnel and implement fire suppression in event fire outbreak occurs.	All personnel	Construction	RCEATO-PLN-K-00A Safety Management Plan Addendum
6.16.5	Fire management and cigarette smoking rules will be covered in site inductions	The Engineers Site Environmental Advisor	Construction	Induction
6.16.6	Fire management within National Parks needs to comply with DEC requirements (RTIO-HSE-0144697)	All personnel	Construction	RTIO-HSE-0144697 Fire management in relation to all RCE projects in the National Parks

6.17 Greenhouse Gas Management

Rio Tinto has established an Environmental Standard for greenhouse gas emissions has been to ensure greenhouse gas (GHG) emissions minimisation for Rio Tinto projects and operations.

Objectives:

- all GHG emission data captured for the Project; and
- Minimise GHG emissions where practicable.

Table 6-17 Management Actions for GHGs

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.17.1	Ensure appropriate monitoring systems (flow meters and construction power usage, calibration records) are established for the recording and reporting of GHG emissions.	The Engineers Project Manager / the Engineers Site Construction Manager		

6.17.2	All personnel must be made aware of any project specific GHG reduction or energy efficiency programmes in place and their associated targets.	The Engineers Site Construction Manager	Construction	
6.17.3	GHG records to be recorded and provided monthly the RTIOEP SEA as per the RTIO environmental monthly report template. Both NGER and Non NGER data is required.	The Engineers Site Construction Manager	Construction	Quarterly Environmental Report Template (RTIO-HSE-0015107)
6.17.4	Energy efficiency to be included as part of environmental design review to ensure energy efficient design and identify improvement opportunities.	The Engineers Engineer	Pre-Construction	Design Reviews

6.18 Concrete Batching

Concrete batching will be required for light and mast footings, culverts, mixing cement stabiliser and associated works.

Concrete production produces highly alkaline wash-out water and aggregate with pH values exceeding pH 12 which can have significant environmental and human health effects if not managed appropriately.

Concrete production uses substances that if introduced to the environment will be present at levels above background concentrations with the potential to present, a risk to harm human health and or the environment. The operation of a concrete batching facility falls under the *Contaminated Sites Act 2003* and *Environmental Protection (Controlled Waste) Regulations 2003*. As a result, concrete batching plants are heavily regulated in Western Australia and familiarity with the *Environmental Protection Act 1986: Environmental Protection (Concrete Batching and Cement Product Manufacturing) Regulations 1998* is essential. All batching plants now require a licence per tenure which is the responsibility of the contractor to organise in liaison with RTIOEP.

Objective:

- to control the run off, dust, waste and other emissions associated with the operation of the concrete batch plant.

Table 6-18 Management Actions for Concrete Batching

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.18.1	All concrete batch plants must comply with the Environmental Protection (Concrete Batching and Cement Product Manufacturing) Regulations 1998. Licence is to be	The Engineers Project Manager / the Engineers Site Construction Manager	Tender / Pre-construction	*Environmental Protection (Concrete Batching and Cement Product Manufacturing) Regulations 1998.*

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
	obtained per tenure.			
6.18.2	Concrete batching must be carried out in such a manner that dust escaping from the premises is minimised and controlled	The Engineers Project Manager / the Engineers Site Construction Manager	At all times	*Environmental Protection (Concrete Batching and Cement Product Manufacturing) Regulations 1998.* / CEMP Section 6.9 Dust Management
6.18.3	Any materials spilt during concrete batching shall be cleaned up and reported immediately as per spill response procedures.	The Engineers Project Manager / the Engineers Site Construction Manager	As required	
6.18.4	All inside areas on the premises must be cleaned regularly to prevent accumulation of dust on any surface.	The Engineers Site Construction Manager	As required	
6.18.5	Aggregate and sand must be stored in storage bins or bays designed to minimise airborne dust where practicable. Aggregate and sand stored in a bin or bay is not to exceed the height of the bin or bay (including any windshields fitted to it),	The Engineers Site Construction Manager	At all times	*Environmental Protection (Concrete Batching and Cement Product Manufacturing) Regulations 1998* / CEMP Section 6.14 Hazardous Waste
6.18.6	If cement is kept on premises, it must be stored in bags or in a cement storage silo. The storage silo must comply with sub-regulation 2 hence it must be fitted with an air cleaning system which complies with regulation 7 and a level indicator or a relief valve which complies with regulation 8.	The Engineers Site Construction Manager	Construction	*Environmental Protection (Concrete Batching and Cement Product Manufacturing) Regulations 1998.*
6.18.7	Waste water draining from the site area where agitators, mixers or moulds are loaded, where concrete is batched and water used for washing agitators mixers, moulds or cleaning split materials, or that is likely to be contaminated in any way, must be collected in a slurry pit or settling pond.	The Engineers Site Construction Manager	Construction	*Environmental Protection (Concrete Batching and Cement Product Manufacturing) Regulations 1998*
6.18.8	Any water removed from or	The Engineers Site	At all times	*Environmental*

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
	which might overflow from, a slurry pit drains into a settling pond.	Construction Manager		*Protection (Concrete Batching and Cement Product Manufacturing) Regulations 1998*
6.18.9	Settled material in a slurry pit must not dry out (except for removal purposes) or be higher than 30 cm below top of slurry pit. Settling ponds must be sized for appropriate waste water retention times. All waste water treatment equipment must be maintained, emptied or cleaned as often as necessary to ensure efficient operation.	The Engineers Site Construction Manager	At all times	*Environmental Protection (Concrete Batching and Cement Product Manufacturing) Regulations 1998.*
6.18.10	No water used in concrete batching or cement product manufacturing is discharged from the premises until approval for discharge to the environment has been obtained and it has been: • through a silt trap; or • contained in a settling pond for long enough to allow particulates to settle out; and • through an oil interceptor if the water is likely to contain hydrocarbons.	The Engineers Site Construction Manager	At all times	*Environmental Protection (Concrete Batching and Cement Product Manufacturing) Regulations 1998*
6.18.11	Waste generated during concrete batching (including material removed from slurry pits, settling ponds, silt traps and oil interceptors) must be recycled or disposed of at an appropriately licensed landfill or waste facility.	The Engineers Site Construction Manager	At all times	*Environmental Protection (Concrete Batching and Cement Product Manufacturing) Regulations 1998*
6.18.12	Vehicles carrying concrete or any of its ingredients must not leave the premises until it has been washed free of cement slurry and dust.	The Engineers Site Construction Manager / All Project personnel	At all times	*Environmental Protection (Concrete Batching and Cement Product Manufacturing) Regulations 1998.*
6.18.13	Audit and inspection of the concrete batching facility will be conducted by the Engineer as required to ensure that it complies with Environmental Protection Regulations and Company Standards.	The Engineers Site Construction Manager / the Engineers Site Environmental Advisor	As required	The Engineers Form 125 – Concrete Batching Plant Audit

6.19 Demobilisation, Rehabilitation and Handover

Rehabilitation enables construction sites to be restored as close to their previous condition as possible. Rehabilitation of all non permanent areas is required including borrow pits, temporary access roads and laydown areas. Post-construction environmental management of the areas associated with the Project will be required to establish and record the success of the CEMP measures and capture any lessons for improvement of future management.

Objectives:

- To rehabilitate all non-permanent disturbed areas to a standard that achieves a safe, stable landform which supports self-sustaining native vegetation, is free draining and non polluting and visually compatible with the surrounding landscape;
- Successful re-establishment of rehabilitated areas;
- Demobilise from the site in an environmentally responsible manner;
- Ensure records and knowledge necessary for the management of environmental risk in the operations phase are handed over to the client through an operations and maintenance manual;
- Assess performance for continual improvement purposes; and
- Compliance with Handover procedure and number of items requiring ongoing management at demobilisation.

Table 6-19 Management Actions for Demobilisation, Rehabilitation and Handover

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.19.1	The Engineers shall develop, document and maintain a demobilisation management strategy. The strategy shall be approved by RTIOEP at least 12 weeks prior to demobilisation.	The Engineers Site Construction Manager	Construction	
6.19.2	Loose debris, litter and other non-hazardous materials will be collected, recycled where possible/practicable, removed and disposed of to landfill.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0058421 Rehabilitation Management Plan
6.19.3	Where practicable, materials shall be made available for recycling and/or reuse.	The Engineers Site Construction Manager	Construction	
Rehabilitation of Temporary Infrastructure Areas				
6.19.5	Unless otherwise specified, any temporary facilities shall be removed prior to demobilisation. All structures not handed over will be dismantled down to footing level and removed.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0011608 Rehabilitation Handbook
6.19.6	Above ground services shall be disconnected, removed and appropriately recycled or disposed of.	The Engineers Site Construction Manager	Construction	
6.19.7	All redundant piping and cabling that	The Engineers Site	Construction	

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
	is buried less than 1 metre below ground level will be removed and appropriately recycled or disposed. Redundant piping and cabling that is buried 1m or more below ground level can be left in situ if disconnected.	Construction Manager		
6.19.8	Disposal of waste concrete shall not proceed without consultation with the site environmental representative and RTIO environmental advisor, or without the following information: • Amount of waste concrete placed by Operations so far in the current year. • Additional volume expected to be disposed of as part of demobilisation. • Disposal location. • Method(s) of disposal.	The Engineers Site Construction Manager	Construction	Concrete volumes are recorded for compliance purposes. Up to 500 tonnes of inert material can be placed without registration.
6.19.9	For disposal of waste concrete and inert material (i.e. bitumen roads) a hierarchy of disposal shall be adopted: 1. Re-use on site (i.e. for road base, engineering fill or similar). 2. Re-use off site where suitable options exist nearby. 3. Disposal on-site within suitable areas such as borrow pits and landfill.	The Engineers Site Construction Manager	Construction	
6.19.10	Footings less than $5m^2$ or less than 2m below ground level will be removed and appropriately disposed of. Other footings may be left in place, cracked and covered by at least 1m of clean fill to achieve a natural ground level.	The Engineers Site Construction Manager	Construction	
6.19.11	Drainage structures (e.g. Pipelines, culverts etc) will be removed and salvaged for reuse or if not possible, appropriately disposed, and natural drainage re-instated.	The Engineers Site Construction Manager	Construction	
Hazardous materials				
6.19.12	Hazardous materials will be drained and/or removed from all storage vessels, fixed plant, sumps and septic tanks and removed from site for appropriate treatment through an appropriately licensed contractor prior to demobilisation.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0035253 Contaminated Sites Management Plan
6.19.13	HDPE or similar lining to be removed from secondary containment bunds and disposed of.	The Engineers Site Construction Manager	Construction	

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.19.14	All exposed faces containing fibrous materials must be sealed and grid coordinates included in the handover package.	The Engineers Site Construction Manager	Construction	
6.19.15	All mineral waste stockpiles containing fibrous material must be: •Capped with at least 1m of clean material. • Covered with topsoil if available. • Signposted. • Documented with the grid coordinates included in the handover package.	The Engineers Site Construction Manager	Construction	
Removal of Contaminated Soil				
6.19.16	Contaminated Sites Register to be developed for the project.	The Engineers Site Construction Manager	Construction	
6.19.17	All hydrocarbon impacted soil shall be removed and taken off site. If available, and agreed, this may be deposited at the sites bioremediation area or approved alternative.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0035253 Contaminated Sites Management Plan
6.19.18	Validation soil testing shall be undertaken in areas identified in the contaminated sites register or where large or repeated spills have occurred, to verify all hydrocarbon impacted material has been removed.	The Engineers Site Construction Manager	Construction	
Environmental Management Records				
6.19.19	A handover meeting with environmental personnel shall occur in accordance with the environmental handover procedure. The procedure outlines all the necessary records that must be provided as part of the handover and is collected through a single operations and maintenance manual.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0016368 Environmental Handover Procedure RTIO-HSE-0016367 Environmental Handover Checklist
6.19.20	Measures shall be undertaken to ensure successful rehabilitation of all areas disturbed for construction that are not handed over to Operations in the operations and maintenance manual.	The Engineers Site Construction Manager	Construction	
6.19.21	Rehabilitation re-shaping as a minimum shall ensure: • Natural drainage is re-instated. • The final profile of a constructed drainage line should consist of a flat base and sloped banks. Depth should not exceed 0.5m.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0058421 Rehabilitation Management Plan

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
	• Rock armouring shall be used where necessary to prevent erosion. • Windrows shall be constructed on the contour every 50m on sloped areas to prevent large sheet water flows. • Areas that are very compacted shall be ripped twice to a depth of 0.5m metres along the contour. A three-tyne attachment has been found to be most effective for this purpose. • Drilling sumps are to be backfilled with suitable material.			
6.19.22	For roads and other access tracks, if a treatment such as a salt or acid base dust suppressant has been used then clean fill may be required over the surface so that plant germination is not suppressed.	The Engineers Site Construction Manager	Construction	
6.19.23	Roads shall be ripped along the contours to minimise erosion potential. Respread Topsoil windrows (according to HSEQMS procedure – Soil Management) and stockpiled vegetation onto the roads. Then the surface shall be ripped to mix the topsoil and vegetation.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0011596 Soil Resource Management
6.19.24	Borrow pits that have come to a end must be closed in accordance with HSEQMS procedure – Borrow Pit Specification and Management and any additional DEC requirements.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0015216 Borrow Pit Specification and Management
6.19.25	Rehabilitation shall be progressive wherever possible and where appropriate, rehabilitated areas shall be demarcated to prevent inadvertent or unauthorised access and appropriate signs placed.	The Engineers Site Construction Manager	Construction	
6.19.26	Rehabilitation areas shall be surveyed and mapped for future monitoring purposes and provided to EP for inclusion in the handover package. Records detailing the rehabilitation methodology will be kept and handed over to operations in one operations and maintenance manual.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0010757 Rehabilitation Monitoring Procedure
6.19.27	Rehabilitated areas shall be inspected and approved as satisfactory prior to the demobilisation of the contractor.	The Engineers Site Environmental Advisor / RTIOEP Specialist Environmental Advisor	Construction	RTIO-HSE-0010757 Rehabilitation Monitoring Procedure

Environmentally Sensitive areas				
6.19.28	All sensitive areas (i.e. heritage sites and environmental exclusion zones), to be protected, signposted and barricading fencing maintained. Ensure all possible weed locations are recorded and weed outbreaks controlled.	The Engineers Site Construction Manager	Construction	See Section 5.5 Weed Management

6.20 Fibrous Materials Management

Geological formations underlying the proposed rail infrastructure corridor have the potential to contain fibrous minerals. Asbestos is a term applied to a group of fibrous (asbestiform) silicate minerals belonging to the serpentine and amphibole mineral groups. Rarely these intersected minerals are asbestiform, and if present usually occur in veins or small veinlets with the occurrences generally being small and isolated and therefore often not noticed. Where asbestiform minerals are encountered, airborne asbestos fibres may appear as a minor/ trace contaminant in the dust produced during blasting, crushing and subsequent handling. Disturbance of fibrous materials such as asbestiform has been associated with serious health effects and has the potential to cause harm to the environment if allowed to enter water courses or if particles become airborne.

Objectives:

- Prevent exposure to staff and contractors from harmful levels of naturally occurring fibrous minerals in the workplace using appropriate controls; and
- Prevent contamination to the environment from fibrous minerals.

Table 6-20 Management Actions for Fibrous Materials

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.20.1	Where there is a risk that asbestiform or fibrous materials may be encountered in material excavated from site, for fill or for the placement of infrastructure, the material will be assessed (and documented) for the likelihood of exposing fibrous materials in accordance with RTIO Fibrous Materials Management Plan – (RTIO-PDE-0062061).	The Engineers Site Construction Manager	Construction	CP-PRO-HSEQ-010 Fibrous Material Management Procedure RTIO-PDE-0062061 Fibrous Materials Management Plan
6.20.2	Management controls should include; implementing drilling methods (i.e. wet drilling) and blasting techniques which minimise dust generation and be in accordance with RTIO Fibrous Materials Management Plan – (RTIO-PDE-0062061).	The Engineers Site Construction Manager	Construction	RTIO-PDE-0062061 Fibrous Materials Management Plan. CEMP Section 6.14 Hazardous materials management
6.20.3	Material containing asbestos shall be labelled with "caution asbestos" in	The Engineers Site Construction	Construction	Refer to Section 6.14. Hazardous

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
	letters not less than 50mm high (where practicable).	Manager		materials management
6.20.4	Wastes contaminated with fibrous materials or asbestiform should be disposed of in accordance with the Environmental Protection (controlled waste) Regulations 2004.	The Engineers Site Construction Manager	Construction	Refer to CEMP Section 6.12 Waste Management CEMP Section 6.14 Hazardous Materials management
6.20.5	Any fibrous materials exposed during construction will be managed as per section 6.20 Demobilisation, Rehabilitation and Handover.	The Engineers Site Construction Manager	Construction	Refer to CEMP Section 6.20

6.21 Management within National Parks

The Project intersects the class “A” Reserves of the Millstream Chichester National Park (MCNP) and Karijini National Park (KNP). The MCNP and KNP contain significant conservation, recreation and heritage values. Construction sites within the MCNP and KNP will require additional management to mitigate the potential risks posed by the Project.

Objective:

- Mitigate environmental impacts within MCNP and KNP.

Table 6-21 Management Actions for works within class ‘A’ Reserve

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
Weed Management				
6.21.1	Weeds and hygiene at sites within the MCNP and KNP will be managed in accordance with Section 6.5 Weed Management.	The Engineers Site Construction Manager	Construction	Section 6.5 Weed Management
6.21.3	Strict weed hygiene controls will be introduced for all ground engaging equipment working within or immediately adjacent to the national park. Ground engaging vehicles will be required to; 1) Present weed hygiene certificates prior to mobilisation. 2) Wash down prior to demobilisation from weed infested areas within the MCNP and KNP. 3) Sumps used for the wash down of equipment will be covered with clean fill material to reduce the likelihood of weeds originating within sumps. Sumps will be	The Engineers Site Construction Manager	Construction	

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
	designed to prevent run off.			
6.21.4	Bunting and signage will demarcate topsoil stockpiles that are found to contain weeds within the MCNP and KNP to prevent LV's from entering the area and potentially spreading weed propagules.	The Engineers Site Construction Manager	Construction	
6.21.5	For project work areas; where topsoil that originates within the MCNP and KNP is infested with weeds, it will be respread in the location it originated and controlled through inclusion in future spraying programs.	The Engineers Site Construction Manager	Construction	
6.21.6	Targeted weed control programs that involve the spraying of serious environmental weeds at construction sites or where ground disturbance activities are occurring within the Project area will be developed in consultation with DEC. At the end of construction, targeted weed control programs will be handed over to rail operations with a commitment to continue consultation with the DEC to manage weeds.	The Engineers Site Construction Manager	Construction	
Fire Management				
6.21.7	Fire fighting equipment shall be available in all vehicles and at designated work area points (such as fast attack units during hot works). Dedicated fire water trucks will be available at all times during work within the NPs.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0144697 Fire management in relation to all RCE projects in the National Parks
6.21.8	Fire breaks shall be established around key facilities and be maintained on a regular basis in consultation with DEC.	The Engineers Site Construction Manager	Construction	
6.21.9	Communication protocols will be developed in consultation with the DEC with respect to fire management in MCNP and KNP.	The Engineers Occupational, Health and Safety Manager	Pre Construction	DEC-RTIO Comms Protocol-MCNP RTIO-CR-0032291)
Ground Disturbance and Clearing				
6.21.10	Ground Clearing and disturbance will be managed in accordance with Section 6.2 Ground Disturbance and Clearing.	The Engineers Site Construction Manager	Construction	Refer to Section 6.2 Ground Disturbance and Clearing.
6.21.11	Engineering designs for construction, laydown areas shall be reviewed by the engineering team with the aim of reducing the Project's disturbance footprint. This review involved investigating the potential to	The Engineers Engineer	Pre Construction	

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
	use existing cleared/disturbed areas and where this was not practicable reviewing the design footprint to minimise land clearing and disturbance.			
6.21.12	Where practical the use of borrow material will be avoided through balancing the cut to fill ratio for construction of the rail foundation and embankments. Where there is insufficient cut to meet the fill requirements, imported materials will be required.	The Engineers Engineer	Pre Construction	Borrow Pit Search Areas (BSA)
Rehabilitation				
6.21.13	Rehabilitation of disturbed areas shall be done as soon as practicable to facilitate fauna habitat restoration.	The Engineers Site Construction Manager	Post Construction	
Community Relations				
6.21.14	A complaints register will be developed which shall record complaints received from the community, MCNP and KNP Rangers and other members of the public.	The Engineers Site Environmental Advisor	Construction	

6.22 Workforce Management and Public Access within MCNP and KNP

The Project construction sites and associated facilities are located on Miscellaneous Mining Act leases and Land Administration Leases that are bound by the MCNP and KNP. The location presents a number of potential threats to aspects of the National Park management. Construction worker use of the park may result in unwanted impacts to the environment and community, with associated impacts to RTIO reputation. Appendix A contains a document AutoHaul™ Project Workforce – Millstream Chichester and Karijini National Parks Impacts and Management, that addresses the impacts and associated management measures to mitigate these impacts in more detail.

Objectives:

- To minimise impacts on the Millstream Chichester and Karijini National Park and to park users resulting from RTIO employees and contractors use of the area; and
- To ensure sustainable use of park facilities; including visitor centre, recreation areas, scenic areas, picnic areas, rest areas and toilets.

Table 6-22 Management Actions for Construction Actions for construction workforce management within the national park.

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.22.1	The site induction shall include rules regarding MCNP and KNP, including access, fire, weeds, rubbish, noise, alcohol and travelling	The Engineers Site Environmental Advisor	Induction	
6.22.2	Additional toolbox topics and educational material will be provided during the duration of the project (including but not limited to information contained in Appendix A)	The Engineers Site Environmental Advisor	Construction	Appendix A: AutoHaul™ project workforce – Millstream Chichester and Karijini National Park Impacts and Management
6.22.3	All site visits within MCNP and KNP will be in accordance with the Appendix A	The Engineers Site Construction Manager	Construction	Appendix A: RCE project workforce – Millstream Chichester National Park Impacts and Management
6.22.4	The MCNP and Mungaroona Range Nature Reserve Draft Management Plan published by the DEC shall be made available to the Engineers construction personnel employed on the job.	The Engineers Site Environmental Advisor	Construction	DEC 2011. Millstream-Chichester National Park and Mungaroona Range Nature Reserve Management Plan.
6.22.5	Any complaints or incidents involving RTIOEP employees and contractors in MCNP and KNP shall be recorded in a Complaints Register, and shall be reported via the RTIOEP incident reporting process, and as soon as possible thereafter to the DEC, to determine suitable corrective actions and measures to avoid the recurrence of similar events	The Engineers Site Environmental Advisor	Construction	RTIO-HSE-0063098 Non Conformance, Incident and Action Management Work Procedure
6.22.6	Any damage or unauthorised access to MCNP and KNP, including off – road vehicle use, shall be reported as per incident reporting procedures	The Engineers Site Construction Advisor	Construction	RTIO-HSE-0063098 Non Conformance, Incident and Action Management Work Procedure
6.22.7	Where existing National Parks roads or fences area intersected by construction activities, suitable warnings, and crossings (by-passes) shall be constructed and maintained.	The Engineers Site Construction Manager	Construction	
6.22.8	Any fences and tracks affected by construction work shall be restored on completion of construction, as directed by RTIOEP and as agreed with DEC	The Engineers Site Construction Manager	Construction	

NO.	ACTION	RESPONSIBILITY	TIMING	REFERENCE DOCUMENTS
6.22.9	Regular consultation shall take place between RTIOEP project personnel and DEC Park Management. This shall include updates regarding construction timing and progress, particularly if works interfere with MCNP and KNP activities/ and or public access.	RTIO Environmental Specialist	Construction	
6.22.10	Corrective actions arising from incidents or public complaints shall be entered into the corrective actions register and monitored and verified before being closed out.	The Engineers Site Environmental Advisor	Construction	RTIO-HSE-0063098 Non Conformance, Incident and Action Management Work Procedure
6.22.11	The number of complaints and corrective actions shall be entered in the Monthly Environmental Reports, as per the reporting procedures.	The Engineers Site Construction Manager	Construction	RTIO-HSE-0015107 Monthly Environmental Report

7.0 FORMS AND PROCEDURES REFERENCED IN CEMP

Calibre Documents

RCEATO-PLN-T-003	*Construction Environmental Management Plan (this document)*
RCEATO-PLN-T-004	*Environmental Objectives and Targets*
RCEATO-PLN-T-005	*Environmental Training Needs Analysis Plan*
RCEATO-LST-K-001	*Incident Management Verbal Notification Reporting Matrix*
CP-PLN-HSEQ-004	*Legal and Other Requirements Register for RTIOEP*
RCEATO-PLN-K-001	*Safety Management Plan Addendum (SMP)*
RCEATO-PLN-K-002	*Emergency Response and Preparedness Plan*
CPJ-02-PRO-PM-07	*Document Control Procedure*
CP-PRO-HSEQ-010	*Fibrous Management Plan*
Calibre Form 038	*Application to bring Hazardous Material to Site*
Calibre Form 106	*Daily HSE Checklist*
Calibre Form 125	*Concrete Batch Plant Audit*
Calibre Form 135	*Mobilisation / Demob. Weed Hygiene Certificate*
Calibre Form 162	*Internal Environmental Assessment Report*
Calibre Form 163	*Weekly Environmental Inspection Checklist*
Calibre Form 188	*Clearing Permit - Take 5*
Calibre Form 198	*Borrow Pit Management Plan*

RTIO Documents

RTIO-HSE-0037074	*RTIO Incident Report Form*
RTIO-HSE- 0043075	*Clearing Permit Procedure*
RTIO-HSE-0043306	*Clearing Permit Form*
RTIO-CR-001029	*Approvals Coordination Emergency Procedure*
RTIO-HSE-0070927	*RTIO Heath, Safety, Environment and Quality Policy*
RTIO-HSE-0072772	*Legal and Other Requirements Work Practice*
RTIO-HSE-0130122	*Environment Legal and Other Requirements Procedure*
RTIO-HSE-0015765	*Expansion Projects Assessment Schedule 2012*
RTIO-HSE-0010849	*Non Mineral Waste Management Procedure*
RTIO-HSE-0010867	*Spill Response Procedure*
RTIO-HSE-0013116	*Wildlife Interaction Policy*
RTIO-HSE-0015107	*Environmental Monthly Report*
RTIO-HSE-0015216	*Borrow Pit Specification and Management*
RTIO-HSE-0016368	*Environmental Handover Procedure*
RTIO-HSE-0016367	*Environmental Handover Checklist*
RTIO-HSE-0062207	*Hazard Identification and Risk Management Work Practice*
RTIO-PDE-0062061	*Fibrous Materials Management Plan.*

RTIO-HSE-0058421	*Rehabilitation Management Plan*
RTIO-HSE-0063098	*Non Conformance, Incident and Action Management Work Practice*
RTIO-HSE-0121790	*HSEQ Guidance Note - Heritage Site Fencing*
RTIO-HSE-0015765	*Expansion Projects Environmental Assessment Program and Register*
RTIO-HSE-0072773	*HSEQ Improvement Planning Work Practice*
RTIO-HSE-0015533	*Construction Environmental Management Guidance Note & Specification*
RTIO-HSE-0068753	*Measuring, Monitoring and Data Analysis Work Practice*
RTIO-HSE-0013610	*Environmental Incident Classification Guide*
RTIO-HSE-0072775	*Record and Data Management and Control Work Practice*
RTIO-HSE-0011596	*Soil Resource Management*
RTIO-HSE-0012439	*Controlled waste work practice*
RTIO-HSE-003842	*Authorisation: Underground Pipelines or Tanks Containing Hazardous Materials*
RTIO-HSE-0109617	*Contaminated Sites Register Work Practice*
RTIO-HSE-0011608	*Rehabilitation Handbook*
RTIO-HSE-0040347	*Mineral Waste Management Plan*
DC-N001	*Design Criteria Permanent Facilities*
DC-N002	*Design Criteria Temporary Facilities*
SS-C101	*Earthworks Specification*
RTIO-HSE-0067793	*Qualitative Risk Assessment Matrix*

7.1 Other References

- Aquaterra, 2011, *Pre Feasibility Study – 320MT Expansion Project Rail Upgrade Water Supply*; unpublished report prepared for RTIO.
- Bureau of Meteorology (BOM), 2012, Climate summary, http://www.bom.gov.au.
- Department of Conservation and Land Management (CALM), 2002, *Bioregional summary of the 2002 Biodiversity Audit for Western Australia.*
- Biota Environmental Sciences, 2006, Hope Downs Rail Coriidor (Juna Downs Section) Fauna Survey, Unpublished Report prepared for Rio Tinto.
- Biota Environmental Services, 2008a, A Vegetation and Flora Survey of the Rio Tinto Rail Duplication Project – Cape Lambert to Emu Siding. Unpublished report prepared for Rio Tinto.
- Biota Environmental Services, 2008b, A Vegetation and Flora Survey of the Rio Tinto Rail Duplication Project – Bellbird Siding to Juna Downs, Unpublished report prepared for Rio Tinto.
- Biota Environmental Sciences, 2008c, Pilbara Iron Rail Duplication Fauna Assessment: Emu Siding to Rosella Siding. Unpublished report prepared for Rio Tinto.
- Biota Environmental Services, 2008d, Rio Tinto Rail Duplication Fauna Assessment: Bellbird Siding to Juna Downs. Unpublished report prepared for Rio Tinto.
- Biota Environmental Services, 2010, A Vegetation and Flora Survey of the Rio Tinto Rail Duplication – Emu Siding to Rosella Siding Development Areas. Unpublished report prepared for Rio Tinto.
- Ecologia, 2004, Tunkawanna Cockatoo Rail Duplication Flora & Fauna Assessment. Unpublished report prepared for Rio Tinto.
- GHD, 2009, Turee Syncline Infrastructure Area Flora, Vegetation and Fauna Surveys. Unpublished report prepared for Rio Tinto.
- Ninox Wildlife Consulting, 2009, A Vertebrate Fauna Survey of the Proposed Hope Down 4 Option 6 Infrastructure Corridor, near Newman, Western Australia. Unpublished report prepared for Rio Tinto.

APPENDIX A AUTOHAUL™ PROJECT WORKFORCE NATIONAL PARK IMPACTS AND MANAGEMENT

The proximity of the AutoHaul™ construction sites to the Millstream Chichester National Park (MCNP) and Karijini National Park (KNP) has the potential to have an impact to aspects of the National Park Management. The Department of Environment and Conservation (DEC) is concerned that construction worker use of the park may result in unwanted impacts to the environment and community with associated impact to Rio Tinto Iron Ore's (RTIO's) reputation.

Previous construction projects in the Pilbara have recorded offsite environmental/ community impact and nuisance events from actions of workers outside of project areas. In light of this history, and due to the location of construction sites within MCNP, the DEC have identified potential issues regarding the impacts of construction workers' behaviour and actions within the park.

Potential impacts from construction workers utilising the National Parks or from inappropriate behaviour within the National Parks are identified below. Potential mitigation strategies have also been identified below.

Potential Impacts

Use of National Park Facilities

The presence of the workforce population within the National Parks may increase usage of existing park infrastructure including: parking facilities, recreation areas, visitor centre, scenic areas, picnic areas, rest areas and toilet facilities.

This increase in visitor numbers may place extra pressure on National Parks facilities, personnel and management. In turn, this may contribute to impacts such as accelerated wear and tear on facilities, reduced visitor enjoyment of the park and increased park management costs.

Biodiversity Impacts

Vehicles entering the National Parks may increase the risk of introduction and spread of weed species and increase dust impacts from roads. Additionally, impacts to vegetation, flora and heritage sites may result if vehicles venturing off existing roads and tracks within the Parks. Inappropriate disposal of waste, including inappropriate toileting has the potential to impact on the amenity of the Parks.

Increase in Bushfire Frequency

Visitors may increase bushfire frequency within the National Parks through incorrect use of cooking facilities, incorrect disposal of some items (e.g. glass bottles), disposal of cigarette butts to the ground and vehicle travel across grass or other vegetation areas. Increase in bushfire frequency has the potential to change species composition and density within the National Parks and increases the demand on DEC staff in the National Parks.

Disorderly, Inappropriate and Nuisance Behaviour

Construction workers utilising the National Parks may interact with the members of the public, DEC staff and police officers from time to time. Potential areas of concern from worker behaviour include; drunken behaviour within the National Parks; disregard for National Parks rules and

requirements; disregard for the direction of park staff, DEC officers or police officers; general disorderly or nuisance behaviour; and abusive or violent actions to others or within groups of workers.
Inappropriate behaviour of this nature has significant potential to damage Calibre and RTIO's reputation within the local community and with local and stage authorities.

Disposal of Rubbish

National Parks visitors, including project workers, are likely to generate rubbish during the course of a visit. Incorrect disposal of rubbish may create environmental impacts through attraction of pests / vermin, injury to native fauna, increase in fire risk and the introduction of non-biodegradable materials into the environment. Additionally, incorrect rubbish disposal may create visual and amenity impacts affecting other park users.

Increased Emergency Events

Workers travelling through the National Parks may be involved in accidents or emergencies. Public emergency response within the National Parks is the responsibility of the DEC and local SES and may be limited, slow or non-existent depending on staffing levels. The location of the park and potential lack of services may put Project staff at risk in the event of emergencies or accidents. Additionally, increased visitor numbers may stretch already limited public emergency response capabilities.

Regular inspections will be conducted on light vehicles through daily pre-start checks, which include an assessment of vehicle cleanliness. Excessively muddy vehicles will be cleaned prior to leaving site.

Rubbish Management

Vehicles travelling to the National Parks will be provided with plastic bags to allow for rubbish to be collected and disposed of outside the National Parks. Littering and correct waste management in the National Parks will be covered in the Site Induction.

Liaison

Engagement with National Parks staff, local and state government agencies and local police may improve the effectiveness of the management measures given above. Regular interaction between project supervisory and environmental staff with local and state bodies, in the form of a regular meeting or phone hook-up will allow for early identification of trends or problems and allow for the identification of areas for improvement.

Contact Details

The AutoHaul™ Project is being managed by Calibre on behalf of Rio Tinto Iron Ore.

Any issues or comments regarding the AutoHaul™ Project and MCNP should be directed to the Project Manager.

Mitigation Strategies

Assess Risks

Potential risks associated with construction workers use of the National Parks will be incorporated into overall project risk assessments and into specialist environmental and safety risk assessments, including the project HSEQ Risk Register. Appropriate actions, including the actions described in this document, will be included in the risk registers, accountabilities assigned and actions tracked. The project Construction Environmental Management Plan will also describe the required actions for use of the National Parks.

Education and Information

Information on requirements for use of the National Parks, expectations regarding behaviour and the penalties associated with breaches of requirements will be included into the project induction.

Environmental and community awareness will be enhanced through the delivery of targeted toolbox topics to the workforce and through posters and information sheets displayed on HSE notice boards.

Organised Activities and Supervision

To reduce uncontrolled use of the National Parks, organised activities will be scheduled with the presence of an appropriate Supervisor. These activities may include organised and controlled trips or tours to scenic and recreation locations within the National Parks on RDOs. The consumption of alcohol will not be permitted within the National Parks during scheduled outings. Additionally, environmental and supervisory staff may conduct occasional inspections of popular areas within the National Parks to monitor usage.
All vehicles should pay entrance fees upon entering the National Parks.

Vehicles

Vehicles will not be permitted to leave the work site areas without permission from the EPCM. Travel plans, which may include details on; personnel, vehicles, destination and expected departure, arrival and return times and the use of sign-in/out boards will apply. Travel plans will include a check for the provision of sufficient fuel, water and food for off-site travel.

APPENDIX B CHECKLIST FOR WORKS WITHIN CATCHMENT AREAS

Hazard	Water Corporation Guideline Control Measures	Actioned Y / N	Commitments for working within drinking water catchments
Microbiological contamination			
Human	Toilet facilities to be outside of catchment where possible.	N	To ensure protection of the catchment portable ablution facilities will be provided at construction sites for the workforce. Facilities at construction camps will be managed through the Department of Environment and Conservation (DEC) Works Approval process
	Any toilet facilities within catchments should be fully sealed. Well maintained and secure.	Y	Portable toilets within the catchment will be regularly inspected/changed to ensure no overflows or leaks. Toilet facilities and WWTPs at the construction camp will be designed in accordance with the Design Criteria – Permanent or Temporary facilities Refer to section 6.13 Construction Environmental Management Plan (CEMP) – Sewage and Wastewater Management and RTIO – HSE-0018422 (WWTP Standard Specification). Camp facilities will be managed through DEC Works Approval process.
	No human contact with the water body.	Y	The site induction will cover appropriate behaviour for construction staff in environmentally sensitive areas including PDWSA
	No Camping	Y	A temporary mining camp will be constructed to accommodate all staff. This is a compatible land use within a PDWSA. No other forms of camping will be approved.
	No Fishing	Y	Construction workers will be advised of the requirements for working in PDWSA through the site induction .
	No Domestic animal allowed onsite	Y	No domestic animals are allowed on site or at the temporary accommodation facility.
Hydrocarbon	Hydrocarbon clean up/ Spill kit to be onsite at all times and to be used moderately in the event of a spill (>5L).	Y	- Refer to section 6.11 CEMP – Hydrocarbon Management & IEMS procedure RTIO-HSE-0010867 - **All hydrocarbon spills will be cleaned up in accordance with the CEMP.**
	Site supervisor to contact catchment coordinator immediately for spills	Y	Refer to section 6.11 CEMP – Hydrocarbon management & RTIO-HSE-0010867 & RTIO-HSE- 0010715.

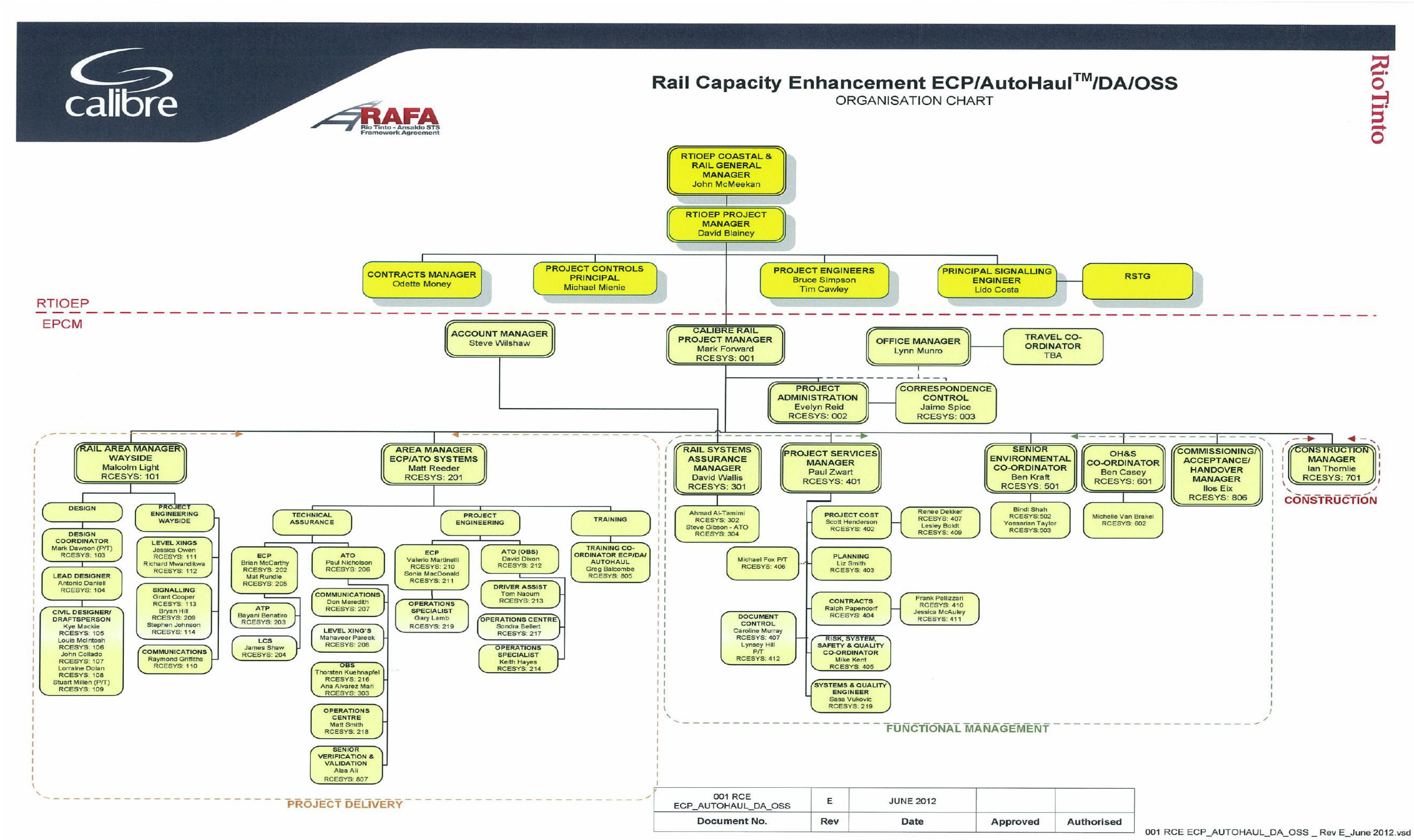
Hazard	Water Corporation Guideline Control Measures	Actioned Y / N	Commitments for working within drinking water catchments
	(>5L) occur or any spills in highly sensitive areas (e.g. streamside)		**Spills >5L will be reported by the site supervisor to the Catchment Coordinator.**
	All machinery to be adequately serviced / maintained and not leaking fluids	Y	-Refer to section 6.11 CEMP Hydrocarbon Management. All vehicles used for the project will be regularly serviced and a maintenance schedule developed. Vehicle pre-starts will occur before each shift and any leaks or faults identified. Additionally any potentially leaking equipment will be identified through regular inspections (6.11.2) of storage areas, vehicle parking areas and wash-down bays. -
	On site maintenance to occur outside of catchment where possible.	N	Temporary maintenance facilities shall be constructed inline with the RTIO Standard Specification – Environmental DC-N002 Environmental Design Principles for Temporary Facilities. Hydrocarbons will be managed in accordance with section 6.11 of the CEMP.
	Any onsite maintenance requires appropriate bunding and removal of chemicals from the catchment immediately after maintenance	Y	Temporary maintenance facilities shall be constructed inline with the RTIO Standard Specification – Environmental DC-N002 Environmental Design Principles for Temporary Facilities. Hydrocarbons will be managed in accordance with section 6.11 of the CEMP.
	Minimal vehicle use on catchments at any one time	N	Construction vehicle movement will be restricted to existing tracks. Water trucks will be used for dust suppression where necessary to prevent excess erosion and potential sediment loading in runoff. Please refer to section 6.9 - Dust Management in the CEMP.
	All refuelling of vehicles /machinery to be done outside of catchment where possible	N	Refer to section 6.11 of CEMP – Hydrocarbon Management. All refuelling facilities will be constructed in line with the RTIO Standard Specification – Environmental DC-N002 Environmental Design Principles for Temporary Facilities. This will ensure necessary controls are in place to minimise risk to the catchment.

Hazard	Water Corporation Guideline Control Measures	Actioned Y / N	Commitments for working within drinking water catchments
	Any refuelling within catchments to occur away from water body and requires appropriate bunding.	Y	Refer to section 6.11 of CEMP – Hydrocarbon Management.
	Bunding onsite for all chemicals as appropriate is required. Refer to MSDS for precautions	Y	Refer to section 6.11 CEMP – Hydrocarbon Management Refer to section 6.14 CEMP – Hazardous Materials Management
	Biodegradable hydraulic fluid to be used where possible.	Y	Consideration will be given to using Biodegradable hydraulic fluid where possible. Risk of hydrocarbon contamination from hydraulic fluid will be managed through section 6.11 of the CEMP hydrocarbon management
	Drip trays to be installed under onsite fixed equipment. Using fuel or hydraulic fluid.	Y	Refer to section 6.11 CEMP – Hydrocarbon Management. All refuelling facilities will be constructed inline with the RTIO Standard Specification – Environmental DC-N002 Environmental Design Principles for Temporary Facilities. This will ensure necessary controls are in place to minimise risk to the catchment.
	Avoid leaving vehicles unattended overnight (vandalism, cut fuel lines)	N	Where possible vehicles will be parked up away from areas accessed by the public.
	Training, if available to be undertaken on how to respond to a hydrocarbon spill	Y	Refer to the section 6.11 CEMP – Hydrocarbon Management and IEMS procedure RTIO-HSE-0010867 – Spill response. All construction employees will be required to take part in spill response training.
	Contaminated soil from minor leaks (<5L) to be removed as soon as possible (minimum daily)	Y	Refer to the section 6.11 CEMP – Hydrocarbon Management and IEMS procedure RTIO-HSE-0010867 – Spill response.
	If any spill does occur, all contaminated soil is to be removed from the catchment and disposed of appropriately in liaison with the Catchment Coordinator and local DoW office	Y	Refer to the section 6.11 CEMP – Hydrocarbon Management and IEMS procedure RTIO-HSE-0010867 – Spill response. (Refer to sections 6.11.17 – 6.11.18). The site environmental advisor shall liaise with the Catchment Coordinator and local DoW office for appropriate disposal of contaminated soils. It is expected that any contaminated soils will be stored and transported to the closest appropriately licensed facility for disposal.

Hazard	Water Corporation Guideline Control Measures	Actioned Y / N	Commitments for working within drinking water catchments
Pesticides and herbicides	Chemicals should not be stored in catchment areas	Y	Refer to Section 6.14 CEMP Hazardous Material Management. With the exception of a few cans of insect spray there will be no need for stored pesticides. There will be stored minor quantities cleaning products. These will have secondary containment.
	Follow guidelines for pesticide applications in drinking water catchments using PSC 88 and State Wide Policy No. 2	Y	Herbicides may be used to control weeds within the catchment area. Please refer to section 6.5 CEMP – Weed Management. The use of pesticides to control weeds within the PDWSA shall be inline with the requirements of PSC88 and State Wide Policy No. 2
	Use only pesticides approved for usage in Catchment Areas according to PSC 88.	Y	Glyphosate (herbicide) is permitted for use according to PSC 88 and shall be used for weed control around construction areas.
Erosion/ Turbidity			
Drainage	There should be no direct run- off of turbid water from site into the drinking water body.	Y	Refer to section 6.10 CEMP – Surface Water, Ground Water and Drainage Management. (6.10.14 – 6.10.17) and IEMS procedure DC-N002). Sediment basins shall be constructed where discharges from the premises are likely to occur. These basins are designed to allow sufficient retention time to reduce suspended solids prior to discharge.
	Construct Culverts in accordance with current design standards and Corporate WI "*Install Culvert*"	N	Camp internal drains and culverts will be designed using; Australian Rainfall and Run off: A guide to flood estimation. This essential design guide, comprised of two volumes, has been published by *Engineers Australia* to provide designers of works and structures subject to floods with the best available information on design flood estimation.
	Ensure all drainage controls are adequate including a buffer around the water body	Y	Refer to section 6.10 CEMP - Surface Water, Ground Water and Drainage Management and IEMS procedure DC-N002.
Rainfall	No work which creates turbidity to be	N	Works which create turbidity are to be contained in sediment basins

Hazard	Water Corporation Guideline Control Measures	Actioned Y / N	Commitments for working within drinking water catchments
Events	conducted during rainfall events or if significant rainfall is forecast for the next 24 hours		(including sump drains or turkeys nests). Contaminated water collection systems shall be established as soon as practicable after the commencement of site works. Refer to section 6.10 CEMP - Surface Water, Ground Water and Drainage Management and IEMS procedure DC-N002).
Tracks	Remain on existing tracks where possible.	Y	Refer to section 6.2 CEMP – Ground disturbing and clearing (6.2.1).
	No new tracks to be created without appropriate drainage	Y	New tracks will be created with appropriate drainage.
	All new tracks to be closed when work is finished	Y	Access tracks established within the PDWSA will be progressively rehabilitated following the closure of the construction camp at the completion of the works.
	Track, road or surface maintenance to be done using water only (potable quality where possible).	Y	Dust suppression on access roads shall be performed using water sourced from an approved Water Corporation water source.
	Dust suppression using water of potable standard, where possible.	Y	Refer to section 6.9 CEMP – Dust Management. Water shall be sourced from the Millstream Link Pipeline or from compatible ground water bores within the vicinity.
Vegetation Buffers	Minimise disturbance to vegetation buffers lining the water body and streams. Rehabilitate disturbed areas immediately after works.	Y	There are no water bodies or associated vegetation buffers within the vicinity of the proposed construction works. All areas that are disturbed and can be rehabilitated will be done so in accordance with section 6.2 of the CEMP
Fire	No camp fires permitted	Y	Refer to section 6.16 CEMP - Fire Management Lighting of fires is not permitted and is included in the mandatory site induction.
Wilderness Degradation	Minimise overall disturbance within catchment wherever possible. Removal of any native assets from the catchment	Y	Refer to section 6.2 CEMP – Management Actions for Ground Disturbance and Clearing and RTIO – HSE-0043075 Clearing Permit Procedure. Where possible existing disturbed areas will be utilised. Appropriate approvals are

APPENDIX C RAIL CAPACITY ENHANCEMENT ECP/AUTOHAUL™/DA/OSS ORGANISATION CHART


calibre
RAFA
Rio Tinto - Ansaldo STS Framework Agreement
Rail Capacity Enhancement ECP/AutoHaul™/DA/OSS
ORGANISATION CHART
RioTinto
RTIOEP COASTAL & RAIL GENERAL MANAGER John McMeekan
RTIOEP PROJECT MANAGER David Blainey
CONTRACTS MANAGER Odette Money
PROJECT CONTROLS PRINCIPAL Michael Mienie
PROJECT ENGINEERS Bruce Simpson Tim Cawley
PRINCIPAL SIGNALLING ENGINEER Lido Costa
RSTG
RTIOEP
EPCM
ACCOUNT MANAGER Steve Wilshaw
CALIBRE RAIL PROJECT MANAGER Mark Forward RCESYS: 001
OFFICE MANAGER Lynn Munro
TRAVEL CO-ORDINATOR TBA
PROJECT ADMINISTRATION Evelyn Reid RCESYS: 002
CORRESPONDENCE CONTROL Jaime Spice RCESYS: 003
RAIL AREA MANAGER WAYSIDE Malcolm Light RCESYS: 101
DESIGN
DESIGN COORDINATOR Mark Dawson (P/T) RCESYS: 103
LEAD DESIGNER Antonio Danieli RCESYS: 104
CIVIL DESIGNER/ DRAFTSPERSON Kye Mackie RCESYS: 105 Louis McIntosh RCESYS: 106 John Collado RCESYS: 107 Lorraine Dolan RCESYS: 108 Stuart Millen (P/T) RCESYS: 109
PROJECT ENGINEERING WAYSIDE
LEVEL XINGS Jessica Owen RCESYS: 111 Richard Mwandikwa RCESYS: 112
SIGNALLING Grant Cooper RCESYS: 113 Bryan Hill RCESYS: 209 Stephen Johnson RCESYS: 114
COMMUNICATIONS Raymond Griffiths RCESYS: 110
AREA MANAGER ECP/ATO SYSTEMS Matt Reeder RCESYS: 201
TECHNICAL ASSURANCE
ECP Brian McCarthy RCESYS: 202 Mat Rundle RCESYS: 205
ATP Bayani Benatiro RCESYS: 203
LCS James Shaw RCESYS: 204
ATO Paul Nicholson RCESYS: 206
COMMUNICATIONS Don Meredith RCESYS: 207
LEVEL XING'S Mahaveer Pareek RCESYS: 208
OBS Thorsten Kuehnapfel RCESYS: 216 Ana Alvarez Mari RCESYS: 303
OPERATIONS CENTRE Matt Smith RCESYS: 218
SENIOR VERIFICATION & VALIDATION Alaa Ali RCESYS: 807
PROJECT ENGINEERING
ECP Valerio Martinelli RCESYS: 210 Sonia MacDonald RCESYS: 211
OPERATIONS SPECIALIST Gary Lamb RCESYS: 219
ATO (OBS) David Dixon RCESYS: 212
DRIVER ASSIST Tom Naoum RCESYS: 213
OPERATIONS CENTRE Sondra Bellert RCESYS: 217
OPERATIONS SPECIALIST Keith Hayes RCESYS: 214
TRAINING
TRAINING CO-ORDINATOR ECP/DA/ AUTOHAUL Greg Balcombe RCESYS: 805
PROJECT DELIVERY
RAIL SYSTEMS ASSURANCE MANAGER David Wallis RCESYS: 301
Ahmad Al-Tamimi RCESYS: 302 Steve Gibson - ATO RCESYS: 304
PROJECT SERVICES MANAGER Paul Zwart RCESYS: 401
PROJECT COST Scott Henderson RCESYS: 402
Renee Dekker RCESYS: 407 Lesley Boldt RCESYS: 409
Michael Fox P/T RCESYS: 406
PLANNING Liz Smith RCESYS: 403
CONTRACTS Ralph Papendorf RCESYS: 404
Frank Pellizzari RCESYS: 410 Jessica McAuley RCESYS: 411
DOCUMENT CONTROL Caroline Murray RCESYS: 407 Lynsey Hill P/T RCESYS: 412
RISK, SYSTEM, SAFETY & QUALITY CO-ORDINATOR Mike Kent RCESYS: 405
SYSTEMS & QUALITY ENGINEER Sasa Vukovic RCESYS: 219
SENIOR ENVIRONMENTAL CO-ORDINATOR Ben Kraft RCESYS: 501
Bindi Shah RCESYS:502 Yossarian Taylor RCESYS:503
OH&S CO-ORDINATOR Ben Casey RCESYS: 601
Michelle Van Brakel RCESYS: 602
COMMISSIONING/ ACCEPTANCE/ HANDOVER MANAGER Ilos Eix RCESYS: 806
FUNCTIONAL MANAGEMENT
CONSTRUCTION MANAGER Ian Thornlie RCESYS: 701
CONSTRUCTION
001 RCE ECP_AUTOHAUL_DA_OSS
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JUNE 2012
Document No.
Rev
Date
Approved
Authorised
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APPENDIX D DEC COMMUNICATIONS PROTOCOL- MCNP (RTIO-CR-0032291)

RioTinto

Iron Ore
152-158 St Georges Terrace
Perth 6000
Western Australia
T + 61 (8) 9327 2000

Communications Protocol with Department of Environment and Conservation on Activities in Millstream-Chichester National Park

The following Communications Protocol (the **Protocol**) with the Department of Environment and Conservation (**DEC**) will be implemented during the proposed rail expansion works within the Millstream Chichester Nation Park (**MCNP**) by the Rio Tinto Iron Ore Group (the **Company**).

Company Contacts:

- Company Representative for works undertaken on site: **Project Manager – Keith Rumsey** ph. (08) 9205 2677 or 0418879015. Keith should be contacted for Project or construction site access information.
- For all environmental queries please contact **Rio Tinto Iron Ore (RTIO) Expansion Projects Specialist Environmental Advisor – Alex Newton** ph. (08) 9205 0490.

Department of Environment and Conservation Contacts:

Primary contacts – Operational incidents (On-site MCNP):

- **Marc McDonald (Senior Ranger)**
  - Email: Marc.McDonald@dec.wa.gov.au
  - Telephone: (08) 9184 5144
- **Stephen Breedveld (Park Ranger)**
  - Email: Stephen.Breedveld@dec.wa.gov.au
  - Telephone: (08) 9184 5144

Secondary contacts – Regional Office, Karratha

- **Duty Officer for fire emergencies**
  - Telephone: (08) 9182 2088
- **Michelle Corbellini (Environmental Impact Assessment for the Pilbara Region – Perth Based)**
  - Email: Michelle.Corbellini@dec.wa.gov.au
  - Telephone: (08) 9334 0260
- **Geoff Passmore (Senior Operations Officer)**
  - Email: Geoffrey.Passmore@dec.wa.gov.au
  - Telephone (08) 9182 2022
- **Cath Rummery (Nature Conservation Leader)**
  - Email: Cath.Rummery@dec.wa.gov.au
  - Telephone: (08) 9182 2005
- **Allisdair MacDonald (Regional Manager)**
  - Email: Allisdair.MacDonald@dec.wa.gov.au

Rio Tinto

- Telephone: (08) 9182 2000

- Prior to works commencing in the MCNP, a monthly update including the proposed works and timing will be forwarded to the Park Rangers.

- At least five working days prior to accessing the MCNP, the Company will provide the DEC Regional Manager with an itinerary and program of the locations of operations on the licence area and informing the Regional Manager at least five days in advance of any changes to the itinerary. All activities and movements shall comply with reasonable access and travel requirements of the Regional Manager regarding seasonal/ground conditions.

- Consistent with the (currently draft) Weed Action Plans for the rail network, an Annual Weed Management Report will be prepared at the conclusion of each 12 month period, a copy of which will be forwarded to the DEC. The DEC will be consulted on the production of the Company Pilbara Weed Strategy and associated Weed Action Plans.

- During the proposed works within the MCNP, an area engineer will prepare a short report on works to date (with appropriate photos) for the endorsement of the Project Engineer which will then be forwarded onto the Park Rangers on a monthly basis.

- The Company will record clearing activities against each lease within the MCNP. These details will be provided to the DEC via required reporting against Ministerial Statement 880 and associated residual impact package and the RTIO Annual Environmental Report for Railways Division.

- The Company will report to the above primary contacts (in the first instance) incidents which involve the following:
  - Fire Emergencies;
  - Chemical/hydrocarbon spills which may impact on the MCNP outside the area in which the Company is working;
  - Chemical/hydrocarbon spills into watercourses;
  - Accidents occurring within the MCNP which will include fauna injuries or mortalities;
  - Negative interactions with MCNP users; and
  - Public or major road closures as a result of the project.

  These incidents will be reported in with the monthly report to the DEC (Pilbara) and will include details on remedial actions taken. Any significant incidents, such as uncontrolled fires, will be reported to the DEC Duty Officer within one hour. All other significant incidents will be reported to the Primary contacts as soon as possible.

- Should the Park Rangers or other relevant DEC staff want to inspect or view the works on site whilst under construction or to conduct works associated with the Rail offset package, for safety reasons, they should contact the Project Engineer on (08) 9205 2677 and they will arrange for safe passage around the construction site.

- A close out inspection will be undertaken by the Company within 21 days of the completion of the works. The DEC staff (including the MCNP Rangers) will be notified of the date of the inspection and given the opportunity to attend. As part of the close out inspection, a reconciliation of the amount of clearing undertaken (in hectares) will be undertaken for the activities undertaken within the MCNP.

APPENDIX 4

RIO TINTO IRON ORE

STANDARD SPECIFICATION - ENVIRONMENTAL
DC-N002 ENVIRONMENTAL DESIGN PRINCIPLES
TEMPORARY FACILITIES

DESIGN PRINCIPLES			
TITLE			NUMBER **DC-N002**
1	Updated by RTIOEP With assistance from KBR.	LP	
0	Issued for use		
Rev	**Description**	**Manager Initials**	**Date**

CONTENTS

SECTION 1 - Scope and Background Information

This document sets out the minimum standards and expectations of Rio Tinto Iron Ore (RTIO) in relation to environmental designs for temporary facilities.

The objective of all environmental design is to achieve design outcomes that comply with company and regulatory requirements and minimise potential environmental impacts.

1.1 Purpose and Application

The purpose of this document is to provide designers of Temporary Facilities with practical advice and guidance in relation to managing risks and meeting environmental requirements. This document shall be applied to the design of any new works and to the upgrade of any existing facilities at RTIO sites.

In preparing this document a number of sources were reviewed including:

- Western Australian environmental legislation and regulations
- Relevant Australian Standards
- Rio Tinto Environment Standards
- The Rio Tinto Iron Environmental Management Systems (IEMS)
- Environmental incident data (supplied by RTIO Expansion Projects)

These information sources were current at the time this document was prepared.

This document sits within the suite of Engineering Design Principles documents (refer Figure 1) and should be used in conjunction with the requisite Engineering Standard Specifications.

This document has been developed with a broad outlook to cover the most common temporary facilities across RTIO sites. Where this document does not address specific facilities or requirements then guidance and approval must be sought from RTIO Environmental Personnel.

For design principles for permanent facilities, refer to Standard Specification DC-N001, Environmental Design Principles – Permanent Facilities.

The use of this document does not exempt designers and engineers from the need to ensure that all relevant regulatory and legal requirements are met prior to construction.

1.2 Document Revision

This document is intended to represent good practice in the area of environmental design and will be periodically revised to reflect changes in such practice. All practitioners with an interest or specialist skill in environmental design are welcome to provide comments and suggest improvements to the document. Comments should be passed to the RTIO Environmental Department.

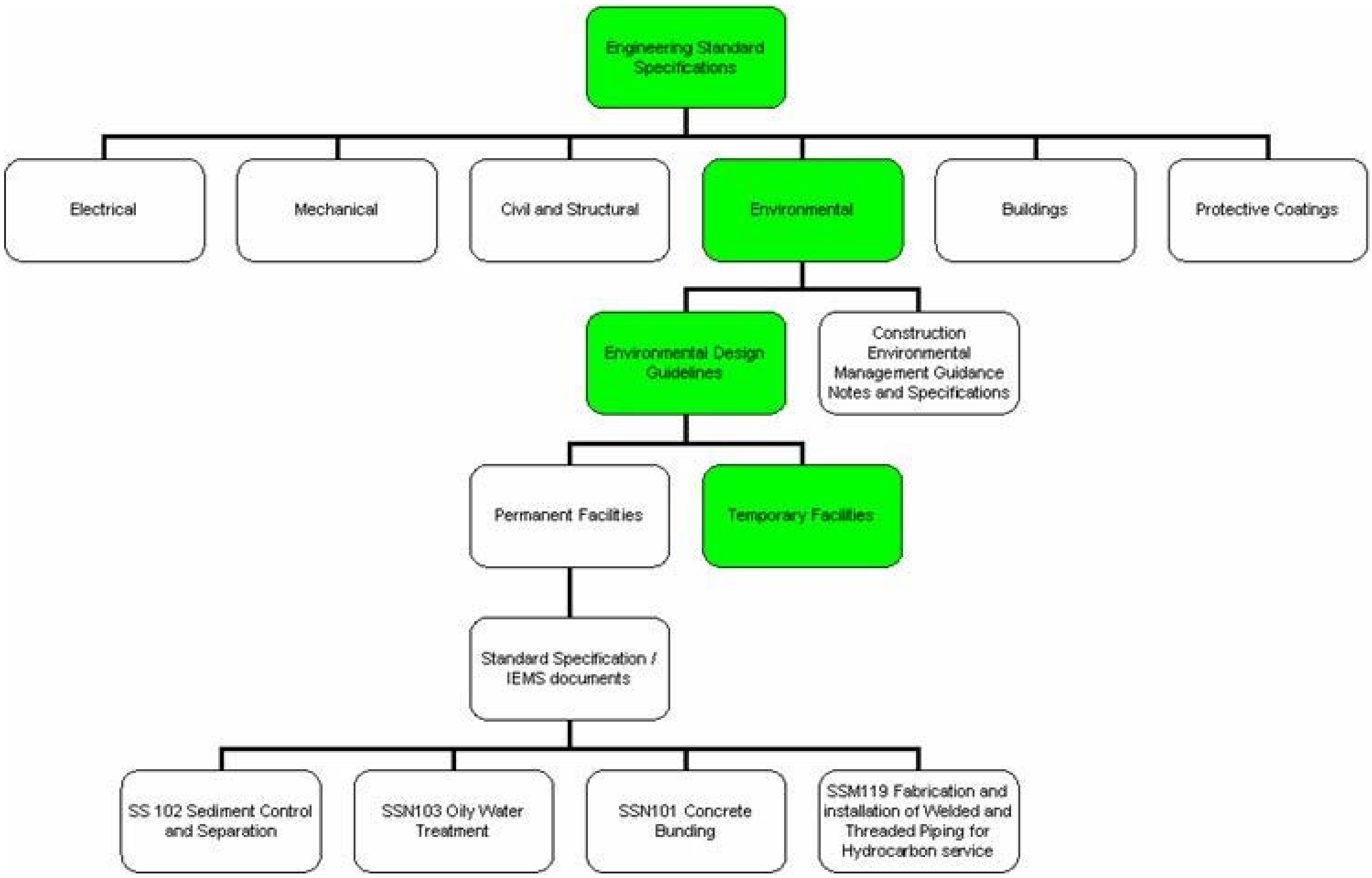

Figure 1. Engineering Design Principles Documentation Summary

1.3 Consideration of the Environment in the Design Process

When designing new facilities or modifying existing facilities, designers tend to focus on achieving the desired performance specification given a certain timeframe and budget. Under such circumstances the effect that the facility may have on the surrounding environment when brought into operation may be overlooked.

Opportunities do exist to reduce the potential environmental risk of a facility throughout its entire life cycle. However, it is more cost effective to implement environmental design changes in the initial concept or detailed design phases, as shown in Figure 2. Consideration of the environment during the design process can avoid the need to make costly modifications to the facility once built and can significantly reduce the likelihood of an environmental incident occurring.

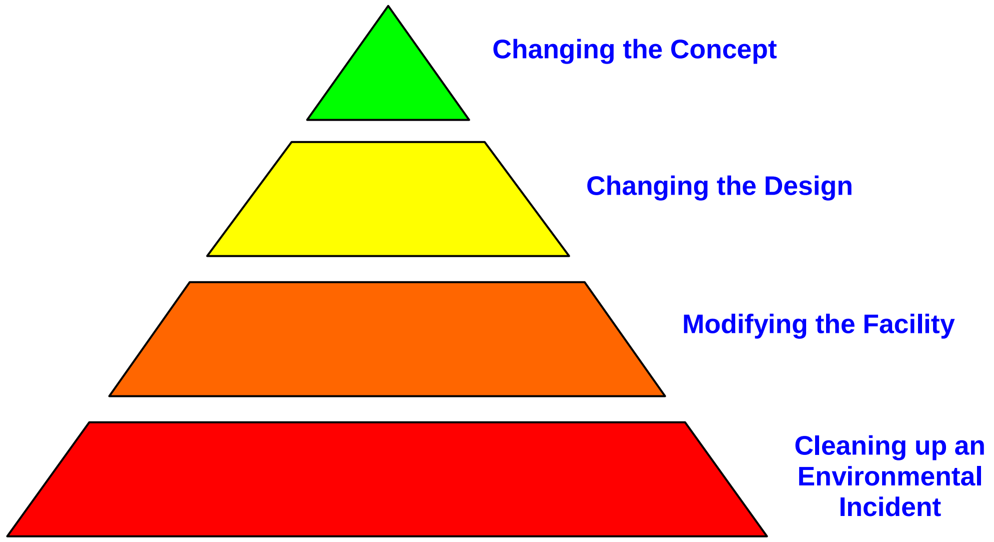

Figure 2: Pyramid Showing Cost of Implementing Environmental Design Changes

Environmental aspects can be taken into account during the design phase by conducting an environmental risk assessment of the proposed facility. Designers must identify and assess environmental risks when preparing engineering designs. This ensures that environmental risks associated with operation of the facility are identified, evaluated and controlled in the design phase. The steps involved in conducting an environmental risk assessment are briefly outlined below.

Where possible, the process of identifying, evaluating and controlling environmental risks should occur within existing processes already established on the project (e.g. Hazard Studies Project Risk Workshops and Design Reviews). This avoids the need for a workshop focusing solely on environment. However, under certain circumstances the Project Manager may decide to conduct a separate environmental risk workshop (e.g. when the facility has been identified as a potentially high risk).

1.3.1 Environmental Hazard Identification

The first step in the environmental design process is to identify potential environmental hazards associated with the proposed design. As a minimum, the nine environmental aspects listed below should be considered in turn and the potential hazards associated with each identified:

- Air Quality
- Acid Rock Drainage
- Greenhouse Gas Emission
- Hazardous Material & Contamination Control
- Noise and Vibration Control
- Non-Mineral Waste Management
- Mineral Waste Management
- Land-use Stewardship
- Water Use and Quality Control

The above environmental aspects are considered where relevant throughout this standard specification.

1.3.2 Assessing Environmental Risk

Environmental risk is assessed by taking into consideration both the probability (likelihood) of an incident occurring and the consequence of such an incident.

The methodology used to conduct an environmental risk assessment must comply with Rio Tinto's risk assessment process.

1.3.3 Adopting Environmental Control Measures

Control measures shall be put in place to manage and mitigate any environmental risks that are identified in the design phase. A general hierarchy of control is shown in Figure 3, with preferred controls shown at the top of the hierarchy.

Note that while control of the hazard by administrative means has been included in the general hierarchy of control, it is usually less effective than the other control measures. Hence it should not be relied on in the design phase to reduce the initial risk to an acceptable level.

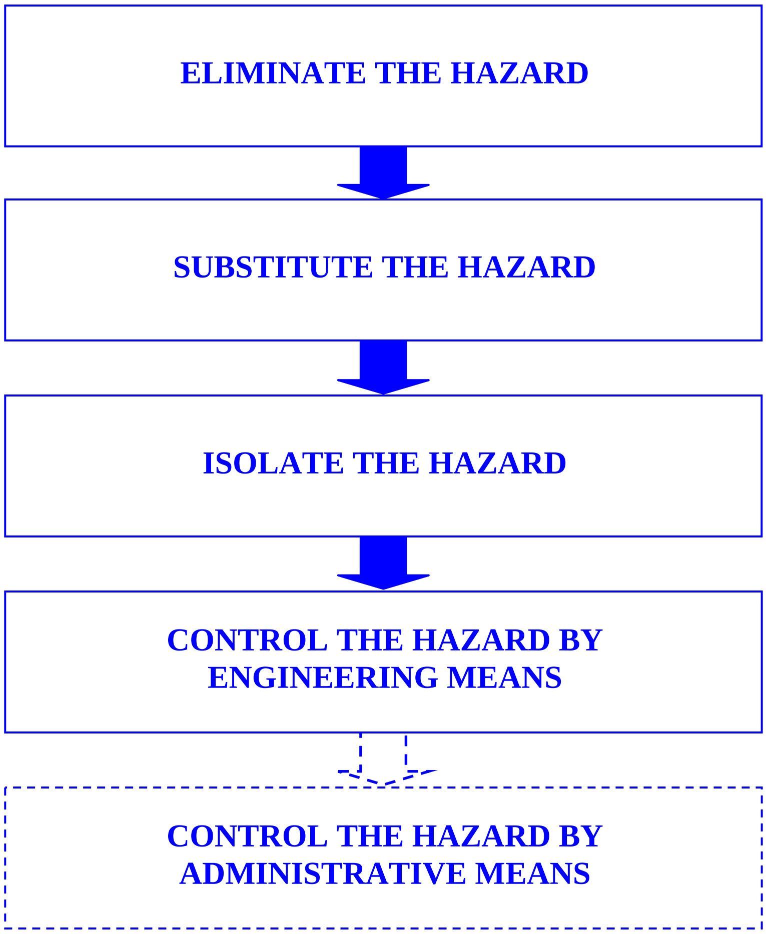

Figure 3: General Hierarchy of Control

1.3.4 Assessing the Residual Risk and Detailing Further Actions Required

Once control measures have been incorporated into the design the residual risk should be assessed to determine if the controls are appropriate. If the residual risk is still unacceptable additional control measures may have to be put in place. It may be appropriate under these circumstances to employ administrative controls, provided all other possibilities have been exhausted. Any administrative controls must be documented in the facility's Operating Manual and brought to the attention of RTIO's Environmental Department.

1.3.5 Developing an Environmental Risk Register

The risk assessment should be used to develop a Construction Environmental Risk Register (CERR) for construction and commissioning of the new facility. At the completion of construction, operating risks should be transferred to an Operational Environmental Risk Register (OERR). All CERR's and OERR's must be submitted to the RTIO Environmental Department for approval.

1.4 Compliance with Statutory Requirements

1.4.1 Government Acts and Regulations

All aspects of the design shall comply with Government Acts and Regulations having jurisdiction over them. This shall include, but not be limited to, those listed in the “Regulations & Standards” Section of each Chapter, and those listed in Section 2 “General Design Principles for All Works” of this standard specification.

1.4.2 Environmental Licence Conditions

All aspects of the design shall comply with the applicable Department of Environment and Conservation (DEC) Licence, and/or Works Approval(s) issued by the DEC.

1.5 Compliance with Company Environmental Requirements

Where Rio Tinto requirements are more exacting than the minimum requirements of the Statutory Regulations and Australian Standards, the former shall prevail in the following order:

1) Rio Tinto Environmental Standards and Associated Guidelines
2) Environmental Design Criteria – Temporary Facilities (this document)
3) IEMS Environmental Management System Documents
4) Pilbara Iron Standard Specifications

Where a conflict occurs between company requirements the most stringent requirement will apply.

The latest revision of the applicable document shall be used in all cases, except for where a new revision to a document in use is issued during the course of the work. In this case, the matter shall be referred to the appropriate RTIO Environmental Department Representative for a determination as to which document version shall be used.

1.5.1 Rio Tinto Environment Standards and Associated Guidelines

The design of all Temporary Facilities must comply with the requirements of Rio Tinto’s Environment Standards and associated Guidelines. The relevant Rio Tinto Environment Standard is listed in the “Applicable Documents” Section of each Chapter.

1.5.2 Environmental Design Criteria – Temporary Facilities (this document)

This document sets out the minimum standards and expectations of RTIO in relation to environmental design. Each chapter addresses different temporary facilities and provides guidance to a Designer. Both design requirements and items to consider and avoid are listed. An explanation of each is given below:

‘Must Have’ requirements are items that the Designer must incorporate into their design or actions that the Designer must take as part of the design process. In some circumstances there may be valid reasons why the requirements cannot be met by the Designer. Under such circumstances any deviation from the requirements must be approved by RTIO.

'Consider' items are items that the Designer should consider, but does not necessarily have to incorporate into the design. They commonly represent good practice in that area.

'Avoid' items are aspects that the Designer should avoid incorporating into the design.

1.5.3 IEMS Documents

Compliance with the requirements of IEMS is mandatory for all EPCM and other contractors designing facilities for RTIO. IEMS documents can be accessed through the RTIO Intranet.

1.6 Compliance with Australian Standards

The Australian Standards listed in the "Regulations & Standards" Section of each Chapter and those listed in Section 2 "General Design Principles for All Works" of this standard specification must be followed where relevant to environmental design.

1.7 Demonstration of Current Good Practice

All Temporary Facilities are expected to demonstrate current good practice in relation to environmental design. Photographs showing examples of current good practice have been included throughout this standard specification for guidance.

1.8 Documents Referenced in this Section

1.8.1 Rio Tinto Documents

Title
Rio Tinto Environment Standards

1.8.2 IEMS Documents

Reference	Title
RTIO-HSE-0015533	Construction Environmental Management Guidance Notes and Specification

1.8.3 PI Standard Specifications

Reference	Title
DC-N001	Environmental Design Principles - Permanent Facilities

1.9 Definitions

Permanent	Facilities constructed for an operating life that exceeds the construction phase of a project and are intended to remain in place for 2 or more years.
Temporary*	Facilities constructed for an operating life of less than or equal to 2 years, or of an inherently temporary nature such as a construction camp (approval from an appropriate person must be given in this case). **If the life of a Temporary facility exceeds, or is expected to exceed, 2 years, the situation should be reviewed and consideration be given to undertaking an upgrade to a Permanent facility.*
RTIO	Rio Tinto Iron Ore
RTIOEP	Rio Tinto Iron Ore Expansion Projects
IEMS	Iron Environmental Management System
Combustible Liquid	*Any liquid other than a flammable liquid that has a flash point and has a fire point that is less than its boiling point. AS 1940-2004 divides combustible liquids are into two classes:* ***Class C1 - A combustible Liquid that has a flash point of 150°C or less** (for example Diesel fuel)* ***Class C2 – A combustible liquid that has a flash point of greater than 150°C** (for example lubricating oil)*
Flammable Liquid	*Liquids or mixtures of liquids, or liquids containing solids in solution or suspension* (eg paints, varnishes, lacquers etc, but not including substances otherwise classified in account of their dangerous characteristics) *which give off a flammable vapour at temperatures of not more than 60.5°C, closed cup test, or not more that 65.6°C, open cup test, normally referred to as the flash point (eg unleaded petrol).*

SECTION 2 - General Design Principles for all Works

2.1 Applicable Documents

The latest revision of the applicable document shall be used in all cases, except for where a new revision to a document in use is issued during the course of the work. In this case, the matter shall be referred to the appropriate RTIO Environment Personnel for a determination as to which document version shall be used.

2.1.1 Design Principles for Permanent Facilities

This document forms DC-N002 Environmental Design Principles for Temporary Facilities. For design principles for permanent facilities, refer to DC-N001 Design Principles for Permanent Facilities.

2.1.2 Standard Specifications

Standard Specifications are applicable for all work unless approved Project Specifications have been developed. Standard Specifications are available from the RTIO intranet or by contacting the RTIO Company Representative.

The following Standard Specifications shall be followed where relevant to environmental design.

No.	Title
SS-N101	Concrete Bunding
SS-N102	Sediment Control and Separation
SS-N103	Oily Water Treatment
SS-M119	Fabrication and Installation of Welded and Threaded Piping for Hydrocarbon Service

2.1.3 Rio Tinto Environmental Standards

Section	Title
E2	Air Quality Control
E4	Greenhouse Gas Emissions
E5	Hazardous Materials and Contamination Control
E7	Non-Mineral Waste Management
E9	Land Use Stewardship
E10	Water Use and Quality Control

2.2 Australian Standards

No.	Title
AS1692 - 2006	Steel tanks for flammable and combustible liquids
AS1940 - 2004	The storage and handling of flammable and combustible liquids
AS3780 - 1994	The storage and handling of corrosive substances
AS4326 - 1995	The storage and handling of oxidizing agents
AS3833 - 1998	The storage and handling of mixed classes of dangerous goods, in packages and intermediate bulk containers

2.3 Other References

- Dangerous Goods related documentation published by the Western Australian Department of Consumer and Employment Protection (DOCEP) is available from website:
- **http://www.docep.wa.gov.au/resourcessafety/Sections/Dangerous_Goods/Guidance_Material/Dangerous_Goods_Stor.html**
- General Requirements for Licensed Premises (GN S303-Rev8)
- General Requirements for Premises Exempt from Licensing (GN S305-Rev3)
- Licensing and Exemptions (GN S301-Rev12)
- List of Companies and Consultants Approved to Examine and Endorse Dangerous Goods Storage Proposals
- Placarding of Stores and Premises (GN S302-Rev10)
- Preparation of an Emergency Plan and Manifests (GN S310-Rev 5)
- Removal and Disposal of Underground Petroleum Storage Tanks (GN S321-Rev7)
- Licence Conditions For The Storage And Handling of Solid Ammonium Nitrate (GN S313-Rev5)
- Storage of Dangerous Goods in Transit (Conditions for Exemption from Licensing) (GN S314-Rev7)
- Design, Operating and Maintenance Guidelines for Marine Refuelling Facilities (GN S317-Rev3)

Guidance notes (below) published by the Western Australian Department of Environment & Conservation (DEC) available via the website:

http://portal.environment.wa.gov.au/portal/page?_pageid=193,68733&_dad=portal&_schema=PORTAL

- Guidelines for mining and mineral processing: Above-ground fuel and chemical storage. Publish Date: 01-JAN-2000. Water Quality Protection Guidelines No. 10
- Guidelines for mining and mineral processing: Liners for waste containment. Publish Date: 01-JAN-2000. Water Quality Protection Guidelines No. 3

- Guidelines for mining and mineral processing: Mechanical servicing and workshop facilities. Publish Date: 01-JAN-2000 04:08 PM Water Quality Protection Guidelines No. 7
- Guidelines for Water Meter Installation. Department of Water, Western Australia, Publish Date: 2007.
- Guidelines for mining and mineral processing: Minesite stormwater. Publish Date: 01-JAN-2000. Water Quality Protection Guidelines No. 6
- Liners for containing pollutants, using synthetic membranes. Water Quality Protection Note WQ26.   Publish Date: 01-JUL-2005.
- Soil liners to contain low-hazard waste. Publish Date: 01-MAR-2003.
- Stormwater management at industrial sites. Publish Date: 01-NOV-2002
- Temporary above ground chemical storage in Public Drinking Water Source Areas. Publish Date: 01-NOV-2000
- Temporary skid mounted fuel transfer and storage in Public Drinking Water Source Areas Publish Date: 01-NOV-1998
- Washdown of Mechanical Equipment. Publish Date: 01-JUL-1998.
- Water Quality Protection Note, “Extractive Industries within Public Drinking Water Source Areas”, July 2000, Department of Environment WA

These information sources were current at the time this document was prepared.

2.4 Statutory Requirements

All aspects of design and construction for project works shall comply with Government Acts and Regulations having jurisdiction over them. These include but are not limited to the following:

- Environmental Protection Act 1986.
- Environmental Protection Amendment Act 2003.
- Environmental Protection Regulations 1987
- Environmental Protection (Concrete Batching and Cement Product Manufacturing) Regulations 1998
- Environmental Protection (Clearing of Native Vegetation) Regulations 2004
- Environmental Protection (Unauthorised Discharges) Regulations 2004
- Environmental Protection (Controlled Waste) Regulations 2001
- Mines Safety and Inspection Act 1994.
- Mines Safety and Inspection Regulations 1995.
- Occupational Safety and Health Act 1984.
- Occupational Safety and Health Regulations 1996.
- Explosives and Dangerous Goods Act 1961.
- Explosives And Dangerous Goods (Dangerous Goods Handling And Storage) Regulations 1992
- Rights in Water and Irrigation Act 1914
- Country Area Water Supply Act 1947
- Contaminated Sites Act 2003

SECTION 3 - Storage of Hazardous Materials

3.1 Introduction

The intent of this Section is to ensure that all Temporary Facilities are designed to store, handle and dispose of hazardous materials in a manner which complies with RTIO and regulatory requirements and takes into account their potential risk to the environment.

A hazardous material is broadly defined as a substance or article (solid, liquid or gas) that poses a specific threat to human health and/or the environment. Typically, hazardous materials may be poisonous, toxic, infectious, carcinogenic, corrosive, oxidizing, water reactive, combustible, flammable, explosive or radioactive. If not stored, handled or disposed of correctly such materials can adversely affect the environment and people's health and safety.

This Section applies to the design of all Temporary Facilities intended for the transportation, handling, and storage of hazardous materials.

The storage of Explosives (e.g. ANFO) is covered in Section 7 of this standard.

3.2 Regulations and Standards

The storage of dangerous goods is regulated by the *Explosives and Dangerous Goods Act* 1961 (WA) and the *Explosives and Dangerous Goods (Dangerous Goods Handling and Storage) Regulations* 1992.

In addition, all discharges from RTIO sites are regulated under the *Environmental Protection Act* (1986) making the accidental or deliberate release of pollutants (such as hydrocarbons and chemicals) in into the natural environment an offence.

The regulations, guidelines and Australian Standards covered in this section provide prescriptive guidance on aspects such as bund design, tank design and fire protection.

3.1.1 Statutory Requirements

Environmental Protection Act 1986 (and associated Regulations)

Mines Safety and Inspection Act 1994 (and associated Regulations)

Occupational Safety and Health Act 1984 (and associated Regulations)

Dangerous Goods Safety Act 2004 (and associated Regulations)

Explosives and Dangerous Goods Act 1961 (and associated Regulations)

3.1.2 Rio Tinto Environment Standards and Guidelines

Reference	Title
Environment Standard (E5)	Hazardous Materials and Contamination Control

3.1.3 Pilbara Iron Standard Specifications

Reference	Title
AM-M101	Diesel and Lubricant Contamination Control Standard
SS-N101	Concrete Bunding for Stored Hydrocarbons
SS-M119	Fabrication and Installation of Welded and Threaded Piping for Hydrocarbon Service

3.1.4 Australian and International Standards

Reference	Title
AS 1692	Steel Tanks for Flammable and Combustible Liquids
AS 1940	The Storage and Handling of Flammable and Combustible Liquids
AS 2430	Classification of Hazardous Areas
AS 3780	The Storage and Handling of Corrosive Substances
AS 4326	The Storage and Handling of Oxidizing Agents
AS/NZS 3833	The Storage and Handling of Mixed Classes of Dangerous Goods in Packages and Intermediate Bulk Containers
API 650	American Petroleum Institute (API) Standard for Welded Steel Tanks for Oil Storage
API CP4	The Australian Institute of Petroleum (AIP) Code of Practice for the Design, Installation and Operation of Underground Petroleum Storage Systems
API CP22	The Australian Institute of Petroleum (AIP) Code of Practice for the Removal and Disposal of Underground Petroleum Storage Tanks
API CP25	The Australian Institute of Petroleum (AIP) Code of Practice for Wastewater Management at Bulk Petroleum Storage Sites
NOHSC: 1015(2001)	National Standard – Storage and Handling of Workplace Dangerous Goods

3.1.5 Other Relevant Documents

NOHSC Hazardous Substances Information System (http://www.nohsc.gov.au/applications/hsis/)

License Conditions for the Storage and Handling of Solid Ammonium Nitrate (DG Class 5.1) – DOCEP Guidance Note S313 Rev5

Guidance Note for Placarding Stores for Dangerous Goods and Specified Hazardous Substances – NOHSC:3009 (1990)

Storage of Dangerous Goods Placarding of Stores and Premises – DOCEP Guidance Note S302 Rev10

Tank Installations for the Storage of Flammable and Combustible Liquids – DOCEP Guidance Note S308 Rev5

Management of Hazardous Substances on Minesites – DOCEP Guideline document no. 7MR978HF

National Code - Storage and Handling of Workplace Dangerous Goods – NOHSC: 2017(2001)

3.3 Environmental Risks

Hydrocarbons and chemicals used on RTIO sites have the potential to cause significant impacts on the environment. Environmental hazards related to hydrocarbon and chemical use can occur as a result of:

- regular spillage of small amounts;
- accidental spillage of large amounts (e.g. through overfill or coupling failure);
- regular discharge of contaminated stormwater or washdown water;
- undetected leakage from storage vessels or transfer pipes.

Hydrocarbons and chemical releases to the environment may have a range of impacts including:

- soil, groundwater or watercourses can become contaminated by hydrocarbons and chemicals, which are toxic to vegetation and wildlife;
- hydrocarbon spills (oil slicks) can cover water bodies and kill fauna such as fish and birds;
- release of corrosive substances can change the acidity or alkalinity of the environment to a level outside the range tolerable to living organisms;
- toxic gases released into the atmosphere can kill, injure or cause long term health problems in human populations and wildlife;
- water sources for nearby towns or camps may become contaminated and may no longer be suitable for human consumption.

Under the Contaminated Sites Act 2003, proponents causing pollution can be directed to undertake extensive clean up and remediation work.

A risk assessment should be performed prior to the design and construction of any temporary facilities for the storage of hazardous materials. The *Operational and Design Environmental Risk Assessment Procedure* (document no. RTIO-HSE-0015225) must be used to guide this process.

Common risks that need to be addressed in relation to the storage of hazardous materials include:

- visible tank and pipe leakage;
- spills during transfer and refuelling;
- compliance with licence requirements;
- oily water generation;
- long term undetected tank or pipe leakage; and
- legal compliance risks.

3.4 Licensing of Dangerous Goods Storage Facilities

Dangerous Goods Storage licensing is managed by the Department of Consumer and Employment Protection (DOCEP).

For the purposes of licensing, Dangerous Goods storage is defined in the *Explosives and Dangerous Goods Act* 1961 (WA) as*: the keeping or retention of dangerous goods on a site for more than 1 hour.*

DOCEP has published a guidance note (GN) on the licensing process for the storage of dangerous goods titled: *Licensing and Exemptions (GN S301-Rev12)*

This document is available via the DOCEP website:

http://www.docep.wa.gov.au/resourcessafety/Sections/Dangerous_Goods/Guidance_Material/Dangerous_Goods_Stor.html

(follow the web link and then select the document *Licensing and Exemptions (GN S301-Rev12)* from the list).

3.4.1 Storage Factors

The requirement for licensing is determined by calculating the 'storage factor'. Once the Storage Factor exceeds 1000 (refer to DOCEP document *Licensing and Exemptions (Guidance Note S301-Rev12)* for further details) a licence is generally required.

It is important to be aware that Storage Factors are cumulative for a site, such that the total site storage should be taken into account when deciding whether or not a licence is required.

The DOCEP document *Licensing and Exemptions (Guidance Note S301-Rev12)* must be referred to when making decisions regarding storage factors and licensing requirements.

3.5 Temporary Hydrocarbon and Chemical Storage

This section addresses the storage of hydrocarbons (such as diesel fuel, petrol, lubricants) and other chemicals such as paints, thinners and acids.

The section is focussed on the storage of hydrocarbons as these are the most commonly stored chemical products on RTIO sites.

3.5.1 Minor Storage

The definition of "Minor Storage" for flammable and combustible liquids is provided in table 2.1 of AS 1940:2004. The definition varies depending on the type of material being stored and the location of the storage facilities.

Based on AS1940:2004, minor storage for hydrocarbons includes:

- Outdoor, above ground diesel tanks (in open land): up to 10,000 litres
- Construction site diesel tanks: up to 10,000 litres.

All Minor Storage of flammable and combustible liquids must comply with the requirements of AS1940:2004.

Designers should refer to other relevant Australian standards when planning minor storage for other types of chemicals including: AS 3870 (for corrosive substances) AS 4326 for oxidising agents and AS 3833 for mixed classes of hazardous materials.

The following criteria must be addressed when considering design requirements for minor storage:

MUST HAVES

☑ A designated storage area must be created.

☑ Transfer points for vehicle refuelling must comply with the requirements in Section 3.9 of this document.

☑ All piping must be above ground.

☑ All tanks and transfer points must be above ground and secondarily contained unless supported by a signed off risk assessment removing this requirement.

☑ All tanks must comply with the requirements of section 3.6 (double skinned tanks), section 3.7 (single skinned tanks) and section 3.8 (bunds) of this standard specification.

☑ Substances in small volume packages (nominally 20 litres or less) must be stored in a self bunded chemical cabinet (refer figure 4) or in a bunded area.

☑ The capacity of bunds around storage tanks must be 100% of the tank capacity (refer AS1940:2004 s 5.8.2).

☑ Containers and drums may be stored on self bunded pallets or in a HDPE lined earthen bund (refer section 3.8). In either case the capacity of the bunding must be at least 100% of the volume of the largest container plus 25% of the storage capacity up 10,000 litres (refer AS1940:2004 s4.4.3 for further details).

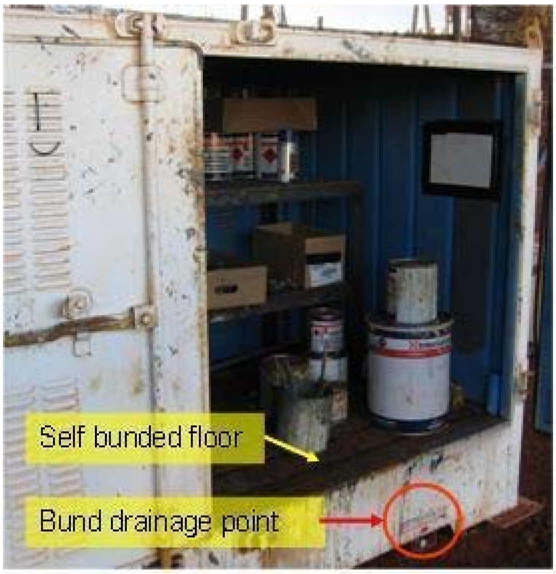

Figure 4. Workshop chemical cupboard (left) and site paint container (right)

3.5.2 Bulk Storage

Bulk storage refers to the storage of flammable and combustible liquids in volumes that exceed “Minor Storage” as defined in the relevant Australian Standards.

Storage of bulk volumes of flammable and combustible liquids must comply with the requirements of AS 1940:2004 and must be stored in either:

(1) A self bunded (double skinned) tank (refer to section 3.6); or

(2) A single skinned tank situated within a bunded compound (refer to sections 3.7 and 3.8 for requirements).

Designers should refer to other relevant Australian standards when planning storage for other types of chemicals including - AS 3870 (for corrosive substances) AS 4326 (for oxidising agents) and AS 3833 for mixed classes of hazardous materials.

3.6 Requirements for Self Bunded (Double Skinned Tanks)

Double skinned tanks may be suitable for bulk storage without additional bunding provided there is an adequate leak detection system, overfill protection and sufficient additional bunding around the discharge and filling points of the tank. The need for external bunding needs to be addressed on a risk assessed basis.

Note that AS 3780 and AS 4362 do not provide design guidelines for double skinned tanks. **Double skinned tanks must not be used for the storage of corrosive substances or oxidising agents.**

The following criteria must be addressed in relation to double skinned tanks for hydrocarbon storage:

MUST HAVES

☑ All double skinned tanks, regardless of capacity, must meet the requirements of AS 1940:2004. Section 5.9 of AS1940:2004 provides a specific list of requirements including (but not limited to):

- The secondary containment shall be designed to contain the entire contents of the primary tank.
- Means shall be provided to establish and monitor integrity of the primary tank.
- Suitable impact or collision protection must be provided (such as bollards or windrows of sufficient height to prevent vehicle impacts).
- Means shall be provided to prevent the release of liquid by siphon flow from the tank.
- Overfill protection shall be provided by an alarm sounding and the flow of liquid being stopped before the tank overflows.
- Each fill point shall be provided with spill containment having a minimum capacity of 15 litres per fill point.
- Piping connections to the tank shall be above the normal maximum fill level.

☑ All piping must be above ground.

☑ A certificate of integrity must be provided to the site environmental manager by the tank vendor prior to the tank being filled and this certificate must be maintained on site files for reference. The person mobilising the tank to site is responsible for obtaining this certificate.

☑ A dip test must be conducted between tank skins prior to the first fill and following the first fill.

3.7 Requirements for Single Skinned Tanks

All single skinned tanks must be stored within a bunded area. The design criteria for bunds are outlined in section 3.8 of this standard specification.

The following criteria must be addressed in relation to single skinned tanks:

MUST HAVES

☑ All single skinned tanks used for the storage of flammable or combustible liquids, regardless of capacity, must meet the requirements of AS 1940:2004. Section 5.9 of AS1940:2004 provides a specific list of requirements including (but not limited to):

- The tank shall be capable of resisting damage from the impact of a motor vehicle, or suitable collision protection shall be provided.

MUST HAVES

- Means shall be provided to prevent the release of liquid by siphon flow from the tank.
- Overfill protection shall be provided by an alarm sounding and the flow of liquid being stopped before the tank overflows.
- Each fill point shall be provided with spill containment having a minimum capacity of 15 litres per fill point.
- All piping connections to the tank shall be above the normal maximum fill level.

☑ All piping must be above ground.

☑ Single skinned tanks for storage of other types of chemicals (*ie* corrosive substances or oxidising agents) must meet the requirement of the respective standard: AS3780 (for corrosive substances) for and AS 4362 for (oxidising agents)

☑ A certificate of integrity must be provided to the site environmental manager by the tank vendor prior to the tank being filled and this certificate must be maintained on site files for reference. The person mobilising the tank to site is responsible for obtaining this certificate.

☑ Single skinned tanks must be stored in a bunded area – refer to Section 3.8 below.

CONSIDER

ⓘ Double skinned tanks are the preferred option for all hydrocarbon storage.

3.8 General Requirements for Bunds

The following requirements must be met in relation to construction of bunds around temporary storage facilities:

MUST HAVES

- ☑ All bunding for flammable and combustible liquids must meet the requirements of AS1940:2004.
- ☑ Bunds for storage of other types of chemicals (eg corrosive substances or oxidising agents) must meet the requirement of the respective standard: AS 3780 (for corrosive substances) and AS 4362 (for oxidising agents).
- ☑ Bunds must be sufficiently impervious to retain the spillage and enable the recovery of such spillage.
- ☑ A low point must be provided in the bund to allow pumping out of spilled material or excess water.

The following additional requirements relate to all earthen bunds (Figure 5 shows an example earthen bund layout):

- ☑ The liner used must be continuous and meet durability requirements (minimum of 0.75mm thick HDPE).
- ☑ The liner must be protected with 200 mm of sand (refer Figure 5).
- ☑ Storage tank support bases must not sit directly on the bund liner.
- ☑ The top of the liner must extend into the top of the bund walls.
- ☑ The bund floor must be compacted and rolled to prevent subsidence and remove sharp edges prior to placement of the liner.
- ☑ The volume of the bund is calculated using the height of the liner within the bund wall - this height may not be the same as the top of the walls.
- ☑ Earthen bunds must have height markers installed to show capacity levels.

CONSIDER

- ⓘ Double skinned or self bunded tanks are the preferred option for all hydrocarbon storage
- ⓘ A 20 litre concrete filled drum with a vertical height marker is easy to stand inside a bund to show capacity levels.

AVOID

! All valves and taps connected to storage tanks must not extend outside the bunded area or penetrate the liner.

! Common builders' plastic is not acceptable as a liner. The only acceptable floor liner for earthen floors is a minimum 0.75mm HDPE continuous liner.

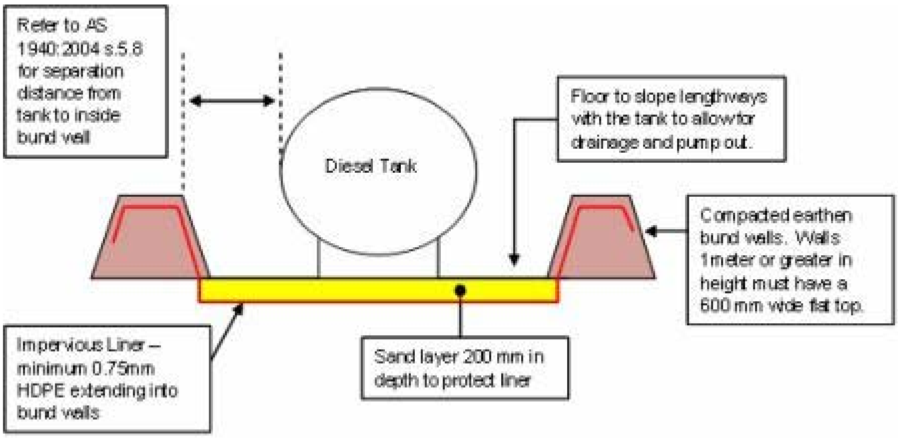

Figure 5. Example layout of an earthen bund for a temporary diesel storage tank.

3.9 Vehicle Refuelling Areas / Hydrocarbon Transfer Points

The following requirements must be met in relation to designing hydrocarbon transfer points**:**

MUST HAVES

- ☑ Vehicle refuelling must occur over a compacted, lined earthen pad or a concrete hardstand area (with the exception of field based machinery refuelling where a drip tray must be used at the transfer point).
- ☑ Concrete hardstand areas must be self contained and allow for oily water collection and treatment. Refer to Figure 6 for a suggested layout.
- ☑ Earthen pads must be set up as follows (refer Figure 7 for suggested layout):
  - ☑ The bund floor must be compacted and rolled to prevent subsidence and remove sharp edges prior to placement of the liner.
  - ☑ Ensure that the liner (min 0.75mm HDPE), is larger than the required hydrocarbon transfer area, and extends under the storage tank and into the roll over bunds.
  - ☑ Cover with earth (cracker dust is ideal) and remove large (i.e. > 20mm) stones that may perforate the liner.
  - ☑ Clearly visible roll over bunds must constructed on all sides of the refuelling area.
- ☑ Drip trays must be provided to capture any possible drips or spills during the transfer of fuel. This may include where there are breaks or joints in the hose.
- ☑ All requirements of the Dangerous Goods Storage licence must be complied with.
- ☑ A method of either measuring fuel delivered into, or dispensed from the tank must be implemented at the transfer point.

CONSIDER

ⓘ Dry break hoses should be used wherever possible.

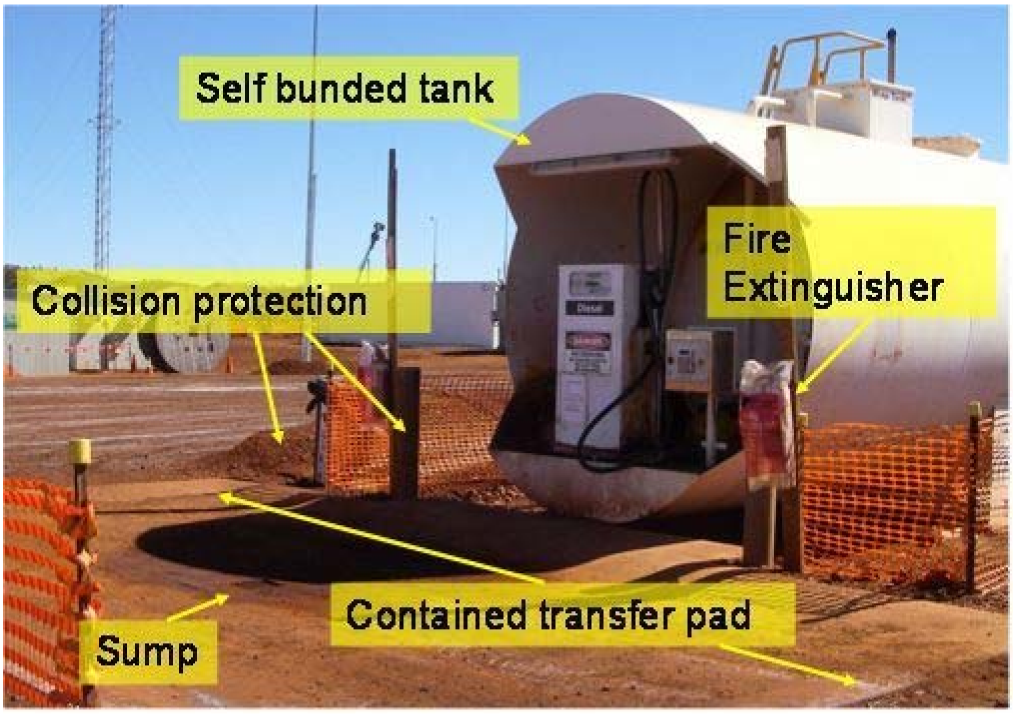

Figure 6. Suggested layout for refuelling point at double skinned tank

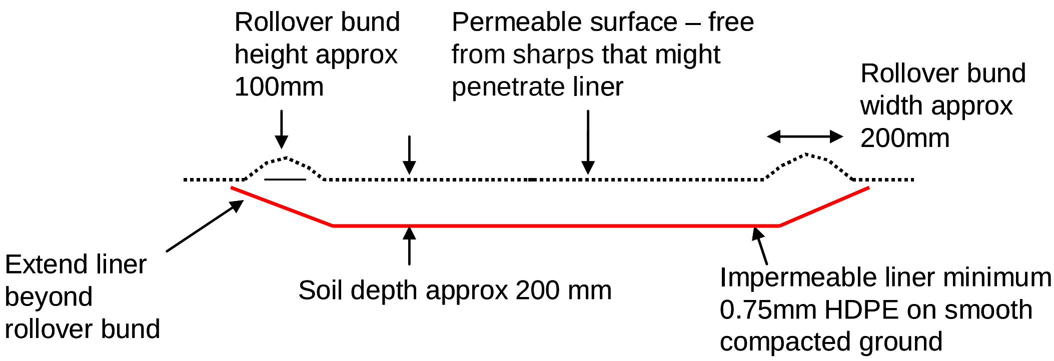

Figure 7. Suggested layout for an earthen rollover bund at refuelling points

3.10 Hydrocarbon Containment and Transfer in the Field

There are many situations in which diesel fuel will be transported in and transferred to various mobile plant and equipment including:

- fuel trailers
- generators
- welder generators
- service trucks
- mobile plant and equipment.

Hydrocarbons are required to be contained during transfer, just as they are in storage.

MUST HAVES

☑ Mobile fuel tanks must be either self bunded or in bunded trailers (refer Figures 8 and 9).

☑ Anti siphon valves must be fitted to the top of the tank.

☑ Transfer points must be secondarily contained in the field using drip trays.

☑ Spill kits must be present.

☑ Operators must be present at all times during fuel transfer.

☑ Dry break couplings must be used for fast fill pumps.

☑ Generators and other stationary equipment with internal fuel tanks must be either self bunded or operating within a bunded area (refer to section 3.8 General Requirements for Bunds).

CONSIDER

ⓘ Consider the use of equipment with spill and drip controls built into the design, for example, equipment with recessed fuel filling points is a good option for reducing minor spills during refuelling. The following generator welders are a good example:

http://www.shindaiwa.com.au/documents/products_start.html#

Figure 8. Example of a double-skinned trailer-mounted storage tank

Figure 9. Integrated storage for hoses and pumps within an enclosed area is good practice for containing drips and leaks.

(Figure 9 used with permission www.fuelequipment.com)

SECTION 4 - Temporary Workshops and Washbays

4.1 Introduction

This Section covers the environmental principles for the design of workshops and washbays. The treatment of contaminated (oily) water from washbays or any other facility on site is covered in Section 6 Oily Water Treatment at Temporary Facilities of this standard specification. This section does not address park up bays.

4.2 Regulations and Standards

There are no specific regulations covering temporary workshops and wash bays.

The storage of dangerous goods and hazardous materials and the release of polluting substances is covered under a range of legislation, regulations and guidelines:

- *Environmental Protection Act 1986*
- *Environmental Protection Regulations 1987*
- *Environmental Protection (Unauthorised Discharges) Regulations 2004*
- *Explosives and Dangerous Goods Act 1961.*
- *Explosives and Dangerous Goods (Dangerous Goods Handling and Storage) Regulations 1992.*
- *Country Areas Water Supply Bylaws 1957*
- *Contaminated Sites Act 2003*
- Department of Environment (WA), Water Quality Protection Note no. WQ 28 June 2005 “Mechanical Servicing and Workshops”
- Department of Environment (WA), Water Quality Protection Note, July 2000 “Extractive Industries within Public Drinking Water Source Areas”
- Rio Tinto Environment Standard E5 – Hazardous Materials and Contamination Control

There may also be project specific legal conditions regarding the controls that need to be implemented at workshops and wash bays related to works approvals, proponent commitments and operating licences.

4.3 Environmental Risks

Environmental risks related to workshops and wash bays include:

- chemical and hydrocarbon storage including used oil storage and related containment risks;
- release or discharge of contaminated stormwater (sediment, lubricants, detergents, coolants and hydrocarbons);
- soil, groundwater and surface water contamination with hydrocarbons and other chemicals, coolants and battery acids due to accidental spillages whilst in storage or during transfer;

- compliance risks related to works approval conditions and proponent commitments;
- generation of non-mineral wastes.

4.4 Temporary Workshop Designs

MUST HAVES

- ☑ Workshops must be constructed on a level surface.
- ☑ Earthen floors must be underlain with an impermeable liner – a minimum 0.75 mm thick HDPE liner is required.
- ☑ Liners must be placed onto a hard compacted surface free from sharp rocks. The top of the liner should be overlain with approximately 200 mm of clean earth.
- ☑ The workshop floor must be bunded with roll over bunds around the edges of the floor area to prevent stormwater ingress.
- ☑ Surface water runoff from workshop floor areas must be contained and treated as per the requirements of Section 6 - *Oily Water Treatment at Temporary Facilities* of this standard specification.
- ☑ Temporary drainage must be installed upslope from workshops to divert clean surface water around the workshop.
- ☑ Facilities must be provided for the collection and segregation of contaminated waste such as oily rags, contaminated soil and oil filters.
- ☑ A laydown area for collection of recyclable materials such as ground engaging tools and used tyres.
- ☑ A bunded area for collection / storage of used lead acid batteries.
- ☑ A bunded area for the storage of all chemicals and hydrocarbons.
- ☑ Spill kits must be available and appropriate for the volume and types of chemicals in use.

CONSIDER

- ⓘ Concrete workshop floors are preferable and do not require a liner.
- ⓘ Chemical cabinets are a safe way to store small quantities of hazardous materials and are preferable to storage in open bunds (refer Figures 4 and 10).
- ⓘ Placing a roof over workshop areas and bunding the workshop floor area (as shown in Figures 10 and 11) should prevent any surface water runoff from the workshop.
- ⓘ Extending workshop roofs over chemical storage bunds means the bunds will not collect stormwater saving on treatment and disposal costs.

AVOID

! Common builders' plastic is not acceptable as a liner. The only acceptable floor liner for earthen floors is a minimum 0.75mm HDPE continuous liner.

Examples of typical workshop layouts are shown in Figures 10 and 11.

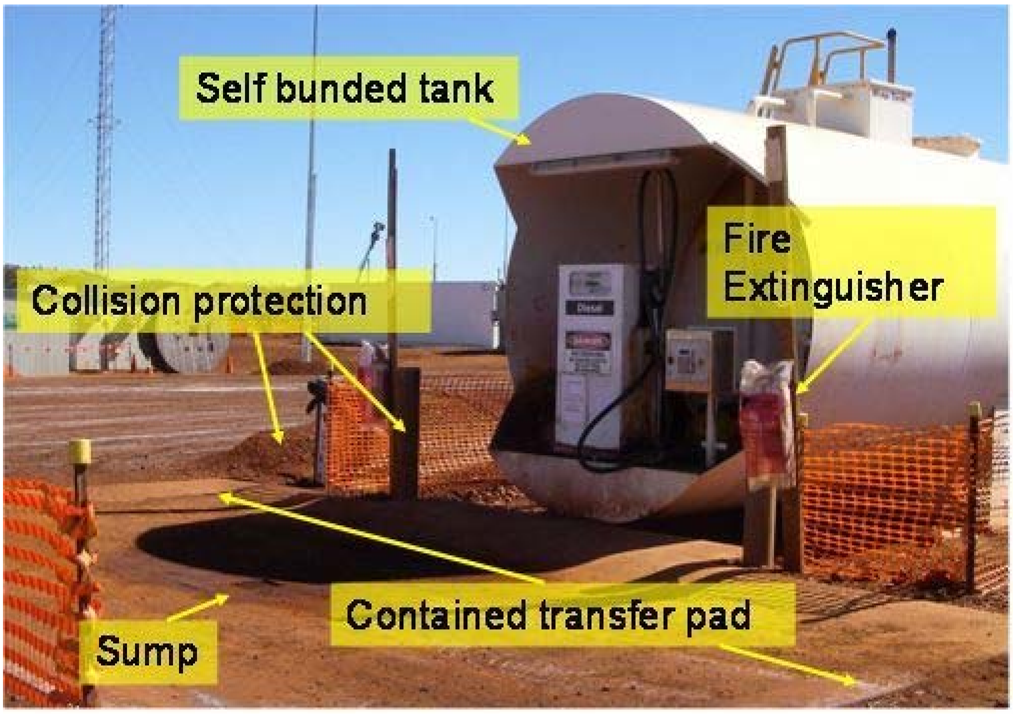

Figure 10: Typical temporary workshop layout

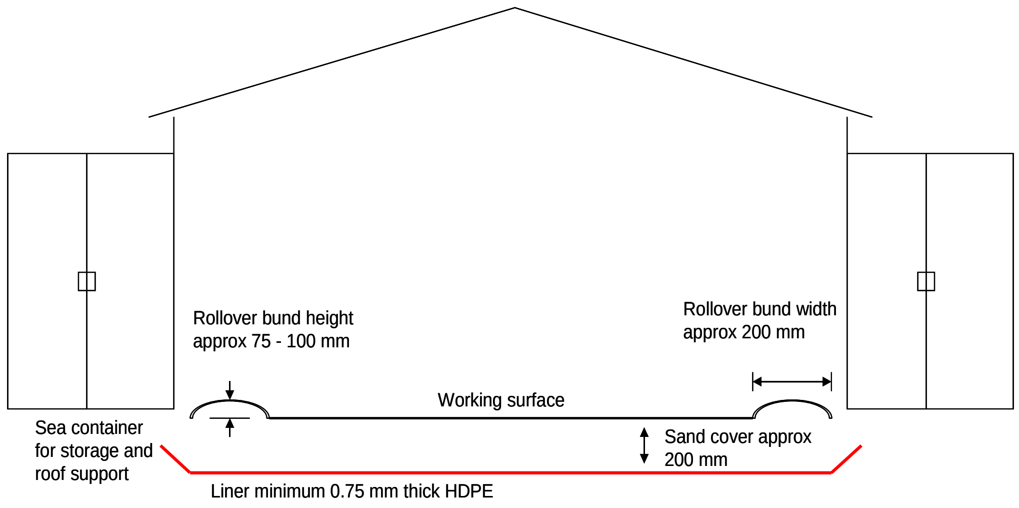

Figure 11: Typical temporary workshop layout (non concrete floor)

4.5 Washbay Designs

MUST HAVES

- ☑ All wash down bays must be bunded and wash down water directed to a collection drain that flows into a silt trap.
- ☑ Silt traps must be concrete, drive in wedge pits to enable easy removal of settled silt material (refer to Figure 12). Exceptions to this must be approved by the RTIOEP Environment Manager.
- ☑ Silt traps with outlet pipes designed for frequent over flow must direct this flow to a filter trench (refer Figure 13). Filter trenches must be placed away from trafficable areas. Note that this requirement applies only to washdown bays that do not generate hydrocarbon contaminated water.
- ☑ Wash down bay pads must be of sufficient dimensions to contain all spray water.
- ☑ Wash down of engine bays and vehicle under bodies is not permitted at wash down bays without an oily water separator.
- ☑ If a wash down bay is intended to be used or is used for washing down vehicle under bodies, engine bays or wash down of any equipment with the potential to generate hydrocarbon contaminated wash water, then the facility design must also incorporate the requirements of Section 6 Oily Water Treatment at Temporary Facilities.
- ☑ Any detergents must be labelled as 'biodegradable' and also conform to AS1792 - *Method for Determining he Biodegradability of Surfactants*. The Australian Standard for a product to carry the 'biodegradable' label is that 80 % of the product must be degraded within 21 days.

CONSIDER

- ⓘ Constructing a centralised wash down bay to a high standard rather than having multiple wash down bays on site.
- ⓘ How many vehicles will be using the facility?
- ⓘ What type and size of vehicles will be using the facility?
- ⓘ How much waste water will be generated ?
- ⓘ What is the seasonal rainfall collection capacity of wash down bay area.
- ⓘ Natural drainage at the proposed site.
- ⓘ Sensitive receptors at the proposed site (typically surface water or groundwater).

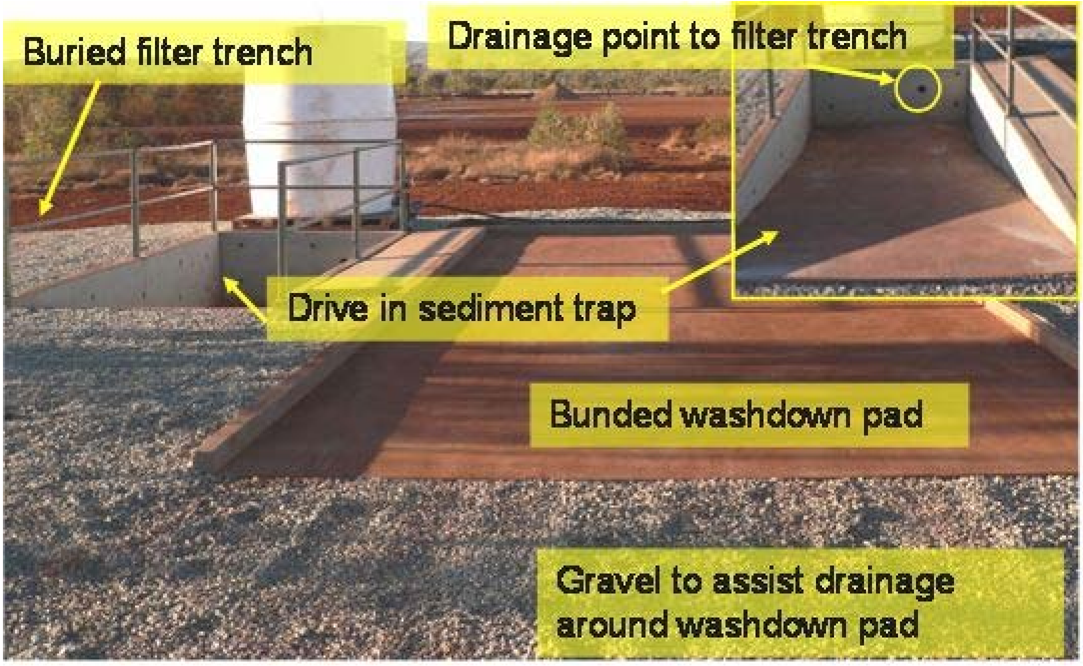

Figure 12. Temporary wash down bay with drive in sediment trap.

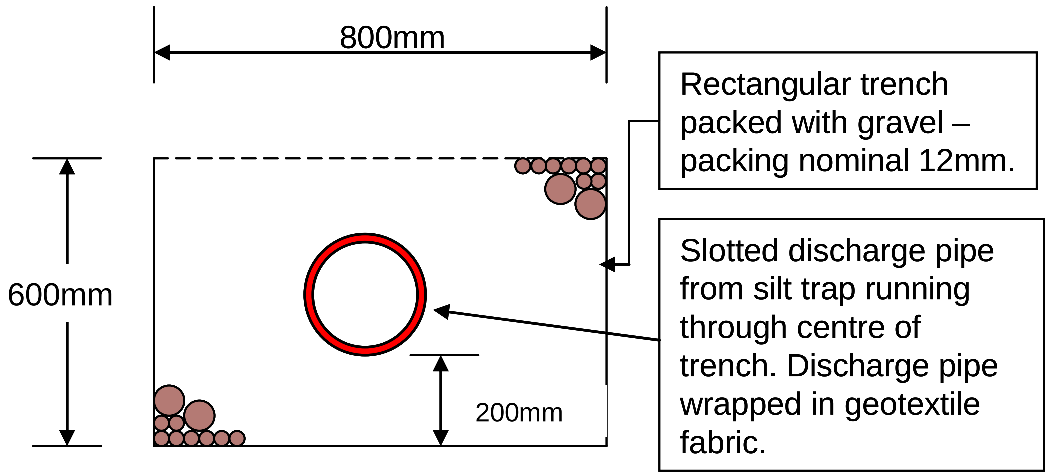

Figure 13. Example filter trench for sediment pond overflow.

SECTION 5 - Erosion & Sediment Control at Temporary Facilities

5.1 Introduction

This section addresses the environmental design requirements for erosion and sediment control structures at temporary facilities.

For sediment controls at wash bays refer to Section 4 – Temporary Workshops and Wash bays.

5.2 Regulations and Standards

Erosion and sediment control is covered under the following legislation and regulations:

- *Environmental Protection Act 1986*
- *Environmental Protection Regulations 1987*
- *Environmental Protection (Unauthorised Discharges) Regulations 2004*
- *Rights in Water and Irrigation Act 1914*

5.3 Environmental Risks

Common risks related to erosion and sediment control include:

- Offsite discharge of silt laden water resulting in potential non-compliance with works approvals, licence conditions
- Offsite discharge of silt laden water resulting in unacceptable impacts on flora and fauna and the siltation of waterways.
- Loss of topsoil resources.

5.4 Design Considerations to Minimise Erosion

Erosion management begins with work planning and ensuring that the following principles are followed before commencing work:

- Always look for opportunities to minimse clearing.
- Implement 'just in time clearing' to avoid opening up areas that are then left exposed to erosion.
- Practice progressive rehabilitation throughout construction.

5.4.1 Temporary Roads and Hardstand Areas

MUST HAVES

- ☑ Haul roads and access tracks must be designed to avoid, where practicable, valleys, drainage lines, dense woodland/mulga flats, creeks/floodplains, rough terrain, rocky outcrops, steep slopes and clay pans.
- ☑ Access tracks must be made to the minimum width necessary for safe passage.
- ☑ Erosion will be minimised by breaking maintenance windrows to allow drainage.
- ☑ Haul roads must be cambered to channel surface water run-off from the road and allow surface water flow through the existing area by using culverts and flood ways.
- ☑ Clearing must not be carried out for tracks intended for one-off use.

CONSIDER

- ⓘ As far as practicable roads should be located along contours and on “gravel” aprons on the lower slopes of hills to minimise earth moving disturbances and erosion potential.
- ⓘ Where practicable, existing roads and tracks should be used in preference to clearing new routes
- ⓘ Consider the catchment area in relation to possible rainfall events and where surface flows will be directed.
- ⓘ Redirection of surface flows around hardstand areas wherever possible.
- ⓘ Consider the relief at hardstand boundaries and whether or not rock armour is required to reduce water velocity and erosion (refer Figure 15).

5.4.2 Temporary Creek Crossings

MUST HAVES

☑ Always locate crossings on stable stream bed material (e.g. rock and gravel) if this is not possible, use stable gravel materials.

☑ Roadways leading up to water crossings must be adequately drained to prevent roadway runoff flowing into the water body.

☑ The use of existing creek crossings must be given priority unless access is blocked or safe working conditions are impeded.

☑ All creek crossings must avoid high velocity areas, bends and areas where banks are unstable or subject to erosion.

☑ Check whether a bed and banks permit is required prior to construction of any creek crossings.

CONSIDER

ⓘ Wherever possible creek crossings shall be approached by gentle slopes, at right angles to the creek.

AVOID

! Do not place any road grading or construction materials where it might enter the stream bed under flood or normal conditions.

5.5 Design Considerations for Erosion and Sediment Control

Sediment traps are a flow through device designed to reduce water flow velocities, promote settling of suspended materials and extend the time it takes runoff to reach downstream areas without long-term storage.

Sediment traps are usually highly effective at trapping coarse-grained sediments, which wash into the basin. Fine-grained sediments such as clays will remain suspended in the water and will travel off-site unless the water is detained for an extended period of time.

In order to maximise retention time and therefore maximise sediment reduction the following must be included in design:

MUST HAVES

- ☑ Construct sediment trap inlets and outlets at a gentle slope using stone / rip rap protection to minimise erosion potential (refer Figure 14).
- ☑ Permeable ground at the base of the sediment trap will allow for seepage of the water, drying out the sediment more quickly.
- ☑ Drainage channels subject to frequent or high flows should consist of rip rap rock protection (refer Figure 15).
- ☑ Site specific licences and permits may place quantitative limits on sediment levels in discharge waters. Sediment trap design must take into account these requirements

CONSIDER

- ⓘ The size of the catchment area and frequency of heavy rainfall events relative to the size of the proposed sediment trap.
- ⓘ The facilities feeding into the sediment trap and the amount of water that is reasonably expected to be generated by these facilities
- ⓘ Minimising the amount of water entering the sediment trap by preventing clean surface runoff from entering cleared or disturbed works areas.

AVOID

- ! Avoid lining sediment traps with material (e.g. HDPE) as they are difficult to clean and are likely to tear. They are also unnecessary for uncontaminated sediment-laden water, which should be collected in a basin that allows seepage.
- ! Sediment basins and silt traps DO NOT WORK when filled with sediment. They must be individually designed to cope with the expected volumes of sediment-laden water from each specific facility and site.

Figure 14. Sediment pond with rock lined inflow and outflow points

Figure 15. Placement of rip rap protection in a flood way.

SECTION 6 - Oily Water Treatment at Temporary Facilities

6.1 Introduction

This Section addresses environmental principles for the design of temporary water treatment systems that remove hydrocarbons from waste water.

6.2 Regulations and Standards

Water quality control is covered under the following legislation and regulations:

- *Environmental Protection Act 1986*
- *Environmental Protection Regulations 1987*
- *Environmental Protection (Unauthorised Discharges) Regulations 2004*
- Pilbara Iron Standard Specification SS-N103 Oily Water Treatment.
- The Australian and New Zealand Environment Conservation Council (ANZECC) Guidelines for Fresh and Marine Water Quality in 2000.
- Rio Tinto Environment Standard E5 – Hazardous Materials and Contamination Control

6.3 Environmental Risks

Risks associated with the discharge of hydrocarbon contaminated water include:

- non compliance with Works Approval Conditions and proponent commitments;
- exposure to the risk of penalties by regulators;
- onsite and offsite pollution;
- clean up costs;
- long term management requirements for contaminated sites;
- reputation related impacts associated with causing pollution;
- wash down and stormwater can become contaminated with hydrocarbons, sediment, detergents or coolants, which can flow into nearby watercourses, impacting on soil, vegetation and wildlife;
- inadequate maintenance of oily water systems will result in out of spec waste water discharge.

6.4 Control of Oily Water Discharges at Temporary Facilities

All temporary facilities at risk of generating oily water discharge must be designed to contain oily water discharge and deliver this discharge to receiving points for either (i) collection or treatment on site or (ii) collection and removal from site as shown in Figure 16.

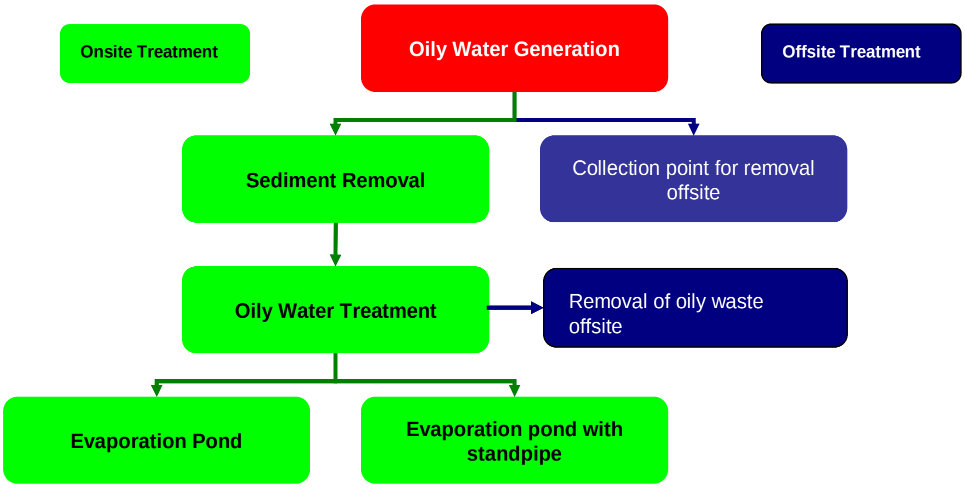

Figure 16. Oily water treatment options.

6.4.1 Oily Water Onsite Treatment

Acceptable systems for onsite oily water treatment include:

- Triple Interceptors
- Plate Separators
- Hydro cyclones

MUST HAVES

☑ A sediment trap must be constructed that allows easy visual inspection and easy removal of accumulating sediments with either hand tools (ie shovels) or machinery (ie bobcats). Refer to section 5 of this standard specification for further details on sediment control.

☑ Systems must be sized in accordance with manufacturers specifications and be capable of dealing with the nature and volume of waste being treated.

☑ Where the treated water is intended for dust suppression, on site a verification of the hydrocarbon concentration must be performed on a regular basis by a NATA certified laboratory.

☑ Evaporation ponds must be correctly sized for the maximum expected inflow and lined with a minimum 0.75mm thick HDPE liner.

☑ Treatment systems for oily water must be approved by RTIO Environmental Personnel prior to construction.

CONSIDER

ⓘ From a water efficiency perspective it is preferable to reuse the treated water if possible for washdown or dust suppression purposes.

6.4.2 Oily Water Offsite Treatment

MUST HAVES

- ☑ Collection ponds must be adequately sized for the volume of water generated.
- ☑ Collection ponds must be adequately sized to cope with expected rainfall.
- ☑ Collection ponds must be constructed with an appropriate liner – minimum 0.75mm thick HDPE liner.
- ☑ A system for logging and retaining all receipts and tracking numbers for all movements of oily waste water off site must be implemented and available for inspection by the organisation responsible for sending the wastes off site. This is a requirement of the *Environmental Protection (Controlled Waste) Regulations 2001.*
- ☑ Treatment systems for oily water must be approved by RTIO Environmental Personnel prior to installation on site.

CONSIDER

ⓘ Designs for all water storage structures should be checked by a civil engineer prior to construction.

SECTION 7 - Temporary Ammonium Nitrate Storage Facilities

7.1 Introduction

This Section covers environmental design requirements for facilities storing ammonium nitrate (AN), specifically AN Prill and AN Emulsion products.

Emulsion based AN can be stored as Dangerous Goods Class 1.1 (Explosive) OR Dangerous Goods Class 5.1 (Corrosive) depending on its formulation. These classes are very different and trigger different storage and handling requirements.

This document only addresses the storage of AN as Dangerous Goods Class 5.1 (Corrosive).

Bulk storage of AN as Class 1.1 (Explosive) is highly hazardous and should be avoided wherever possible. Advice from explosives experts must be sought prior to storage of this material.

7.1.1 Definitions

ANFO

ANFO is a generic name for an explosive mixture of Ammonium Nitrate (AN) and Fuel Oil (FO) used for blasting. The FO component can be diesel, fuel oil or a mixture of diesel and other oils. The two components (AN and FO) are transported to site separately and mixed immediately prior to use.

AN Prill

The AN component of ANFO is typically stored as AN Prill, which is a solid granular form of AN. In temporary facilities this is typically transported in bulka bags.

AN Emulsion

AN emulsion is an emulsified form of AN prill produced by explosive manufacturers and AN suppliers such as Orica. It is typically supplied as an opaque pumpable paste or as a gel and is transported and stored in tanks. Typical components are emulsified AN prill (>60%) and oils, water and/or other emulsifiers (<40%).

7.2 Regulations and Standards

This standard specification must be applied in conjunction with the Department of Consumer and Employment Protection (DOCEP) Licence Conditions for the Storage and Handling of Solid Ammonium Nitrate (Dangerous Goods Class 5.1) Guidance note S313 Rev. 5.

The following regulations and guidance materials are also relevant:

- Australian Standard AS 4326:1995 The storage and handling of oxidizing agents
- Australian Code for the Transport of Dangerous Goods by Road and Rail (ADG Code), Sixth Edition
- Explosives and Dangerous Goods (Dangerous Goods Handling and Storage) Regulations 1992
- AS 4326:1995 The storage and handling of oxidising agents

- Storage of Dangerous Goods - Licensing and Exemptions – Guidance Note GN S301- Rev 12 (DOCEP)
- Storage of Dangerous Goods - Placarding of Stores and Premises - Guidance Note S302 - Rev 10 (DOCEP)
- Storage of Dangerous Goods - General Requirements for Licensed Premises – Guidance Note GN S303 - Rev 8 (DOCEP)
- Guidelines for the Preparation of an Emergency Plan and Manifests – Guidance Note GN S310 - (DOCEP)

BEWARE

! DOCEP Guidance Notes must be checked prior to proceeding as the guidance note *Licence Conditions for the Storage and Handling of Solid Ammonium Nitrate (Dangerous Goods Class 5.1) Guidance note S313 Rev. 5* is subject to change due to ongoing national security issues

! Licensing requirements must be checked with DOCEP **prior** to construction of temporary ANFO storage facilities.

7.3 Environmental Risks

The following common risks relate to AN storage and usage:

- AN is a powerful oxidising agent and can react strongly with incompatible products
- Spills of AN into water bodies may cause algal blooms.
- When spilled onto soil surfaces in bulk AN causes temporary sterilisation of the soil. AN residue remaining in the soil post removal may become mobile acting as a fertiliser and causing surface and groundwater contamination.
- Spills are common in transfer from storage into ANFO trucks.

7.4 Temporary AN Storage Design Considerations

The design of temporary AN storage facilities is highly regulated and must comply with the regulations, standards and guidance material outlined in Section 7.2 of this standard specification.

The following considerations must be addressed in temporary ANFO storage design:

MUST HAVES

- ☑ The design, construction and maintenance of a storage facility must prevent the ingress of groundwater and surface water runoff.
- ☑ The storage and loading area must be designed to allow spillages to be dry cleaned. Collected material must be kept dry until disposal occurs.
- ☑ Temporary AN Emulsion Storage facilities must be contained on a hardstand to ensure spillage can be collected and disposed of appropriately.
- ☑ Loading of AN from the AN transport truck into the storage area must occur in an area that allows containment of spillage.
- ☑ Loading of AN into the ANFO truck must be over hardstand (preferably concrete) to allow dry collection of spillage.
- ☑ Storage and transfer operations should be designed to prevent discharge.

AVOID

- ! The floor of an AN storage facility must not contain bunds, drains, traps, pits or other depressions in which molten ammonium nitrate may become confined in the event of a fire.

7.4.1 Compatibility Considerations in Design

AN is a strong oxidising agent and is incompatible with a range of materials. Designs must ensure that adequate separation and segregation from a range of materials including (but not limited to):

- All flammable and combustible liquids;
- All flammable solids;
- Organic chemicals, acids, alkalis, and other corrosive materials;
- Compressed flammable gases;
- Other contaminating materials including:
    - Baled rags, scrap paper, bleaching powder, cotton bags, caustic soda, fish oils, foam rubber, lubricating oils, linseed oil, or an other oxidisable or drying oils, naphthalene, oiled clothing, oiled textiles, vegetable oils and cement.
- Blasting agents, explosives, sulphur and finely divided combustible solids must not be stored in the same store as ammonium nitrate.

It is best **not** to assume that all class 5.1 oxidising agents are compatible as many are **not** (e.g. ammonium nitrate is also incompatible with sodium nitrate, another oxidising agent). Refer to the product MSDS for guidance.

Incompatible goods must never be permitted to come into contact with the ammonium nitrate and physical barriers must be in place to prevent this from occurring. In order to achieve this, it may be necessary to provide bunds, kerbing or sloping floors to ensure that molten or liquid incompatible products cannot mix.

7.4.2 Housekeeping Considerations in Design

Temporary AN storage designs need to facilitate efficient house keeping to minimise the risk associated with ammonium nitrate storage. The following precautions are strongly recommended:

- Prior to placing ammonium nitrate in a storage area, the area should be cleaned;
- Walls, floors, access ways and surrounding areas and equipment must be kept clear and free of build up of combustible debris, including ammonium nitrate;
- Regular inspections should occur to ensure leaks are detected promptly and cleaned up;
- Ammonium nitrate storage areas should be kept clean and separated from any unnecessary objects by at least 5m in all directions, (e.g. unused wooden pallets);
- All dry vegetation should be cleared away from the store for a distance of at least 5m;
- Filled bags and intermediate bulk containers (IBCs) should be stored in stable stacks;
- Spillage of ammonium nitrate and other materials stored nearby should be cleaned up immediately;
- Damaged bags should be kept in overpacks or slip-over bags to prevent additional spillage;
- Contaminated products must be safely disposed of or made free of ammonium nitrate; and
- Pallets, ropes, slings, covers, machinery, or combustible items must not be allowed to become contaminated with a build up of ammonium nitrate.

SECTION 8 - Concrete Batch Plants

8.1 Introduction

This Section applies to temporary and mobile concrete batch plants on all RTIO sites.

8.2 Regulations and Standards

Batch plants are regulated by the *Environmental Protection (Concrete Batching and Cement Product Manufacturing) Regulations 1998.*

Batch plant registrations (issued by the DEC) must be provided to the site environmental manager prior to mobilisation to site.

8.3 Environmental Risks

Some of the more common risks that need to be managed at Concrete Batch Plants on RTIO sites relate to:

- Inadequate management of contaminated (high pH) wash down water and sediment laden surface water.
- Inadequate management of sediment laden surface water.
- Dust emission from aggregates and stockpiles.
- Cement dust emissions.
- Poor storage practices for chemicals (retardants, acids etc).
- Resource wastage (non recycling of wash down water).

Site specific risk assessments should be conducted to ensure all relevant risks are identified and managed.

8.4 Batch Plant Design Considerations

This section outlines batch plant design considerations to be addressed on all RTIO sites.

8.4.1 Batch Plant Siting

The location and orientation of batch plants is critical to managing environmental impacts.

The following should be considered when selecting a batch plant site:

- The chosen site should be above 100 year flood levels.
- The site should be downwind from sensitive receptors (including buildings).
- The site should be sheltered from prevailing winds – product bunkers, conveyors and hoppers should be sited in the leeward direction.

- A minimum buffer distance of (i) 50 metres is recommended between a mobile batch plant and sensitive environmental areas and (ii) 100 metres for human receptors (*eg.* camps, offices etc) *(Recommended Buffer Distances for Industrial Residual Air Emissions, Victorian EPA Publication AQ 2/86, July 1990).*

8.4.2 Water and Sediment Management

Potential pollutants in batch plant wastewater and stormwater include cement, sand, aggregates, chemical admixtures, fuels and lubricants. Site water management is critical to ensuring these materials are retained at the batch plant site.

MUST HAVES

☑ The batch plant site must be designed and constructed such that clean stormwater, including roof runoff, is diverted away from contaminated areas.

☑ Above-ground fuel and chemical additives must be stored as outlined in Section 3 of this document.

☑ A wastewater collection and recycling system must be designed to collect contaminated water from:

- agitator washout
- truck washing
- yard washdown
- slump stand
- any other process wastewater from the batching plant operation.

☑ Contaminated water must be diverted to a settling pond for evaporation or re-use in the truck washdown process.

☑ Sediments must be periodically collected from settling ponds and placed in the site landfill.

☑ Truck washout must occur in a bunded area. The bunded area must be large enough to collect overspray from truck washout. Truck washouts must be constructed from concrete with the exception of short term batch plants (nominally less than 3 months) where a fully bunded earthen structure is acceptable.

☑ Wash down overspray may be required to be controlled with a temporary fence structure in windy areas.

☑ Liners under earthen bunded washout areas must be a minimum thickness of 0.75mm HDPE (builders plastic is unacceptable)

☑ Water produced during the washout process must pass through a settling pond to allow solids to be removed before being stored in a pond that allows for re-use of the washout water.

Best practice examples are shown in Figures 17 and 18 below.

The site requirements for the truck washout and water recycling facility must be agreed with the Environmental Manager prior to mobilisation to site.

Figure 17. Batch Plant Sediment Collection & Water Recycling

Figure 18. Batch plant truck wash down

8.4.3 Dust Management

Cement, sand and aggregates at a batch plant can produce dust when dry.

MUST HAVES

☑ Water sprinklers must be fitted to product storage bunkers to minimise dust (refer Figure 19) and stockpiles must be adequately sprayed with reticulated sprinklers or covered to minimise dust emissions.

☑ Hoppers, conveyors, chutes, bucket elevators and transfer points must be either:

- Enclosed;
- Fitted with wind shields, water sprays or a dust extraction system; or
- Otherwise designed and operated so as to prevent the escape of any visible dust.

CONSIDER

ⓘ The prevailing wind direction should be considered to ensure that bunkers, conveyors and stockpiles are sited in a leeward position to minimise the effects of the wind.

ⓘ Coarse and fine aggregates should be stored in three sided bunkers wherever possible. Product should not be stored above the bunker walls consistent with regulatory requirements.

ⓘ All coarse and fine aggregate products should be conditioned prior to use.

Figure 19. Concrete bunkers with automate water sprays to control dust.

8.4.4 Cement Silos

MUST HAVES

- ☑ Filters and level alarms must be fitted to all cement storage silos as specified in the Environmental Protection (Concrete Batching and Cement Product Manufacturing) Regulations 1998.

SECTION 9 - Turkeys Nests

9.1 Introduction

This section outlines the environmental design requirements for temporary turkeys nests. This section does not provide civil design specifications for dam structures.

9.2 Regulation and Standards

Rio Tinto Environmental Standard E10.

Site works approvals and operating licences often include proponent water management and water efficiency commitments that must be adhered to.

9.3 Environmental Risks

Stand pipes and turkeys nests carry the risk of loss of containment and poor water management practices including inefficient water transfer, leaking pipes, pumps and liners.

Turkeys nests also present hazards to fauna due to the use of plastic liners that makes egress from the ponds very difficult.

The permanent presence of water also attracts wildlife and cattle to standpipe splash areas increasing the likelihood of weed colonisation in these areas through transport of seeds.

9.4 Design Considerations

Critical to turkeys nest design is the requirement to install the liner system correctly to minimise the risk of leaks. Critical design requirements are addressed in the following table:

MUST HAVES

- ☑ Turkeys nests must be lined with minimum 1.0 mm welded HDPE liner.
- ☑ Prior to placement of the liner on the turkeys nest floor, the floor must be compacted or rolled to remove any sharp edges.
- ☑ The liner must extend to the top of the turkeys nest wall. A trench must be dug along the top of the wall and the top of the liner buried in the trench.
- ☑ Standpipes (or connected bore pumps) must be fitted with a calibrated flow meter that complies with the *Rights in Water and Irrigation (Approved Meters) Order 2003*. The flow meter reading must be recorded at commencement of operation.
- ☑ Standpipes must be fitted with a flexible delivery hosing to minimise splash losses (refer Figure 20).
- ☑ Turkeys nests must be fenced off to prevent fauna access (refer Figure 20).

☑ Turkey nests must have egress devices suitable for fauna to egress the turkeys nest (refer to Figure 21).

☑ Abrasion prevention devices such as old tyres must be used under inflow and outflow pipes to prevent liner wear (refer to Figure 21).

CONSIDER

ⓘ Tanks are preferred to turkeys nests wherever possible (refer figure 22).

ⓘ Fencing off standpipe compounds to prevent cattle access to overflow areas.

ⓘ Excavating a shallow overflow chute in the top of the turkeys nest wall and lining it with a piece of HDPE liner to direct overflow water over the edge of the turkeys nest without scouring out the wall.

Examples of temporary turkeys nest layouts are shown in Figure 20 and 21 below.

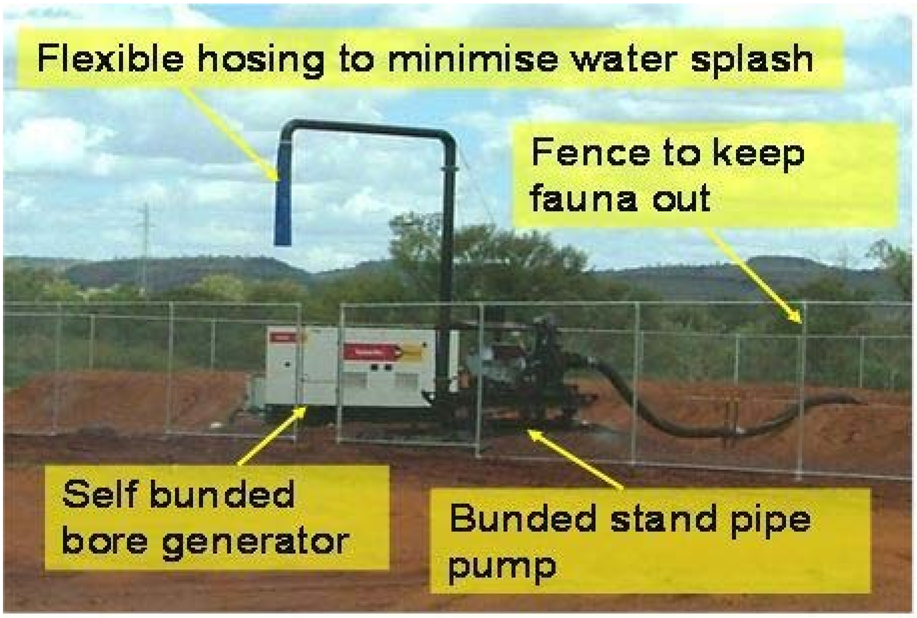

Figure 20. Turkeys nest layout

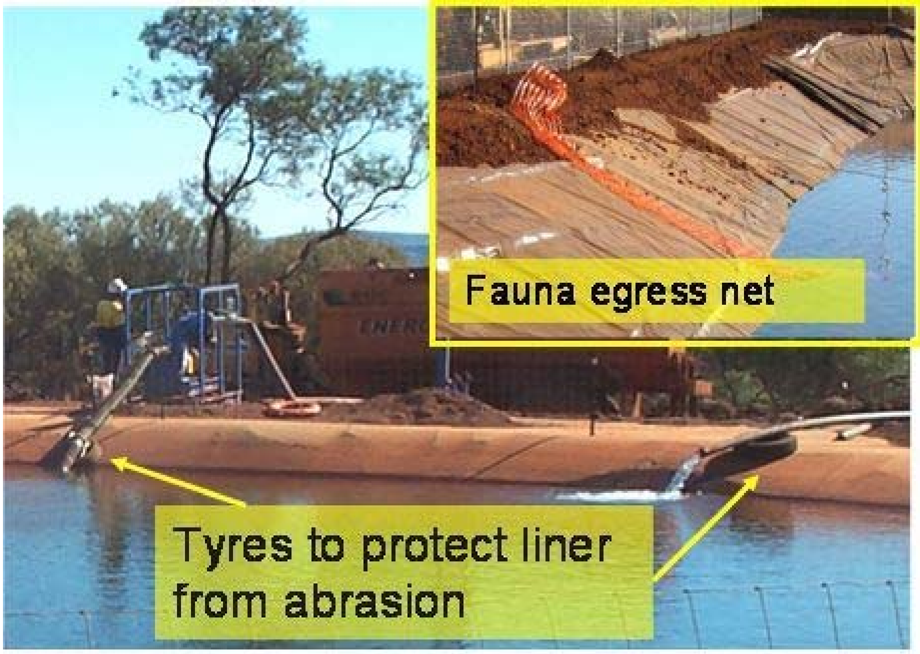

Figure 21. Turkeys nest liner protection and fauna egress net

Figure 22. Example of a tank in place of a turkeys nest.

SECTION 10 - Portable Ablutions and Septic Systems

10.1 Introduction

This Section applies to temporary and portable ablutions on all RTIO sites. Waste water treatment plants are not covered in this scope.

10.2 Regulation and Standards

Portable ablutions are subject to the following regulations and standards:

- *Health (Treatment of sewage and disposal of effluent and liquid waste) Regulations 1974*
- *Environmental Protection (Controlled Waste) Regulations 2001*
- Local council regulations.

10.3 Environmental Risks

Common environmental risks related to temporary / portable ablutions include:

- Health – untreated sewage carries a high level of harmful bacteria.
- Licensed carriers must be used and as sewage is a controlled waste, full documentation is required to be kept on file in relation to the movement of controlled waste.
- Loss of amenity due to odours associated with the poor placement (ie upwind of occupied buildings) and/or maintenance of facilities.
- Excessive water use.

10.4 Design Considerations

The key design requirements for portable ablution systems with holding tanks are addressed below.

MUST HAVES

☑ Portable ablutions facilities must have fully enclosed sewage holding tanks with inspection ports for level measurement.

☑ Low water use cisterns must be installed.

☑ Overflow sumps must be installed to capture any accidental spills.

CONSIDER

- ⓘ Prevailing winds and potential odours when placing a toilet facility.
- ⓘ Hire units are available with level indicators and flashing lights that indicate when the holding tank is 75% full.
- ⓘ Placing sewage contractors on a regular collection schedule to empty the holding tank – this should commence as soon as a system is commissioned on site.
- ⓘ Waterless urinals are an excellent way to minimise water use. Regular urinals may be converted to waterless urinals using Desert Cubes, a proprietary product that minimises water use and reduces odours (refer Figure 23 below): Internet: http://www.desert.com.au/
- ⓘ Hire units with integrated holding tanks (refer Figure 24 below) minimise the risk of sewage pipe failures.
- ⓘ Alternative systems are available such as composting toilets and chemical toilets which may be suitable for low volume sewage treatment.

Figure 23. Desert Cubes are an excellent way to minimise water usage in urinals.

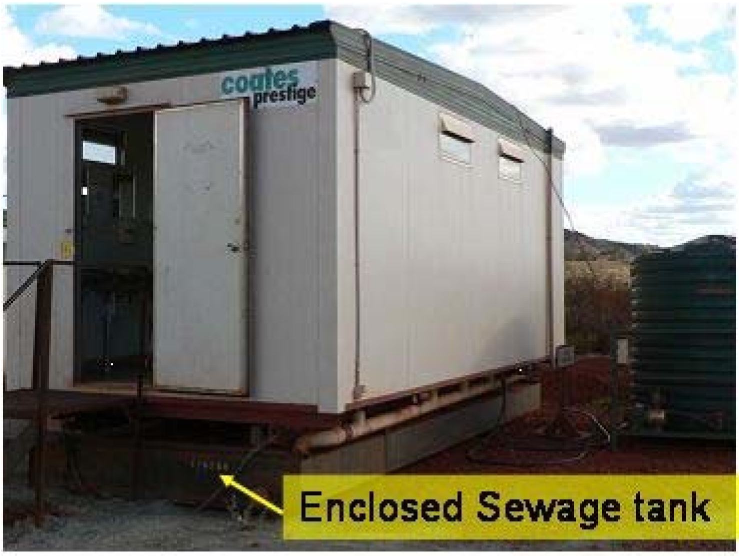

Figure 24. Portable ablution facility with inbuilt holding tank in base.

10.5 Septic Systems

Septic tank installations result in the release of sewage to the environment, and hence are not as environmentally friendly as enclosed holding tank systems or Waste Water Treatment Plants.

In determining whether a septic system is required the following points must be considered:

- Will the facility be used intermittently or only for a short period of time? For example, will the system be servicing a short-term construction site or camp area with only intermittent occupation such as for maintenance activities?
- Is it unlikely that the facility will need to cater for additional people in the future (e.g. it will not need to be used for an operational purpose in the future)?
- Is the area requiring the septic system remote from planned or existing sewage treatment systems (e.g. remote or satellite offices) and hence connection to such facilities is not viable?
- Is a closed system with contents regularly evacuated not practical (e.g. the area cannot be readily serviced from a town or the capacity is not practical)?
- Will the number of people serviced by the system be sufficiently small such that a septic system can provide adequate treatment/storage?
- Is a package treatment plant at this location not viable from an economic point of view ?

Each of these points must be addressed and discussed with RTIO Environmental Personnel prior to designing septic tank systems.

MUST HAVES

☑ Any decision to install a septic tank installation over a WWTP must ensure that an unacceptable environmental risk is not posed by the facility. All septic tank installations must be approved by RTIO Environment Personnel.

☑ All septic tank installations must be constructed and approved in compliance with the requirements of the Health (Treatment of Sewage and Disposal of Effluent and Liquid Waste) Regulations 1974. Note that DEC approval is not required, however, local council approval is likely to be required.

☑ Decommissioning of septic tanks must be carried out in compliance with Division 3 of the Health (Treatment of Sewage and Disposal of Effluent and Liquid Waste) Regulations 1974.

CONSIDER

- ⓘ Access for vehicles for pumping and sludge removal.
- ⓘ The ground conditions (for excavation of tanks and piping as well as for water infiltration).
- ⓘ The depth of groundwater to avoid contamination.
- ⓘ Proximity to potable water bores.
- ⓘ Potential damage to the septic tank and drainfield from traffic and wheel loads on the system.
- ⓘ Locating the drainfield area away from trees to avoid tree root problems.

SECTION 11 - Temporary Power Stations

11.1 Introduction

This Section applies to temporary power stations on all RTIO sites.

11.2 Regulation and Standards

Temporary power stations are subject to the following regulations:

- *Environmental Protection Act 1986*
- *Environmental Protection (Noise) Regulations 1997*

If the power station generating capacity or the combined site power generation capacity exceeds 10 MW then a works approval must be approved by the Department of Environment and Conservation prior to the commencement of construction.

11.3 Environmental Risks

Common environmental risks related to temporary power stations include:

- Generation of contaminated wastes.
- Fuel spills during refuelling.
- Continual oil drips and leaks through engine breathers.
- Transformer oil spills.
- Emission of greenhouse gases.

11.4 Design Considerations

MUST HAVES

☑ Fuel storage must comply with the requirements outlined in Section 3 of this specification.

☑ External oil filled transformers must be bunded (refer to Figure 25). Bunding must comply with the requirements outlined in Section 3.8.

☑ Refuelling facilities must comply with the requirements of section 3.10 of this specification.

☑ Generators with internal fuel tanks must be either self bunded or operating within a bunded area (refer to section 3.8 General Requirements for Bunds).

☑ Noise emissions must comply with the requirements of the Environmental Protection (Noise) Regulations 1997.

☑ Engine breathers and radiator overflows must be contained within bunded areas.

☑ Spill kits must be provided at all temporary power stations.

CONSIDER

ⓘ Location of the temporary power station in relation to sensitive receptors.

ⓘ Placement of contaminated waste bins adjacent for oil filters, fuel filters and used oil and oily rags.

Figure 25. Bunding around a temporary power station transformer.

SECTION 12 - Mobile Crushing and Screening Plants

12.1 Introduction

This Section applies to mobile crushing and screening plants on all RTIO sites.

12.2 Regulation and Standards

Mobile crushing and screening plants are subject to the following regulations:

- *Environmental Protection Act 1986*
- *Environmental Protection (Noise) Regulations 1997*

Mobile crushing and screening plants may require a licence from the Department of Environment and Conservation depending on their intended production volumes. This must be confirmed with RTIO Environment Personnel prior to mobilisation of such machinery to site.

12.3 Environmental Risks

Common environmental risks related to mobile crushing and screening plants include:

- Generation of excessive dust.
- Generation of excessive noise.
- Fuel and oil leaks.
- Generation of sediment laden runoff.

12.4 Design Considerations

MUST HAVES

☑ Fuel storage must comply with the requirements outlined in Section 3 of this specification.

☑ Refuelling must comply with the requirements of section 3.10 of this standard specification.

☑ Noise emissions must comply with the requirements of the Environmental Protection (Noise) Regulations 1997.

- ☑ Drip trays must be placed beneath potential leak sources for all stationary machines.
- ☑ The works area must be drained through a sediment control trap as per Section 5.5 of this standard specification.
- ☑ Water sprays must be fitted to all mobile crushing and screening plant to minimise dust emissions during operations.
- ☑ Wind protection devices such as skirts and shrouds must be fitted wherever possible to control wind blown cost.
- ☑ Siting of crushing and screening plants must consider sensitive receptors located downwind.

CONSIDER

- ⓘ Location of the plant in relation to sensitive receptors.
- ⓘ Placement of contaminated waste bins adjacent for oil filters, fuel filters and used oil and oily rags.
- ⓘ Wherever possible crushing and screening plant should be located on the leeward side of structures or geographic features.

APPENDIX 5

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Communications Protocol with Department of Environment and Conservation on Activities in Millstream-Chichester National Park

The following Communications Protocol (the **Protocol**) with the Department of Environment and Conservation (**DEC**) will be implemented during the proposed rail expansion works within the Millstream Chichester Nation Park (**MCNP**) by the Rio Tinto Iron Ore Group (the **Company**).

Company Contacts:

- Company Representative for works undertaken on site: **Project Manager – Keith Rumsey** ph. (08) 9205 2677 or 0418879015. Keith should be contacted for Project or construction site access information.

- For all environmental queries please contact **Rio Tinto Iron Ore (RTIO) Expansion Projects Specialist Environmental Advisor – Alex Newton** ph. (08) 9205 0490.

Department of Environment and Conservation Contacts:

Primary contacts – Operational incidents (On-site MCNP):

- **Marc McDonald (Senior Ranger)**
  - Email: Marc.McDonald@dec.wa.gov.au
  - Telephone: (08) 9184 5144
- **Stephen Breedveld (Park Ranger)**
  - Email: Stephen.Breedveld@dec.wa.gov.au
  - Telephone: (08) 9184 5144

Secondary contacts – Regional Office, Karratha

- **Duty Officer for fire emergencies**
  - Telephone: (08) 9182 2088
- **Michelle Corbellini (Environmental Impact Assessment for the Pilbara Region – Perth Based)**
  - Email: Michelle.Corbellini@dec.wa.gov.au
  - Telephone: (08) 9334 0260
- **Geoff Passmore (Senior Operations Officer)**
  - Email: Geoffrey.Passmore@dec.wa.gov.au
  - Telephone (08) 9182 2022
- **Cath Rummery (Nature Conservation Leader)**
  - Email: Cath.Rummery@dec.wa.gov.au
  - Telephone: (08) 9182 2005
- **Allisdair MacDonald (Regional Manager)**
  - Email: Allisdair.MacDonald@dec.wa.gov.au

- Telephone: (08) 9182 2000

- Prior to works commencing in the MCNP, a monthly update including the proposed works and timing will be forwarded to the Park Rangers.

- At least five working days prior to accessing the MCNP, the Company will provide the DEC Regional Manager with an itinerary and program of the locations of operations on the licence area and informing the Regional Manager at least five days in advance of any changes to the itinerary. All activities and movements shall comply with reasonable access and travel requirements of the Regional Manager regarding seasonal/ground conditions.

- Consistent with the (currently draft) Weed Action Plans for the rail network, an Annual Weed Management Report will be prepared at the conclusion of each 12 month period, a copy of which will be forwarded to the DEC. The DEC will be consulted on the production of the Company Pilbara Weed Strategy and associated Weed Action Plans.

- During the proposed works within the MCNP, an area engineer will prepare a short report on works to date (with appropriate photos) for the endorsement of the Project Engineer which will then be forwarded onto the Park Rangers on a monthly basis.

- The Company will record clearing activities against each lease within the MCNP. These details will be provided to the DEC via required reporting against Ministerial Statement 880 and associated residual impact package and the RTIO Annual Environmental Report for Railways Division.

- The Company will report to the above primary contacts (in the first instance) incidents which involve the following:
  - Fire Emergencies;
  - Chemical/hydrocarbon spills which may impact on the MCNP outside the area in which the Company is working;
  - Chemical/hydrocarbon spills into watercourses;
  - Accidents occurring within the MCNP which will include fauna injuries or mortalities;
  - Negative interactions with MCNP users; and
  - Public or major road closures as a result of the project.

  These incidents will be reported in with the monthly report to the DEC (Pilbara) and will include details on remedial actions taken. Any significant incidents, such as uncontrolled fires, will be reported to the DEC Duty Officer within one hour. All other significant incidents will be reported to the Primary contacts as soon as possible.

- Should the Park Rangers or other relevant DEC staff want to inspect or view the works on site whilst under construction or to conduct works associated with the Rail offset package, for safety reasons, they should contact the Project Engineer on (08) 9205 2677 and they will arrange for safe passage around the construction site.

- A close out inspection will be undertaken by the Company within 21 days of the completion of the works. The DEC staff (including the MCNP Rangers) will be notified of the date of the inspection and given the opportunity to attend. As part of the close out inspection, a reconciliation of the amount of clearing undertaken (in hectares) will be undertaken for the activities undertaken within the MCNP.

APPENDIX 6

Licence Conditions for L47/47

Cond No	Condition	Compliance
1	The licensee notifying the holder of any underlying pastoral lease by telephone or in person, or by registered post if contact cannot be made, prior to undertaking airborne geophysical surveys or any ground disturbing activities utilising equipment such as scrapers, graders, bulldozers, backhoes, drilling rigs; water carting equipment or other mechanised equipment.	Noted
2	The licensee or transferee, as the case may be, shall within thirty (30) days of receiving written notification of: • the grant of the licence; and • registration of a transfer introducing a new licensee. advise, by registered post, the holder of any underlying pastoral lease details of the grant or transfer.	Noted
3	The terms and conditions contained within: • Statement No. 000514 issued by the Minister for the Environment on 28 June 1999 and entitled "Statement that a proposal may be implemented (pursuant to the provisions of the Environmental Protection Act 1986) - West Angelas Iron Ore Project - Shires of East Pilbara, Ashburton and Roebourne"; and • the approval by the Department of Environmental Protection on 13 August 2001 of the referral dated August 2001 under Part IV of the Environmental Protection Act 1986 and Section 1.3 of State No. 000514.	Noted
4	The area of the licence to be reduced as soon as practicable after construction, to a minimum for safe maintenance and operation of the railway line.	Noted
5	No interference with Geodetic Survey Stations W32, MD35 and W40 and mining within 15 metres thereof being confined to below a depth of 15 metres from the natural surface.	Noted – Project is aware of the location of the survey stations and will not interfere or disturb these locations.
6	No interference with the transmission line or the installations in connection therewith, and the rights of ingress to and egress from the facility being at all times preserved to the owners thereof.	Noted
7	Unless otherwise agreed to by the owners, mining on a strip of land 20 metres wide with any pipeline as the centreline being confined to below a depth of 31 metres from the natural surface and no mining material being deposited upon such strip and the rights of ingress to and egress from the facility being at all times preserved to the owners thereof.	Noted
and in respect to those portions of land designated "PWR/5" and "PCA/1" in Tengraph, being for a proposed water reserve and proposed catchment area, also subject to:		
8	Adequate pollution prevention safeguards being implemented, including the following: • The use and storage of designated substances shall be in accordance with the Water and Rivers Commission "Water Quality Protection Note" (Toxic and Hazardous Substances in Public Drinking Water Source Areas) which may be obtained from the Commission's Regional Office. • The holder of the miscellaneous licence is to be responsible for the immediate cleanup of any spillage of diesel, petrol or other polluting substances. The cleanup will be undertaken to the	Noted

Cond No	Condition	Compliance
	satisfaction of the Water and Rivers Commission and will be done so at the expense of the tenement holder. • The site must be kept free of any materials that may pose a threat to water quality after decommissioning.	
Consent to mine on Water Supply Reserve 38991, Water Supply and Pipeline Reserve 36991 and the Millstream Water Reserve grant subject to:		
9	Adequate pollution prevention safeguards being implemented, including the following: • The use and storage of designated substances shall be in accordance with the Water and Rivers Commission "Water Quality Protection Note" (Toxic and Hazardous Substances in Public Drinking Water Source Areas) which may be obtained from the Commission's Regional Office. • The holder of the miscellaneous licence is to be responsible for the immediate cleanup of any spillage of diesel, petrol or other polluting substances. The cleanup will be undertaken to the satisfaction of the Water and Rivers Commission and will be done so at the expense of the tenement holder. • The site must be kept free of any materials that may pose a threat to water quality after decommissioning.	Noted
10	No activity being carried out that will in any manner affect or compromise the physical integrity, access, operation and maintenance of any of the Water Corporation's infrastructure, assets or operations.	Noted
11	No activity being carried out within 10 metres of any part of the Water Corporation's infrastructure, assets or operations without the written consent of the Water Corporation, which consent may be subject to such conditions as the Water Corporation sees fit.	Noted
12	For the purpose of Conditions 10 and 11, the infrastructure and assets of the Water Corporation includes all current and future assets and infrastructure owned, occupied or operated by the Water Corporation, its agents, contractors or assigns. They include but are not limited to: • plant, facilities and equipment for treatment, communication, monitoring, storage, transport and construction; • access roads, routes or tracks; • pipes, meters, apparatus used for in connection with the supply of water; • excavations, structures, buildings, plant and equipment; and • reservoirs, storage dams, weirs, wells, bores, tanks and buried pipework.	Noted
13	The licensee giving written notice to the Water Corporation of its intention to carry out any exploration or mining operations, 21 days prior to the commencement of any such operations.	Noted
Consent to mine on Repeater Station Site Reserve 40091 granted subject to:		
14	Reasonable access to the reserve being at all times maintained in favour of the vested authority.	Noted
Consent to mine on Resting Place for Travellers & Stock Reserve 27915 granted subject to:		
15	No mining operations being carried out on Resting Place for Travellers and Stock Reserve 27915 which restrict the use of the reserve.	Noted
Consent to mine on Millstream-Chichester National Park Reserves 30071 and 38333 granted subject to:		

Cond No	Condition	Compliance
16	Prior to any environmental disturbance the licensee will prepare detailed environmental management plans for both construction and maintenance phases and obtain the written approval of the Regional Manager, Department of Conservation and Land Management (CALM).The plan is to include but not limited to: • Maps and aerial photographs showing in detail the plan of the project. • The purpose, specifications and life of all proposed disturbances including drainage management design, material requirements. • Proposals which may disturb any declared rare or geographically restricted flora and fauna. • Details of Environmental safeguards incorporated into project design to ensure damage to the environment is minimised especially in regard to drainage modification and shadow impacts on vegetation. • Programs and commitments for ongoing weed and drainage management. • Techniques, prescriptions and target dates for the rehabilitation of all proposed disturbances. • Measures that will be undertaken to protect assets from fire.	A copy of this Mining Proposal has been provided to the DEC. Activities will not commence until approval has been received in writing from the DEC.
17	Prior to any environmental disturbance, the licensee will provide a bank guaranteed unconditional performance bond in favour of the Minister for State Development for due compliance with the approved plan for the sum specified in the approval.	Noted
18	The licensee to provide an annual report to the Regional Manager, CALM detailing environmental performance and any disturbances and rehabilitation plans for the following year.	Noted
19	The licensee is to pay to the Executive Director, CALM, compensation for all areas cleared of vegetation for mining, in support of mining and degraded by clearing as a result of mining, in association with this licence. Provided that the Executive Director of CALM and the licensee may agree from time to time that land shall be transferred or works undertaken by the licensee in lieu of payment under this condition. Compensation shall also cover CALM's administrative costs for inspections and condition compliance monitoring of works. The rate of compensation for land clearing shall be $2,736.54 per hectare and be adjusted annually by the Executive Director of CALM. The CPI for Perth will provide the basis of such adjustments. The rate of compensation for CALM's administrative costs resulting from mining or mining related works shall be the actual CALM full costs of necessary inspections and monitoring and associated office work. The rate for this administrative costs compensation is to be estimated in advance of CALM's work being undertaken and is to be agreed by the licensee. If no agreement can be reached, a default standard compensation rate for administrative costs, equal to the calculated land clearing rate, shall apply. The Executive Director of CALM shall issue a notice for the amount of compensation due and the licensee shall pay the amount within two (2) months of the date of such notice. A penalty of interest at the Commonwealth bond rate will be charged for late payment.	Noted
20	(a) The licensee will at all times indemnify and keep indemnified CALM, the Executive Director and its servants and agents from and against all actions claims suits and demands which may be made by any person in respect of loss, damages or personal injury (including death) of any person or persons arising out of or in relation to or incidental to the use of the specified area by the licensee except to the extent caused by the negligent act or omission of CALM, the Executive Director or its servants or agents or arising in connection with paragraphs (a) or (b).(b) The right shall be hereby reserved to the Executive Director, his agents, employees	Noted

Cond No	Condition	Compliance
	or workmen to enter upon and carry out such duties and exercise such powers upon the licence and adjacent areas as it may be necessary or expedient to carry out or exercise in the administration or for the purposes of the Conservation and Land Management Act, 1984 or any other enactment or any regulation made thereunder and the licensee shall not be entitled to any compensation by reason of any inconvenience or disturbance or loss occasioned by any action on the part of the Executive Director, his agents, contractors or employees except to the extent caused by the negligent act or omission of CALM, the Executive Director or its servants or agents.(c) This licence provides for: • use by CALM staff of any roads or tracks maintained by the licence holder; • use of agreed sections of any roads within the licence area for members of the public to gain access to the park; • roads and tracks to be constructed by the licensee, at no cost to other parties, across the licence area to provide vehicular access for the licensee's or the State's management purposes and for public access subject to prior agreement between the licensee and CALM.	
21	The construction and operation of the project and measures to protect the environment being carried out in accordance with the document titled:- • "NOI 4620 Pilbara Rail Additional Proposal Rail Realignment" dated 15 March 2004 and retained on Department of Industry and Resources File No. E2779/200301; • "Notice of Intent to Establish a Fibre Optic Cable Underground between 7 Mile and Kanjenjie" dated 19 January 2006 (NOI 5224) and retained on Department of Industry and Resources File No. E2779/200301; • "Installation of Safe Working Optic Regenerator Unit Adjacent to Pilbara Irons Rail Network - Mining Proposal - Miscellaneous Licence 47/47" (MP 20511) dated 26 September 2008, signed by Alistair Conn and retained on Department of Mines and Petroleum File No. E2780/200303; • "Dampier to Tom Price Railway, Installation of Radio Base Station at 83.5km mark - L47/47, Mining Proposal" (Reg ID 21134) dated 28 January 2009, signed by Jeremy English and retained on Department of Mines and Petroleum File No. E2780/200303; • (Reg ID 30136) "Rail Capacity Expansion Project - Construction of Ibis and Koala Turnout Pads - Early works Mining Proposal L47/47" dated 17 March 2010 signed by Alistair Conn and retained on Department of Mines and Petroleum File No. EARS-MP-30136. • (Reg ID: 31005) "Programme of Work on L47/47 for Rio Tinto Iron Ore" dated 16 May 2011 signed by David Blainey and retained on Department of Mines and Petroleum File No. EARS-POW-31005. • (Reg ID: 31082) "Rail Capacity Enhancement Project, Railway Crossover Construction at Ibis and Koala Sidings, L47/47" dated 30 May 2011, signed by Shreya Shah and letter of the same title, dated 02 August 2011 signed by Shreya Shah, both retained on Department of Mines and Petroleum file No. EARS-MP-31082. Where a difference exists between the above document(s) and the following conditions, then the following conditions shall prevail.	Noted
~~21~~	~~The construction and operation of the project and measures to protect the environment being carried out generally in accordance with the document~~	

Cond No	Condition	Compliance
	~~titled:-~~ - ~~"NOI 4620 Pilbara Rail Additional Proposal Rail Realignment" dated 15 March 2004 and retained on Department of Industry and Resources File No. E2779/200301;~~ - ~~"Notice of Intent to Establish a Fibre Optic Cable Underground between 7 Mile and Kanjenjie" dated 19 January 2006 (NOI 5224) and retained on Department of Industry and Resources File No. E2779/200301;~~ - ~~"Installation of Safe Working Optic Regenerator Unit Adjacent to Pilbara Irons Rail Network - Mining Proposal - Miscellaneous Licence 47/47" (MP 20511) dated 26 September 2008, signed by Alistair Conn and retained on Department of Mines and Petroleum File No. E2780/200303;~~ - ~~"Dampier to Tom Price Railway, Installation of Radio Base Station at 83.5km mark - L47/47, Mining Proposal" (Reg ID 21134) dated 28 January 2009, signed by Jeremy English and retained on Department of Mines and Petroleum File No. E2780/200303;~~ - ~~(Reg ID 30136) "Rail Capacity Expansion Project - Construction of Ibis and Koala Turnout Pads - Early works Mining Proposal L47/47" dated 17 March 2010 signed by Alistair Conn and retained on Department of Mines and Petroleum File No. EARS-MP-30136.~~ - ~~(Reg ID: 31005) "Programme of Work on L47/47 for Rio Tinto Iron Ore" dated 16 May 2011 signed by David Blainey and retained on Department of Mines and Petroleum File No. EARS-POW-31005.~~ ~~Where a difference exists between the above document(s) and the following conditions, then the following conditions shall prevail.~~	
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~~21~~	~~The construction and operation of the project and measures to protect the~~	

Cond No	Condition	Compliance
	~~environment being carried out generally in accordance with the document titled:-~~ • ~~"NOI 4620 Pilbara Rail Additional Proposal Rail Realignment" dated 15 March 2004 and retained on Department of Industry and Resources File No. E2779/200301.~~ ~~Where a difference exists between the above document(s) and the following conditions, then the following conditions shall prevail.~~	
~~22~~	~~The licensee arranging lodgement of an Unconditional Performance Bond executed by a Bank or other approved financial institution in favour of the Minister for State Development for due compliance with the environmental conditions of the licence in the sum of:~~	
~~22~~	~~The licensee arranging lodgement of an Unconditional Performance Bond executed by a Bank or other approved financial institution in favour of the Minister for State Development for due compliance with the environmental conditions of the licence in the sum of:~~ • ~~$328,500~~	
~~22~~	~~The licensee arranging lodgement of an Unconditional Performance Bond executed by a Bank or other approved financial institution in favour of the Minister for State Development for due compliance with the environmental conditions of the licence in the sum of:~~ • ~~$308,500~~	
23	The development and operation of the project being carried out in such a manner so as to create the minimum practicable disturbance to the existing vegetation and natural landform.	Noted
24	All topsoil being removed ahead of all mining operations from sites such as pit areas, waste disposal areas, ore stockpile areas, pipeline, haul roads and new access roads and being stockpiled for later respreading or immediately respread as rehabilitation progresses.	Noted
25	At the completion of operations, all buildings and structures being removed from site or demolished and buried to the satisfaction of the Director, Environment Division, DoIR.	Noted
26	All rubbish and scrap is to be progressively disposed of in a suitable manner.	Noted
27	At the completion of operations, or progressively where possible, all access roads and other disturbed areas being covered with topsoil, deep ripped and revegetated with local native grasses, shrubs and trees to the satisfaction of the Director, Environment Division, DoIR.	Noted
28	Any alteration or expansion of operations within the lease boundaries beyond that outlined in the above document(s) not commencing until a plan of operations and a programme to safeguard the environment are submitted to the Director, Environment, DoIR for his assessment and until his written approval to proceed has been obtained.	Noted
~~29~~	~~The licensee arranging lodgement of an Unconditional Performance Bond executed by a Bank or other approved financial institution in favour of the Minister responsible for the Mining Act 1978 for due compliance with the environmental conditions of the lease in the sum of:~~ • ~~$308,500.~~	
30	The Licensee submitting to the Executive Director, Environment Division,	Noted

Cond No	Condition	Compliance
	DMP, a brief annual report outlining the project operations, minesite environmental management and rehabilitation work undertaken in the previous 12 months and the proposed operations, environmental management plans and rehabilitation programmes for the next 12 months. This report to be submitted each year in: • April.	
~~30~~	~~The Licensee submitting to the Director, Environment Division, DMP, a brief annual report outlining the project operations, minesite environmental management and rehabilitation work undertaken in the previous 12 months and the proposed operations, environmental management plans and rehabilitation programmes for the next 12 months. This report to be submitted each year in:~~ • ~~March.~~	

Licence Conditions for L47/228

L47/228	Condition	Compliance
1	The licence is granted in accordance with the provisions of Clause 8(1)(b)(i) of the *Iron Ore (Robe River) Agreement Act 1964* and construction and operation of infrastructure and other facilities on this licence proceeding in accordance with proposals approved pursuant to this Agreement.	Noted.
2	The Licensee submitting a plan of proposed operations and measures to safeguard the environment to the Director, Environment, DMP for assessment and written approval prior to commencing any development or construction.	Noted – this Mining Proposal has been submitted to DMP.
3	The rights of ingress to and egress from Miscellaneous Licences 47/47, 47/102 and 47/103 being at all times preserved to the licensee and no interference with the purpose or installations connected to the licence.	Noted.
4	No interference with Geodetic Survey Stations SSM-HRE 152-154, SSM-PYR 58-63 AND SSM-W 32 and mining within 15 metres thereof being confined to below a depth of 15 metres from the natural surface.	Noted – the proposed works will avoid the referenced stations.
5	No interference with the transmission line or the installations in connection therewith without the prior written consent of the owners thereof, and the rights of ingress to and egress from the facility being at all times preserved to the owners thereof.	Noted.
6	Mining on a strip of land 20 metres wide with any pipeline as the centreline being confined to below a depth of 31 metres from the natural surface and no mining material being deposited upon such strip and the rights of ingress to and egress from the facility being at all times preserved to the owners thereof.	Noted.
7	No mining on a strip of land 60 metres wide with the Tom Price and Pannawonica Railway Line as the centreline and no materials being deposited or machinery or buildings being erected on such strip of land without the prior written consent of the owners thereof.	Noted.
8	Blasting operations being controlled so that no damage or injury can be caused by fly rock, concussion, vibration or other means.	Noted.
	Consent to commence any activities in respect to licence purposes on Catchment Area 48 shown in red in Tengraph and Water Supply Reserve 35798 given subject to:-	
9	Written notification, where practicable, of the time frame, type and extent of proposed ground disturbing activities being forwarded to the Department of Water Karratha seven days prior to commencement of those activities.	Noted. This Mining Proposal has also been submitted to DoW.
10	Any significant waterway (flowing or not), wetland or its fringing vegetation that may exist on site not being disturbed or removed without prior written approval from the Department of Water.	Noted.
11	The rights of ingress to and egress from the Licence being at all reasonable times preserved to officers of the Department of Water for inspection	Noted.

	and investigation purposes.	
12	The storage and disposal of hydrocarbons, chemicals and potentially hazardous substances being in accordance with the Department of Water's Guidelines and Water Quality Protection Notes.	Noted – refer to Sections 6.5, 6.8, 6.9 and 6.10.
13	All proposed exploration activities within Public Drinking Water Source Areas complying with the Department of Water's Water Quality Protection Note Land Use Compatibility in Public Drinking Water Source Areas.	Noted – refer to Section 6.5.
14	All Mining Act tenement activities within Public Drinking Water Source Areas being prohibited unless the prior written approval has been obtained from the Department of Water.	Noted. This Mining Proposal has also been submitted to DoW.
15	All Mining Act tenement activities are prohibited within 2 kilometres of the maximum storage level of a reservoir including the reservoir itself, unless the prior written approval of the Department of Water is first obtained.	Noted. This Mining Proposal has also been submitted to DoW.
16	Storage and use of hydrocarbons and potentially hazardous substances requiring the prior written approval or appropriate permits from the Department of Water.	Noted – refer to Sections 5.5, 6.5, 6.8, 6.9 and 6.10.
17	All hydrocarbon or other pollutant spillage being reported to the Department of Water. Remediation being carried out to the satisfaction of the Department of Water.	Noted – refer to Section 6.9.
18	All Mining Act tenement activities are prohibited within a 500-metre radius in a P1 area or a 300-metre radius in a P2 or P3 area of any Public Drinking Water Source production well or dam, unless the written approval of the Department of Water is first obtained.	Noted.
19	All Mining Act tenement activities are prohibited within a 300-metre radius of any observation well in a Public Drinking Water Source Priority P1, P2 & P3 Areas unless the written approval of the Department of Water is first obtained.	Noted.
	Consent to any activities in respect to licence purposes on Millstream-Chichester National Park Reserve 300071 given subject to:-	
20	Prior to lodgement of a proposal for mining, or other ground-disturbing activities including infrastructure development and operation, the licencee preparing a Conservation Impact Assessment and Management Plan (CIAMP) to identify, address and suitably mitigate and offset where appropriate, the impacts of the proposed mining on the conservation purpose and use of the affected land and submitting the CIAMP to the director General, DEC for approval. The CIAMP shall be prepared in consultation with DEC in a manner consistent with written guidance to be provided by the Director General, DEC. The CIAMP and DEC's decision on its acceptability and any recommended conditions to apply to the related mining proposal is to accompany the lodgement of the Mining Proposal application with the Department of Mines and Petroleum. The process requirements, documentation and conditions of approval relating to a formal environmental assessment of the proposed	Noted – DEC EMB have agreed that this Mining Proposal can be lodged (e-mail dated 26 October 2012).

	developments or activities under Part IV of Environmental Protection Act 1986 may be accepted by the Director General, DEC, as partly or fully meeting the requirements for development and approval of a CIAMP.	
21	At least five working days prior to accessing the reserve, unless otherwise agreed with the DEC Regional Manager, the holder providing the Regional Manager with an itinerary and program of the locations of operations on the licence/lease area and informing the Regional Manger at least five days in advance of any changes to that itinerary. All activities and movements shall comply with reasonable access and travel requirements of the Regional Manager regarding seasonal/ground conditions.	Noted – refer to Section 4.8 and Appendix 5.
22	Rights being reserved to persons authorised by the Director General, DEC, to enter the licence area and carry out land management operations and other duties and exercise such powers as may be necessary or expedient for the administration of the Conservation and Land Management Act 1984 and Regulations, the Wildlife Conservation Act 1950 and Regulations, the Bush Fires Act 1954 and Regulations and the Emergency Management Act 2005 and Regulations provided that such entry shall not, in the opinion of the State Mining Engineer and the Director General, DEC, unduly prejudice or interfere with the company's operations.	Noted.
23	The construction and operation of the project and measures to protect the environment and decommissioning and rehabilitation of the project areas being carried out in accordance with the approved mining proposal and approved CIAMP. The lessee is to submit to the Director of Environment, Department of Mines and Petroleum and the director General, DEC, or delegate, a brief annual report outlining the project operations, environmental management plans and rehabilitation programs for the next 12 months. This report to be submitted in February of each year that on-ground activities (excepting rail transport operations) are conducted.	Noted.